8	7	6	5	4	3	2	1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
			2012-01-12

D7 MLB

LAST_MODIFIED=Thu Jan 12 10:24:09 2012

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9509	1	SCH,MLB,D7	SCH	CRITICAL	
820-3302	1	PCBF,MLB,D7	PCB	CRITICAL	

DRAWING
TITLE=D7
ABBREV=DRAWING
LAST_MODIFIED=Thu Jan 12 10:24:09 2012

DRAWING TITLE: SCH,D7,MLB

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

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8	7	6	5	4	3	2	1

System Block diagram can be found on Kismet

PATH: Kismet > K70/72 > Block Diagrams > K70 Block Diagram

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

System Block Diagram

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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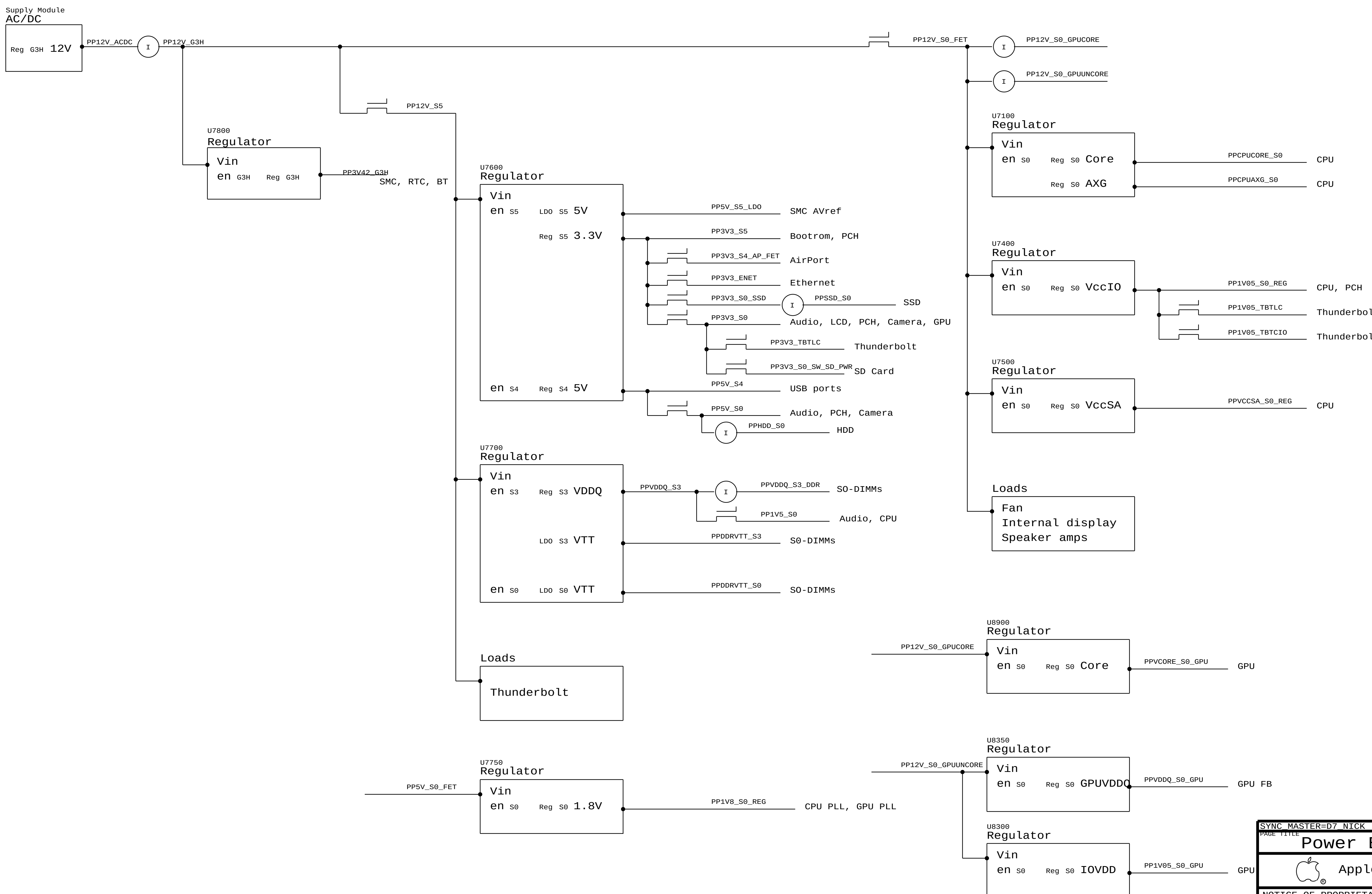

SYNC_MASTER=D7_NICK SYNC_DATE=01/03/2012

PAGE TITLE
Power Block Diagram

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-4441	PCBA,MLB,DEV,D7	DEVELOPMENT,D7_DEVEL
639-3566	PCBA,MLB,D7,GSA,GOOD	D7_COMMON,CPU:GOOD,GPU:GSA,GS,FBA,SSD:N,EEEE:DF98
639-3668	PCBA,MLB,D7,GSB,GOOD	D7_COMMON,CPU:GOOD,GPU:GSB,GS,FBB,SSD:N,EEEE:F117
639-3567	PCBA,MLB,D7,GTX,BETTER	D7_COMMON,CPU:BETTER,GPU:107GTX,FBA,FBB,SSD:Y,EEEE:DT42
639-3665	PCBA,MLB,D7,GTX,CTO	D7_COMMON,CPU:CTO,GPU:107GTX,FBA,FBB,SSD:Y,EEEE:F116

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL,48.0X4.8	EEEE_DF98	CRITICAL	EEEE:DF98
825-7122	1	MLB LABEL,48.0X4.8	EEEE_DT42	CRITICAL	EEEE:DT42
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F116	CRITICAL	EEEE:F116
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F117	CRITICAL	EEEE:F117

BOM Groups

BOM GROUP	BOM OPTIONS
D7_COMMON	COMMON,ALTERNATE,D7_COMMON1,D7_COMMON2,D7_PROGPARTS
D7_COMMON1	XDP,RSMRST:SMC,SPEAKERID,TBTHV:P12V
D7_COMMON2	SNS_CPUCORE:3PHASE,CPUCOREDRV:ISL6612,IG:N,GPU_ROM:YES,SNS_GPUS0:K70,SNS_VDDQS3_DDR:Y
D7_PROGPARTS	SMC:PROTO1,BOOTROM:PROG,T29ROM:PROG,CIVROM:PROG,CAMROM:PROG
D7_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM,DEVEL_SENSORS,DEVEL_AUDIO
DEVEL_SENSORS	SNS_VDDQS0_GPU:Y,SNS_VDDQS3:Y,TEMPSNSDEV
D7_PRODUCTION	SNS_VDDQS0_GPU:N,SNS_VDDQS3:N,VREFMRGN:N

Add 'K70_PRODUCTION' at RevA release

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4240	1	IVB,QC13,QS,E0,2.8G,65W,4+1,1.10,6M,LGA	CPU	CRITICAL	CPU:GOOD
337S4258	1	IVB,QC48,QS,E1,2.9G,65W,4+2,1.10,6M,LGA	CPU	CRITICAL	CPU:BETTER
337S4246	1	IVB,SR0PN,PRQ,E1,3.1G,65W,4+2,1.15,8M,LG	CPU	CRITICAL	CPU:CTO

Replace with 65W part

Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4234	1	IC,PCH,PPT-DT,Z77,QS,C1	U1800	CRITICAL	
338S1047	1	IC,TBT,CR-4C,ES1,288 FCBGA,12X12MM	U3600	CRITICAL	
337S4221	1	IC,GPU,NV,GK107-GS-2/1-QS-A	U8000	CRITICAL	GPU:GSA
337S4220	1	IC,GPU,NV,GK107-GS-2/1-QS-A	U8000	CRITICAL	GPU:GSB
337S4239	1	IC,GPU,NV,GK107-GTX-QS-A2	U8000	CRITICAL	GPU:107GTX
343S0592	1	IC,BCM57766,CIV+,A0,8X8	U3900	CRITICAL	
607-9432	1	K70,GDDR5,SAMSUNG	VRAM	CRITICAL	GPU:107GTX

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3493	1	IC,CR,V24.2,D7/D7I	U3690	CRITICAL	T29ROM:PROG
335S0865	1	IC,EEPROM,SERIAL,256KB,MLP8	U3690	CRITICAL	T29ROM:BLANK
335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK
341S3480	1	IC,PROGRMD,EFI ROM,K70	U5110	CRITICAL	BOOTROM:PROG
335S0862	1	IC,SERIAL FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	CIVROM:BLANK
341S3487	1	IC,ENET 1MBITFLASH,CIV,PVT,J40	U3990	CRITICAL	CIVROM:PROG
338S1098	1	IC,SMC12-A3,BLANK,D7	U4900	CRITICAL	SMC:BLANK
341S3484	1	IC,SMC,PROGRMD,PROTO1,D7	U4900	CRITICAL	SMC:PROTO1
341S3388	1	IC,SMC,PROGRMD,EVT,D7	U4900	CRITICAL	SMC:EVT
341S3389	1	IC,SMC,PROGRMD,DVT,D7	U4900	CRITICAL	SMC:DVT
341S3390	1	IC,SMC,PROGRMD,PVT,D7	U4900	CRITICAL	SMC:PVT
341S3409	1	IC,SMC,PROGRMD,PROD,D7	U4900	CRITICAL	SMC:PROD
341S3453	1	IC,CAMERA FLASH,K70/K72	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK

Alternate: 335S0812

Alternate: 335S0854

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0107	377S0126		ALL	USB diodes
157S0055	157S0058		ALL	Enet magnetics
376S1081	376S0975		ALL	P/NCh dual FET
341S3486	341S3487		ALL	P/NCh dual FET

CPU Socket

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET,MOLEX,LGA1155,CPU-LF	U1000	CRITICAL	

CPU Socket Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

VRAM BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
607-9432	K70,GDDR5,SAMSUNG	FB:BOTH_SAMSUNG
607-9435	K70,GDDR5,HYNIX	FB:BOTH_HYNIX
607-9433	K70,GDDR5,SAMSUNG_CH1	FB:CH1_SAMSUNG
607-9436	K70,GDDR5,HYNIX_CH1	FB:CH1_HYNIX
607-9434	K70,GDDR5,SAMSUNG_CH2	FB:CH2_SAMSUNG
607-9437	K70,GDDR5,HYNIX_CH2	FB:CH2_HYNIX

VRAM Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S0619	4	IC,SGRAM,GDDR5,32MX32,1.5GHz,G-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB:BOTH_SAMSUNG
333S0620	4	IC,GDDR5,32MX32,1.5GHZ,VEGA 44NM,B-DIE	U8400,U8450,U8500,U8550	CRITICAL	FB:BOTH_HYNIX
333S0631	2	IC,SGRAM,GDDR5,64MX32,4.2GBPS,D-DIE,HF	U8400,U8450	CRITICAL	FB:CH1_SAMSUNG
333S0630	2	IC,GDDR5,2GB,M-DIE,170B FBGA	U8400,U8450	CRITICAL	FB:CH1_HYNIX
333S0631	2	IC,SGRAM,GDDR5,64MX32,4.2GBPS,D-DIE,HF	U8500,U8550	CRITICAL	FB:CH2_SAMSUNG
333S0630	2	IC,GDDR5,2GB,M-DIE,170B FBGA	U8500,U8550	CRITICAL	FB:CH2_HYNIX

VRAM Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
607-9435	607-9432		VRAM	GDDR5_BOTH
607-9436	607-9433	GPU:GSA	VRAM	GDDR5_CH1
607-9437	607-9434	GPU:GSB	VRAM	GDDR5_CH2

GPU Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
607-9433	1	K70,GDDR5,SAMSUNG_CH1	VRAM	CRITICAL	GPU:GSA
607-9434	1	K70,GDDR5,SAMSUNG_CH2	VRAM	CRITICAL	GPU:GSB

Programmable Parts (unused)

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S0724	1	IC,1 MBIT SERIAL FLASH	U8701	CRITICAL	GPUROM:BLANK

SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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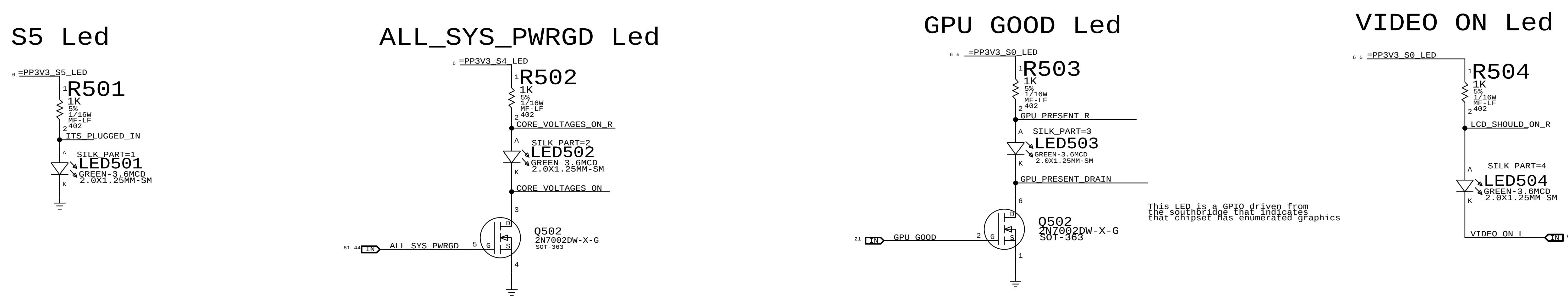

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

DEBUG LEDS

Apple Inc.

DRAWING NUMBER	SIZE
051-9509	D
REVISION 4.2.0	
BRANCH	
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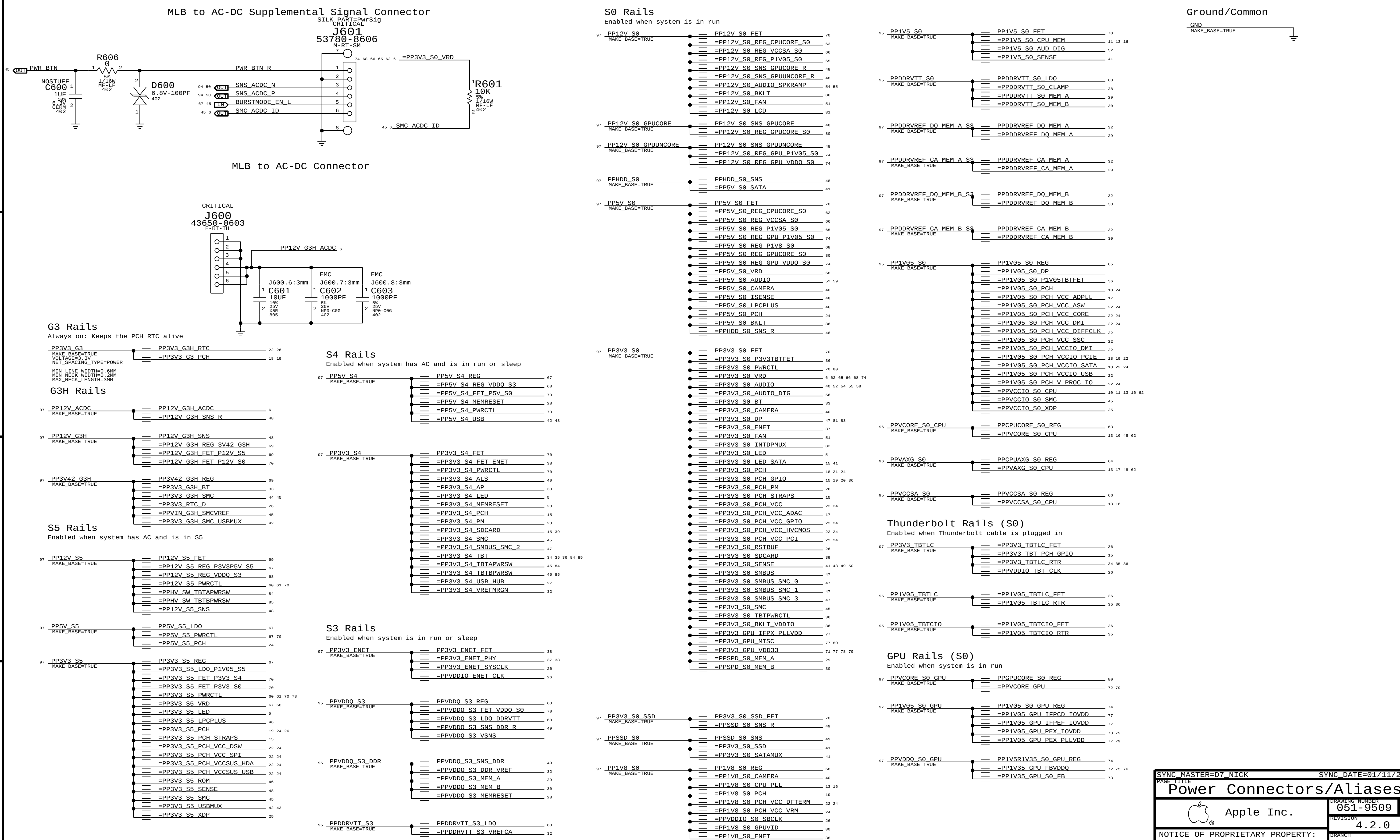
SHEET 5 OF 100	



SYNC_MASTER=D7_NICK SYNC_DATE=01/11/2012

PAGE TITLE

Power Connectors/Aliases

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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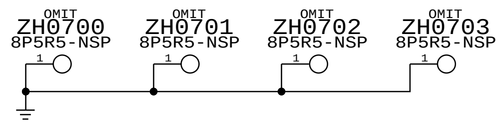
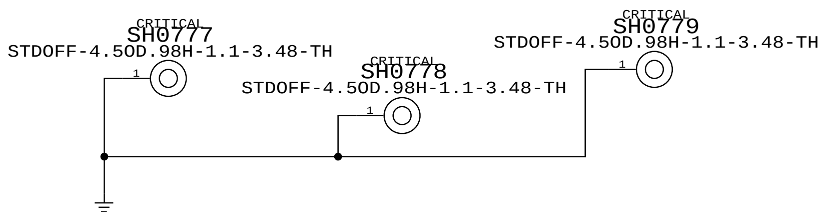
CPU Heatsink

4mm Plated Holes (998-0850)



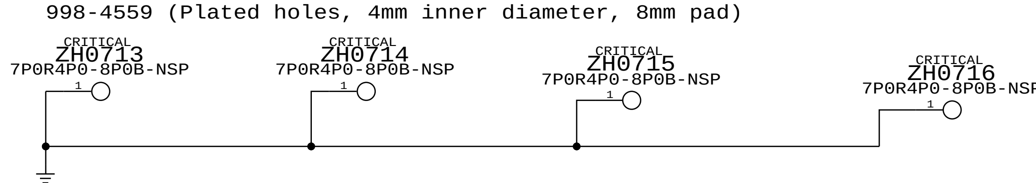
GPU HEATSINK MOUNTING FEATURES

(860-0988)



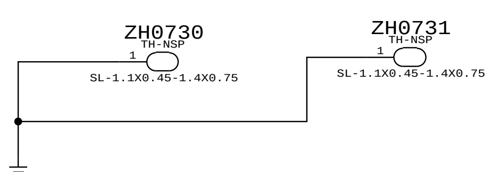
Rear Cover

998-4559 (Plated holes, 4mm inner diameter, 8mm pad)



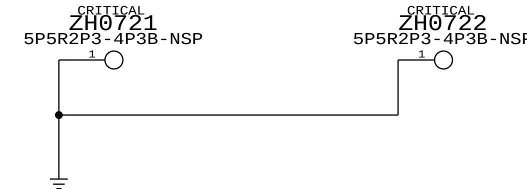
USB Can holes

998-3975 (Plated slot holes, 1.10mm x 0.45mm)



WIRELESS CARD MTG HOLES

998-4560 (Plated holes, 2.3mm inner diameter, 4.3mm pad)

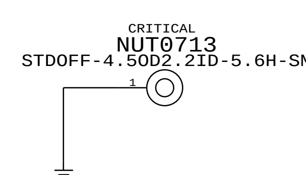


POGO PINS

APN: 870-1939

SSD STANDOFF

APN: 860-1461



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE
Holes/PD parts

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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8	7	6	5	4	3	2	1

CPU Reserved

Page	Net	Alias	Properties
10	TP_CPU_RSVD<16..1>	NC_CPU_RSVD<16..1>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_RSVD<46..19>	NC_CPU_RSVD<46..19>	MAKE_BASE=TRUE NO_TEST=TRUE
91 10	CPU_CFG<15..12>	TP_CPU_CFG<15..12>	MAKE_BASE=TRUE

CPU Memory

Page	Net	Alias	Properties
12	TP_MEM_A_DQ_CB<7..0>	NC_MEM_A_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQ_CB<7..0>	NC_MEM_B_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_A_CLK_N<2..3>	NC_MEM_A_CLKN<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_A_CLK_P<2..3>	NC_MEM_A_CLKP<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_A_CS_L<2..3>	NC_MEM_A_CS_L<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_A_CKE<2..3>	NC_MEM_A_CKE<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_B_CLK_N<2..3>	NC_MEM_B_CLKN<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_B_CLK_P<2..3>	NC_MEM_B_CLKP<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_B_CS_L<2..3>	NC_MEM_B_CS_L<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_B_CKE<2..3>	NC_MEM_B_CKE<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_A_ODT<2..3>	NC_MEM_A_ODT<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
88 12	MEM_B_ODT<2..3>	NC_MEM_B_ODT<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH USB

Page	Net	Alias	Properties
20	USB_PCH_4_N	NC_USB_PCH_4_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_4_P	NC_USB_PCH_4_P	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_N	NC_USB_PCH_5_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_P	NC_USB_PCH_5_P	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_N	NC_USB_PCH_6_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_P	NC_USB_PCH_6_P	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_N	NC_USB_PCH_11_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_P	NC_USB_PCH_11_P	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_N	NC_USB_PCH_12_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_P	NC_USB_PCH_12_P	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_N	NC_USB_PCH_13_N	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_P	NC_USB_PCH_13_P	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PLL

Page	Net	Alias	Properties
22	PP1V05_S0_PCH_VCCAPLLDMI2	NC_PP1V05_S0_PCH_VCCAPLLDMI2	MAKE_BASE=TRUE NO_TEST=TRUE
22	PP1V05_S0_PCH_VCCAPLL_EXP	NC_PP1V05_S0_PCH_VCCAPLL_EXP	MAKE_BASE=TRUE NO_TEST=TRUE
22	PP1V05_S0_PCH_VCCAPLL_SATA	NC_PP1V05_S0_PCH_VCCAPLL_SATA	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Clocks

Page	Net	Alias	Properties
18	TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4N	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4P	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6N	NC_PCIE_CLK100M_PE6N	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6P	NC_PCIE_CLK100M_PE6P	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7N	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PE_TX_N<3..0>	NC_PE_TXN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PE_TX_P<3..0>	NC_PE_TXP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PE_RX_N<3..0>	NC_PE_RXN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PE_RX_P<3..0>	NC_PE_RXP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_N	NC_DMI_MIDBUS_CLK100N	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_P	NC_DMI_MIDBUS_CLK100P	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CLKOUT_PEG_A_N	NC_CLKOUT_PEG_AN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CLKOUT_PEG_A_P	NC_CLKOUT_PEG_AP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CLKOUT_DPN	NC_PCH_CLKOUT_DPN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CLKOUT_DPP	NC_PCH_CLKOUT_DPP	MAKE_BASE=TRUE NO_TEST=TRUE
18	PCH_CLK25M_XTALOUT	NC_PCH_CLK25M_XTALOUT	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO64_CLKOUTFLEX0	NC_PCH_GPIO64_CLKOUTFLEX0	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO65_CLKOUTFLEX1	NC_PCH_GPIO65_CLKOUTFLEX1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO66_CLKOUTFLEX2	NC_PCH_GPIO66_CLKOUTFLEX2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO67_CLKOUTFLEX3	NC_PCH_GPIO67_CLKOUTFLEX3	MAKE_BASE=TRUE NO_TEST=TRUE

PCH and CPU FDI

Page	Net	Alias	Properties
91 10	CPU_FDI_TX_N<7..0>	NC_CPU_FDI_TXN<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
91 10	CPU_FDI_TX_P<7..0>	NC_CPU_FDI_TXP<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_N<7..0>	NC_PCH_FDI_RXN<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_P<7..0>	NC_PCH_FDI_RXP<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
91 10	CPU_FDI_FSYNC<1..0>	NC_CPU_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
91 10	CPU_FDI_LSYNC<1..0>	NC_CPU_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
91 10	CPU_FDI_INT	NC_CPU_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_FSYNC<1..0>	NC_PCH_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_LSYNC<1..0>	NC_PCH_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_INT	NC_PCH_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Unused Display

Page	Net	Alias	Properties
19	TP_CRT_IG_RED	NC_CRT_IG_RED	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_GREEN	NC_CRT_IG_GREEN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_BLUE	NC_CRT_IG_BLUE	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_HSYNC	NC_CRT_IG_HSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_VSYNC	NC_CRT_IG_VSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_CLK	NC_CRT_IG_DDC_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_DATA	NC_CRT_IG_DDC_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLN<3..0>	NC_DP_IG_B_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLP<3..0>	NC_DP_IG_B_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_N	NC_DP_IG_B_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_P	NC_DP_IG_B_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_HPD	NC_DP_IG_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_CLK	NC_DP_IG_B_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_DATA	NC_DP_IG_B_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLN<3..0>	NC_DP_IG_C_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLP<3..0>	NC_DP_IG_C_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_N	NC_DP_IG_C_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_P	NC_DP_IG_C_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_HPD	NC_DP_IG_C_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_CLK	NC_DP_IG_C_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_DATA	NC_DP_IG_C_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLN<3..0>	NC_DP_IG_D_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLP<3..0>	NC_DP_IG_D_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXN	NC_DP_IG_D_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXP	NC_DP_IG_D_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_HPD	NC_DP_IG_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_CLK	NC_DP_IG_D_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_DATA	NC_DP_IG_D_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINN	NC_SDVO_TVCLKINN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINP	NC_SDVO_TVCLKINP	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLN	NC_SDVO_STALLN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLP	NC_SDVO_STALLP	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTN	NC_SDVO_INTN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTP	NC_SDVO_INTP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTCTL	NC_PCH_L_BKLTCTL	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTEN	NC_PCH_L_BKLTEN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_VDD_EN	NC_PCH_L_VDD_EN	MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED GRAPHICS ALIASES

Page	Net	Alias	Properties
	TP_DVPCNTL_M<0..1>	NC_DVPCNTL_M<0..1>	MAKE_BASE=TRUE NO_TEST=TRUE
	TP_DVPCNTL<0..2>	NC_DVPCNTL<0..2>	MAKE_BASE=TRUE NO_TEST=TRUE
	TP_DVPCLK	NC_DVPCLK	MAKE_BASE=TRUE NO_TEST=TRUE
	TP_DVPDATA<4..23>	NC_DVPDATA<4..23>	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_CLK_C_P	NC_HDMI_EG_CLK_C_P	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_CLK_C_N	NC_HDMI_EG_CLK_C_N	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_DDC_CLK	NC_HDMI_EG_DDC_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_DDC_DATA	NC_HDMI_EG_DDC_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_DATA_C_P<0..2>	NC_HDMI_EG_DATA_C_P<0..2>	MAKE_BASE=TRUE NO_TEST=TRUE
	HDMI_EG_DATA_C_N<0..2>	NC_HDMI_EG_DATA_C_N<0..2>	MAKE_BASE=TRUE NO_TEST=TRUE
77	GPU_TDIODE_P	NC_GPU_TDIODE_P	MAKE_BASE=TRUE NO_TEST=TRUE
77	GPU_TDIODE_N	NC_GPU_TDIODE_N	MAKE_BASE=TRUE NO_TEST=TRUE
78	EG_LCD_PWR_EN	NC_EG_LCD_PWR_EN	MAKE_BASE=TRUE NO_TEST=TRUE

PCH SATA

Page	Net	Alias	Properties
18	TP_SATA_C_R2D_CN	NC_SATA_C_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_R2D_CP	NC_SATA_C_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RN	NC_SATA_C_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RP	NC_SATA_C_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CN	NC_SATA_D_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CP	NC_SATA_D_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RN	NC_SATA_D_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RP	NC_SATA_D_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CN	NC_SATA_E_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CP	NC_SATA_E_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RN	NC_SATA_E_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RP	NC_SATA_E_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CN	NC_SATA_F_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CP	NC_SATA_F_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RN	NC_SATA_F_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RP	NC_SATA_F_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Reserved

Page	Net	Alias	Properties
19	TP_PCH_RESERVE_0	NC_PCH_RESERVE_0	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_1	NC_PCH_RESERVE_1	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_2	NC_PCH_RESERVE_2	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_3	NC_PCH_RESERVE_3	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_4	NC_PCH_RESERVE_4	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_5	NC_PCH_RESERVE_5	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_6	NC_PCH_RESERVE_6	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_7	NC_PCH_RESERVE_7	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_8	NC_PCH_RESERVE_8	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_9	NC_PCH_RESERVE_9	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_10	NC_PCH_RESERVE_10	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_11	NC_PCH_RESERVE_11	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_12	NC_PCH_RESERVE_12	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_13	NC_PCH_RESERVE_13	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_14	NC_PCH_RESERVE_14	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_15	NC_PCH_RESERVE_15	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_16	NC_PCH_RESERVE_16	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_17	NC_PCH_RESERVE_17	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_18	NC_PCH_RESERVE_18	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_19	NC_PCH_RESERVE_19	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_20	NC_PCH_RESERVE_20	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_21	NC_PCH_RESERVE_21	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_22	NC_PCH_RESERVE_22	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_23	NC_PCH_RESERVE_23	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_24	NC_PCH_RESERVE_24	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_25	NC_PCH_RESERVE_25	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_26	NC_PCH_RESERVE_26	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_27	NC_PCH_RESERVE_27	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_28	NC_PCH_RESERVE_28	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Test Points

Page	Net	Alias	Properties
21	TP_PCH_TP1	NC_PCH_TP1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP2	NC_PCH_TP2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP3	NC_PCH_TP3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP4	NC_PCH_TP4	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP5	NC_PCH_TP5	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP6	NC_PCH_TP6	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP7	NC_PCH_TP7	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP8	NC_PCH_TP8	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP9	NC_PCH_TP9	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP10	NC_PCH_TP10	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP11	NC_PCH_TP11	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP12	NC_PCH_TP12	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP13	NC_PCH_TP13	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP14	NC_PCH_TP14	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP15	NC_PCH_TP15	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP16	NC_PCH_TP16	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP17	NC_PCH_TP17	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP18	NC_PCH_TP18	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP19	NC_PCH_TP19	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP20	NC_PCH_TP20	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCI

Page	Net	Alias	Properties
20	TP_PCI_AD<31..0>	NC_PCI_AD<31..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_C_BE_L<3..0>	NC_PCI_C_BE_L<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_PAR	NC_PCI_PAR	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_RESET_L	NC_PCI_RESET_L	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_INIT3V3_L	NC_PCH_INIT3V3_L	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_LPC_DREQ0_L	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Miscellaneous

Page	Net	Alias	Properties
18	TP_HDA_SDIN1	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN2	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN3	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM0	NC_PCH_PWM0	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM1	NC_PCH_PWM1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM2	NC_PCH_PWM2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM3	NC_PCH_PWM3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_SST	NC_PCH_SST	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_CLK1	NC_PCH_CL_CLK1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_DATA1	NC_PCH_CL_DATA1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_RST1	NC_PCH_CL_RST1	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT2	NC_PCI_CLK33M_OUT2	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT3	NC_PCI_CLK33M_OUT3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_GPIO8	NC_PCH_GPIO8	MAKE_BASE=TRUE NO_TEST=TRUE
22	PP1V05_S0_PCH_FDIPLL	NC_PP1V05_S0_PCH_FDIPLL	MAKE_BASE=TRUE NO_TEST=TRUE
22	PP1V05_S0_PCH_VCC_A_CLK	NC_PP1V05_S0_PCH_VCC_A_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
22	TP_PPVOUT_PCH_DCPSUSBYP	NC_PPVOUT_PCH_DCPSUSBYP	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

Unused Signal Aliases

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8	7	6	5	4	3	2	1

Whistler aliases

89 71 PEG_D2R_N<0..15> MAKE_BASE=TRUE =PEG_D2R_N<15..0> 10

89 71 PEG_D2R_P<0..15> MAKE_BASE=TRUE =PEG_D2R_P<15..0> 10

89 71 PEG_R2D_C_N<0..15> MAKE_BASE=TRUE =PEG_R2D_C_N<15..0> 10

89 71 PEG_R2D_C_P<0..15> MAKE_BASE=TRUE =PEG_R2D_C_P<15..0> 10

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

Signal Aliases

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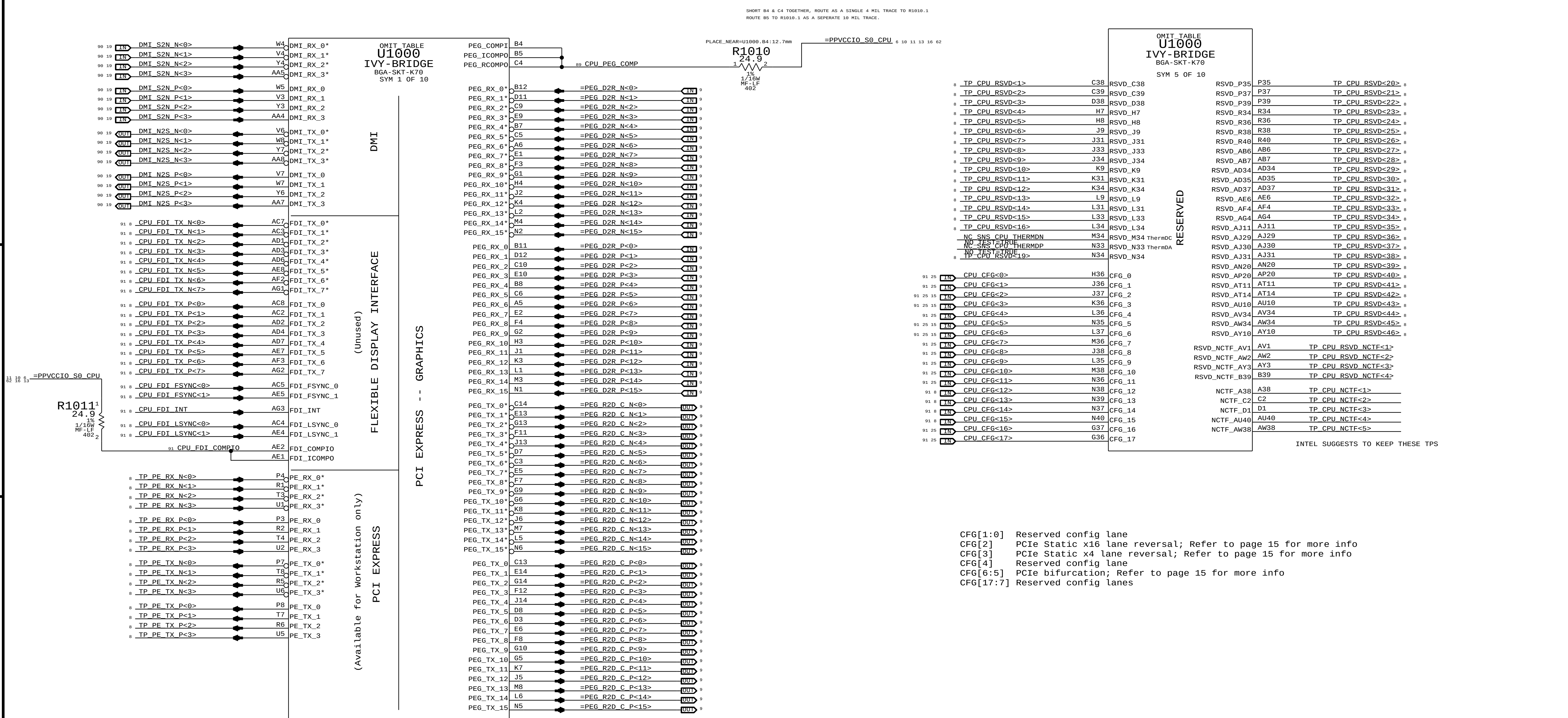

CFG[1:0] Reserved config lane
CFG[2] PCIe Static x16 lane reversal; Refer to page 15 for more info
CFG[3] PCIe Static x4 lane reversal; Refer to page 15 for more info
CFG[4] Reserved config lane
CFG[6:5] PCIe bifurcation; Refer to page 15 for more info
CFG[17:7] Reserved config lanes

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

CPU DMI/PEG/FDI/RSVD

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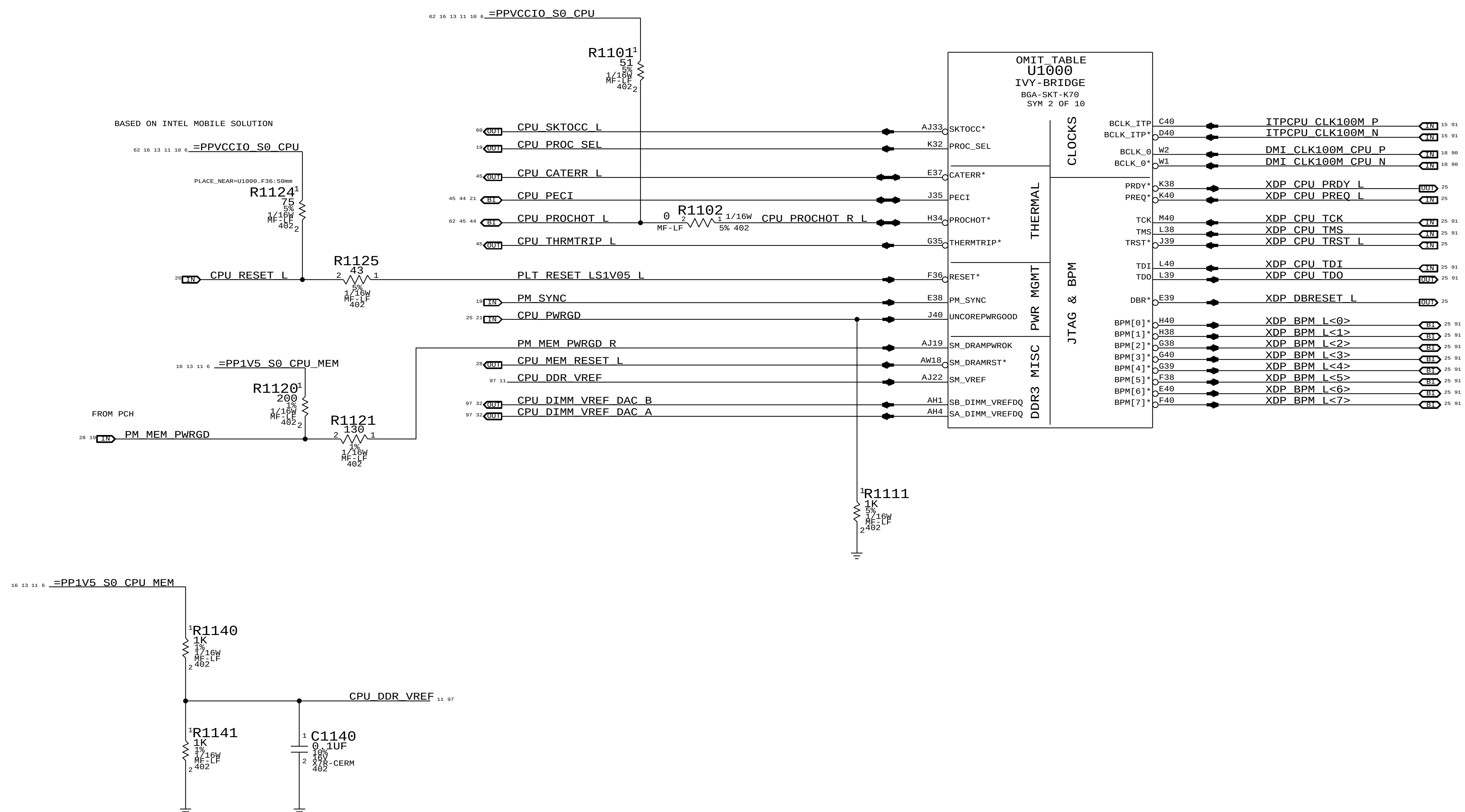

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

CPU CLOCK/MISC/JTAG

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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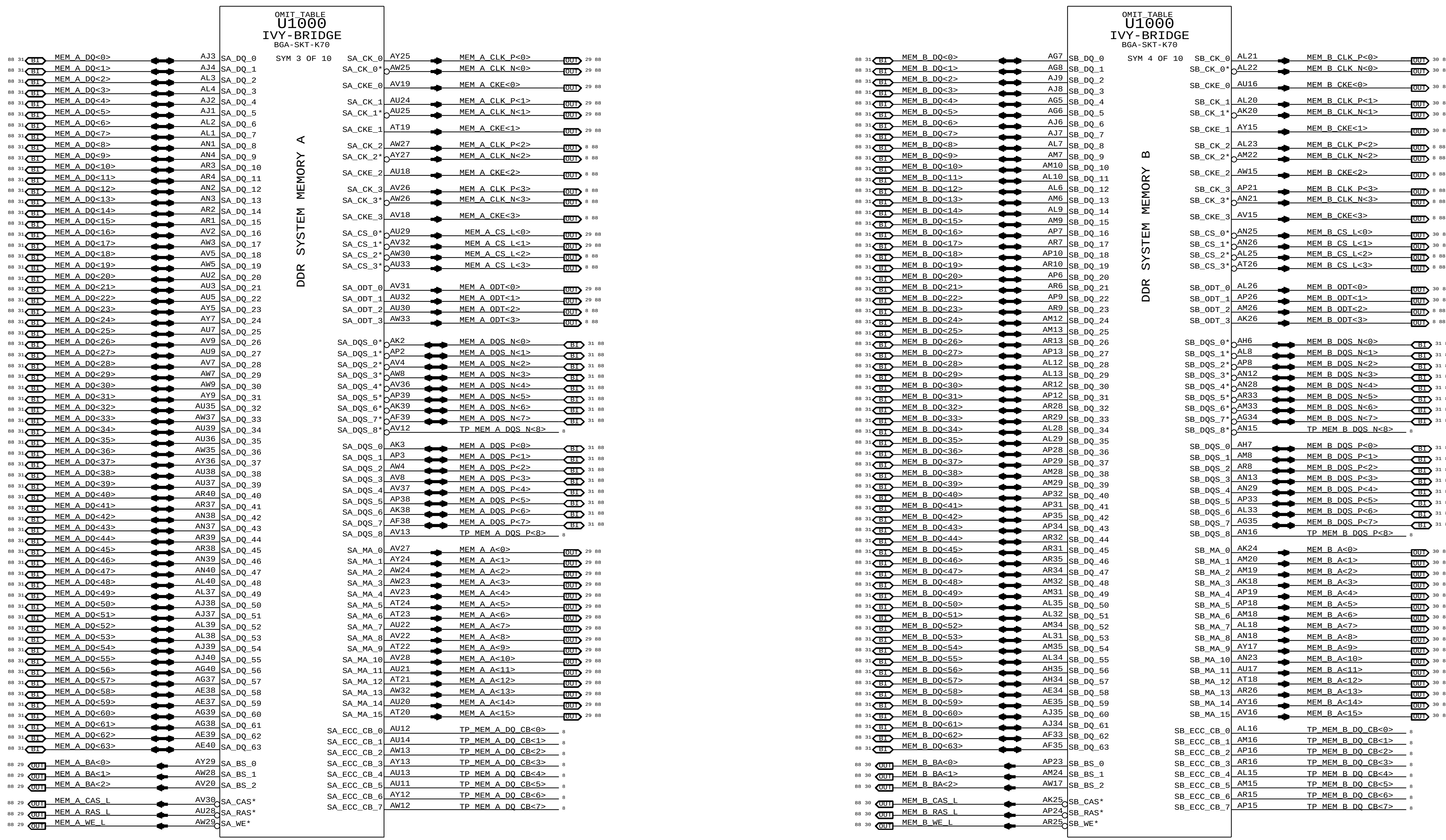

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

CPU DDR3 INTERFACES

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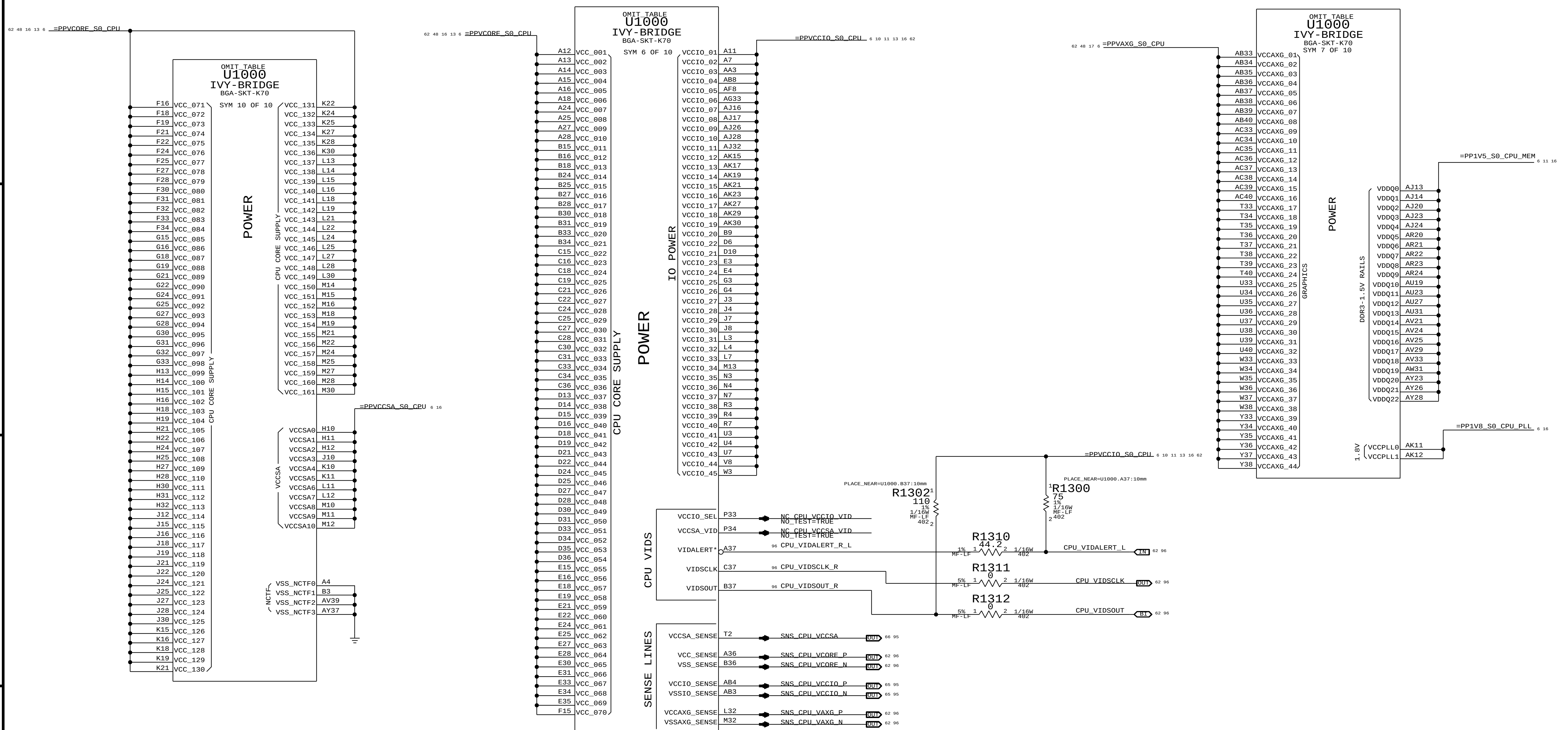

SYNC_MASTER=K70_MLB	SYNC_DATE=11/30/2011
PAGE TITLE: CPU POWER	
Apple Inc.	DRAWING NUMBER: 051-9509 SIZE: D
	REVISION: 4.2.0
NOTICE OF PROPRIETARY PROPERTY:	BRANCH
	PAGE: 13 OF 113
	SHEET: 13 OF 100

8 7 6 5 4 3 2 1

D C B A

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VSS

A17	VSS_001	VSS_091	AM27
A23	VSS_002	VSS_092	AM3
A26	VSS_003	VSS_093	AM30
A29	VSS_004	VSS_094	AM36
A35	VSS_005	VSS_095	AM37
AA33	VSS_006	VSS_096	AM38
AA34	VSS_007	VSS_097	AM39
AA35	VSS_008	VSS_098	AM4
AA36	VSS_009	VSS_099	AM40
AA37	VSS_010	VSS_100	AM5
AA38	VSS_011	VSS_101	AN10
AA6	VSS_012	VSS_102	AN11
AB5	VSS_013	VSS_103	AN14
AC1	VSS_014	VSS_104	AN17
AC6	VSS_015	VSS_105	AN19
AD33	VSS_016	VSS_106	AN22
AD36	VSS_017	VSS_107	AN24
AD38	VSS_018	VSS_108	AN27
AD39	VSS_019	VSS_109	AN30
AD40	VSS_020	VSS_110	AN31
AD5	VSS_021	VSS_111	AN32
AD8	VSS_022	VSS_112	AN33
AE3	VSS_023	VSS_113	AN34
AE33	VSS_024	VSS_114	AN35
AE36	VSS_025	VSS_115	AN36
AF1	VSS_026	VSS_116	AN5
AF34	VSS_027	VSS_117	AN6
AF36	VSS_028	VSS_118	AN7
AF37	VSS_029	VSS_119	AN8
AF40	VSS_030	VSS_120	AN9
AF5	VSS_031	VSS_121	AP1
AF6	VSS_032	VSS_122	AP11
AF7	VSS_033	VSS_123	AP14
AG36	VSS_034	VSS_124	AP17
AH2	VSS_035	VSS_125	AP22
AH3	VSS_036	VSS_126	AP25
AH33	VSS_037	VSS_127	AP27
AH36	VSS_038	VSS_128	AP30
AH37	VSS_039	VSS_129	AP36
AH38	VSS_040	VSS_130	AP37
AH39	VSS_041	VSS_131	AP4
AH40	VSS_042	VSS_132	AP40
AH5	VSS_043	VSS_133	AP5
AH8	VSS_044	VSS_134	AR11
AJ12	VSS_045	VSS_135	AR14
AJ15	VSS_046	VSS_136	AR17
AJ18	VSS_047	VSS_137	AR18
AJ21	VSS_048	VSS_138	AR19
AJ25	VSS_049	VSS_139	AR27
AJ27	VSS_050	VSS_140	AR30
AJ36	VSS_051	VSS_141	AR36
AJ5	VSS_052	VSS_142	AR5
AK1	VSS_053	VSS_143	AT1
AK10	VSS_054	VSS_144	AT10
AK13	VSS_055	VSS_145	AT12
AK14	VSS_056	VSS_146	AT13
AK16	VSS_057	VSS_147	AT15
AK22	VSS_058	VSS_148	AT16
AK28	VSS_059	VSS_149	AT17
AK31	VSS_060	VSS_150	AT2
AK32	VSS_061	VSS_151	AT25
AK33	VSS_062	VSS_152	AT27
AK34	VSS_063	VSS_153	AT28
AK35	VSS_064	VSS_154	AT29
AK36	VSS_065	VSS_155	AT3
AK37	VSS_066	VSS_156	AT30
AK4	VSS_067	VSS_157	AT31
AK40	VSS_068	VSS_158	AT32
AK5	VSS_069	VSS_159	AT33
AK6	VSS_070	VSS_160	AT34
AK7	VSS_071	VSS_161	AT35
AK8	VSS_072	VSS_162	AT36
AK9	VSS_073	VSS_163	AT37
AL11	VSS_074	VSS_164	AT38
AL14	VSS_075	VSS_165	AT39
AL17	VSS_076	VSS_166	AT4
AL19	VSS_077	VSS_167	AT40
AL24	VSS_078	VSS_168	AT5
AL27	VSS_079	VSS_169	AT6
AL30	VSS_080	VSS_170	AT7
AL36	VSS_081	VSS_171	AT8
AL5	VSS_082	VSS_172	AT9
AM1	VSS_083	VSS_173	AU1
AM11	VSS_084	VSS_174	AU15
AM14	VSS_085	VSS_175	AU26
AM17	VSS_086	VSS_176	AU34
AM2	VSS_087	VSS_177	AU4
AM21	VSS_088	VSS_178	AU6
AM23	VSS_089	VSS_179	AU8
AM25	VSS_090	VSS_180	AV10

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VSS

AV11	VSS_181	VSS_271	G8
AV14	VSS_182	VSS_272	H1
AV17	VSS_183	VSS_273	H17
AV3	VSS_184	VSS_274	H2
AV35	VSS_185	VSS_275	H20
AV38	VSS_186	VSS_276	H23
AV6	VSS_187	VSS_277	H26
AW10	VSS_188	VSS_278	H29
AW11	VSS_189	VSS_279	H33
AW14	VSS_190	VSS_280	H35
AW16	VSS_191	VSS_281	H37
AW36	VSS_192	VSS_282	H39
AW6	VSS_193	VSS_283	H5
AY11	VSS_194	VSS_284	H6
AY14	VSS_195	VSS_285	H9
AY18	VSS_196	VSS_286	J11
AY35	VSS_197	VSS_287	J17
AY4	VSS_198	VSS_288	J20
AY6	VSS_199	VSS_289	J23
AY8	VSS_200	VSS_290	J26
B10	VSS_201	VSS_291	J29
B13	VSS_202	VSS_292	J32
B14	VSS_203	VSS_293	K1
B17	VSS_204	VSS_294	K12
B23	VSS_205	VSS_295	K13
B26	VSS_206	VSS_296	K14
B29	VSS_207	VSS_297	K17
B32	VSS_208	VSS_298	K2
B35	VSS_209	VSS_299	K20
B38	VSS_210	VSS_300	K23
B6	VSS_211	VSS_301	K26
C11	VSS_212	VSS_302	K29
C12	VSS_213	VSS_303	K33
C17	VSS_214	VSS_304	K35
C20	VSS_215	VSS_305	K37
C23	VSS_216	VSS_306	K39
C26	VSS_217	VSS_307	K5
C29	VSS_218	VSS_308	K6
C32	VSS_219	VSS_309	L10
C35	VSS_220	VSS_310	L17
C7	VSS_221	VSS_311	L20
C8	VSS_222	VSS_312	L23
D17	VSS_223	VSS_313	L26
D2	VSS_224	VSS_314	L29
D20	VSS_225	VSS_315	L8
D23	VSS_226	VSS_316	M1
D26	VSS_227	VSS_317	M17
D29	VSS_228	VSS_318	M2
D32	VSS_229	VSS_319	M20
D37	VSS_230	VSS_320	M23
D39	VSS_231	VSS_321	M26
D4	VSS_232	VSS_322	M29
D5	VSS_233	VSS_323	M33
D9	VSS_234	VSS_324	M35
E11	VSS_235	VSS_325	M37
E12	VSS_236	VSS_326	M39
E17	VSS_237	VSS_327	M5
E20	VSS_238	VSS_328	M6
E23	VSS_239	VSS_329	M9
E26	VSS_240	VSS_330	N8
E29	VSS_241	VSS_331	P1
E32	VSS_242	VSS_332	P2
E36	VSS_243	VSS_333	P36
E7	VSS_244	VSS_334	P38
E8	VSS_245	VSS_335	P40
F1	VSS_246	VSS_336	P5
F10	VSS_247	VSS_337	P6
F13	VSS_248	VSS_338	R33
F14	VSS_249	VSS_339	R35
F17	VSS_250	VSS_340	R37
F2	VSS_251	VSS_341	R39
F20	VSS_252	VSS_342	R8
F23	VSS_253	VSS_343	T1
F26	VSS_254	VSS_344	T5
F29	VSS_255	VSS_345	T6
F35	VSS_256	VSS_346	U8
F37	VSS_257	VSS_347	V1
F39	VSS_258	VSS_348	V2
F5	VSS_259	VSS_349	V33
F6	VSS_260	VSS_350	V34
F9	VSS_261	VSS_351	V35
G11	VSS_262	VSS_352	V36
G12	VSS_263	VSS_353	V37
G17	VSS_264	VSS_354	V38
G20	VSS_265	VSS_355	V39
G23	VSS_266	VSS_356	V40
G26	VSS_267	VSS_357	V5
G29	VSS_268	VSS_358	W6
G34	VSS_269	VSS_359	Y5
G7	VSS_270	VSS_360	Y8

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

CPU GROUNDS

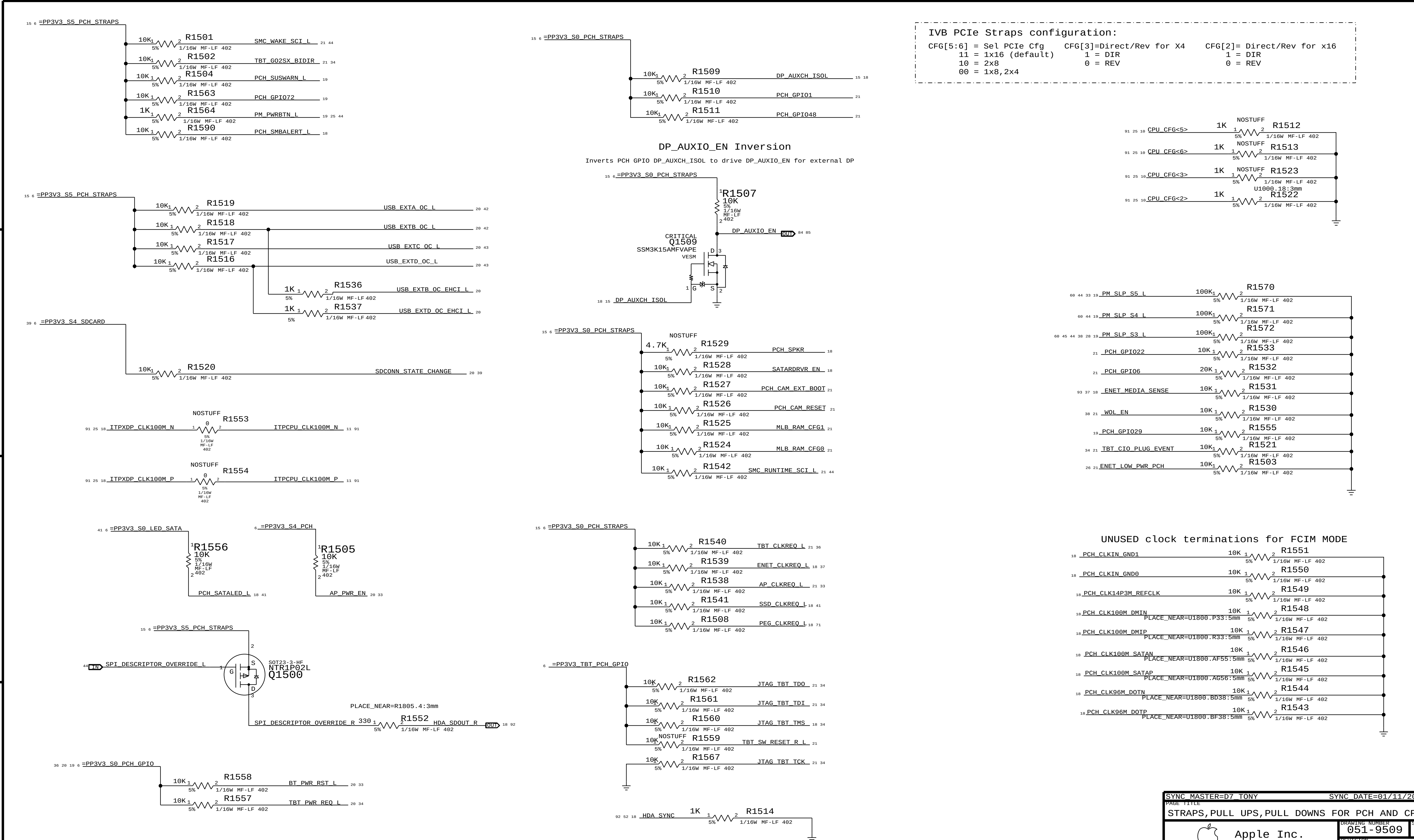
Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

8 7 6 5 4 3 2 1



SYNC_MASTER=D7_TONY SYNC_DATE=01/11/2012

PAGE TITLE

STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

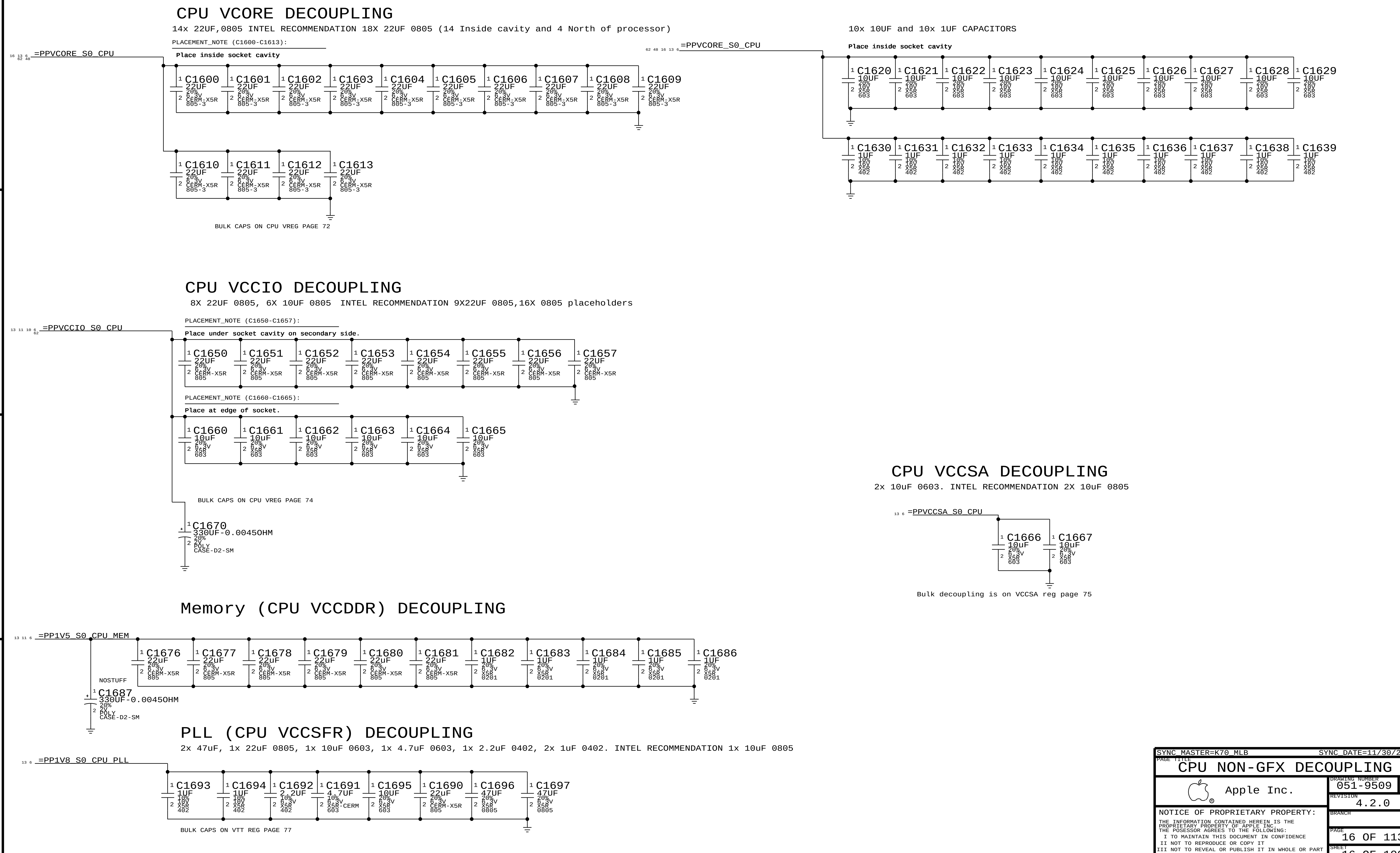
Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

8	7	6	5	4	3	2	1



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

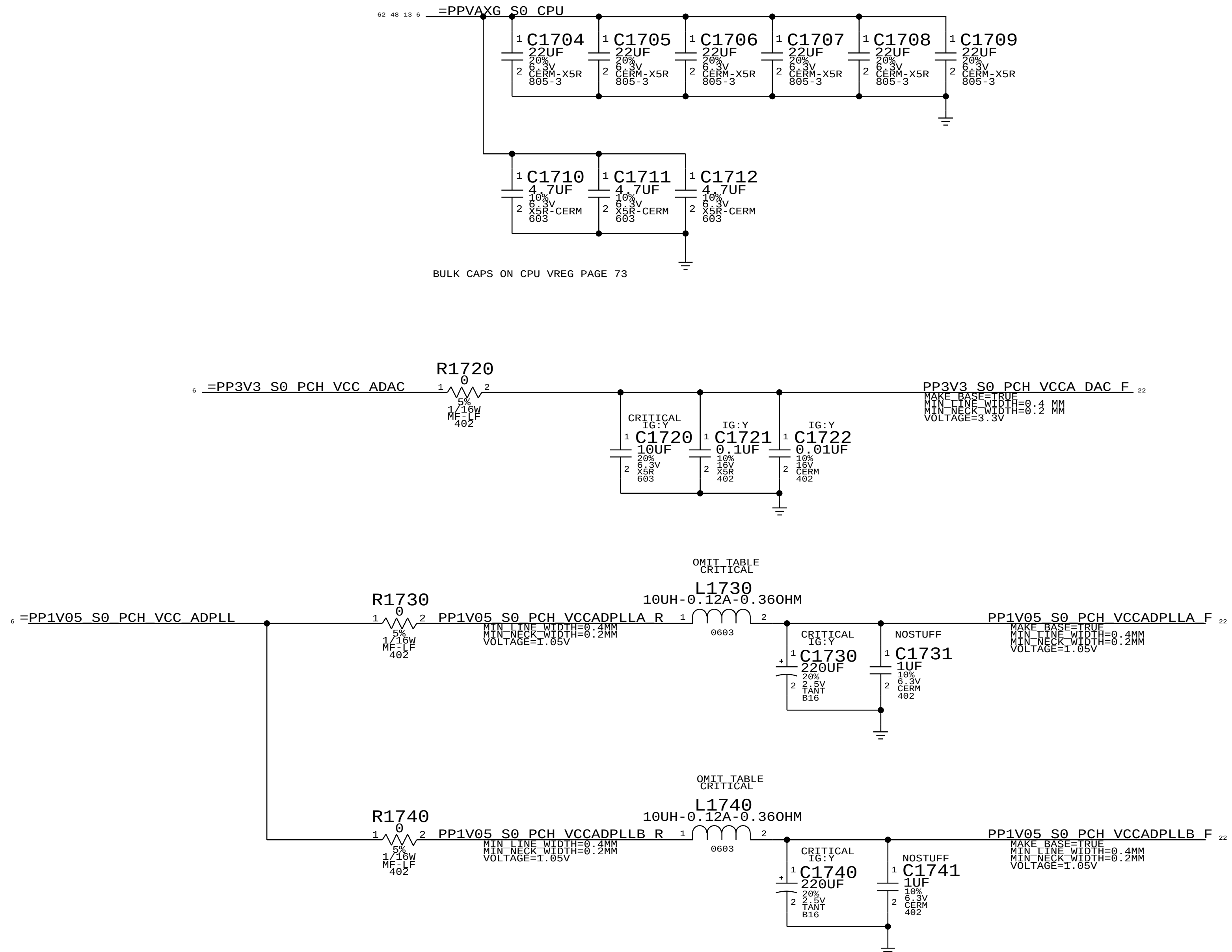
8	7	6	5	4	3	2	1

VAXG DECOUPLING

INTEL RECOMMENDATION 4X22UF 0805,3X 4.7UF

PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1070	2	IND,WW,10UH,20%.120MA,0.36OHMS	L1730,L1740	CRITICAL	IG:Y
113S0022	2	RES,MF,1/10W,0OHM,5,0603,SMD,LF	L1730,L1740		IG:N

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

GFX DECOUPLING & PCH PWR ALIAS

Apple Inc.

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REVISION 4.2.0

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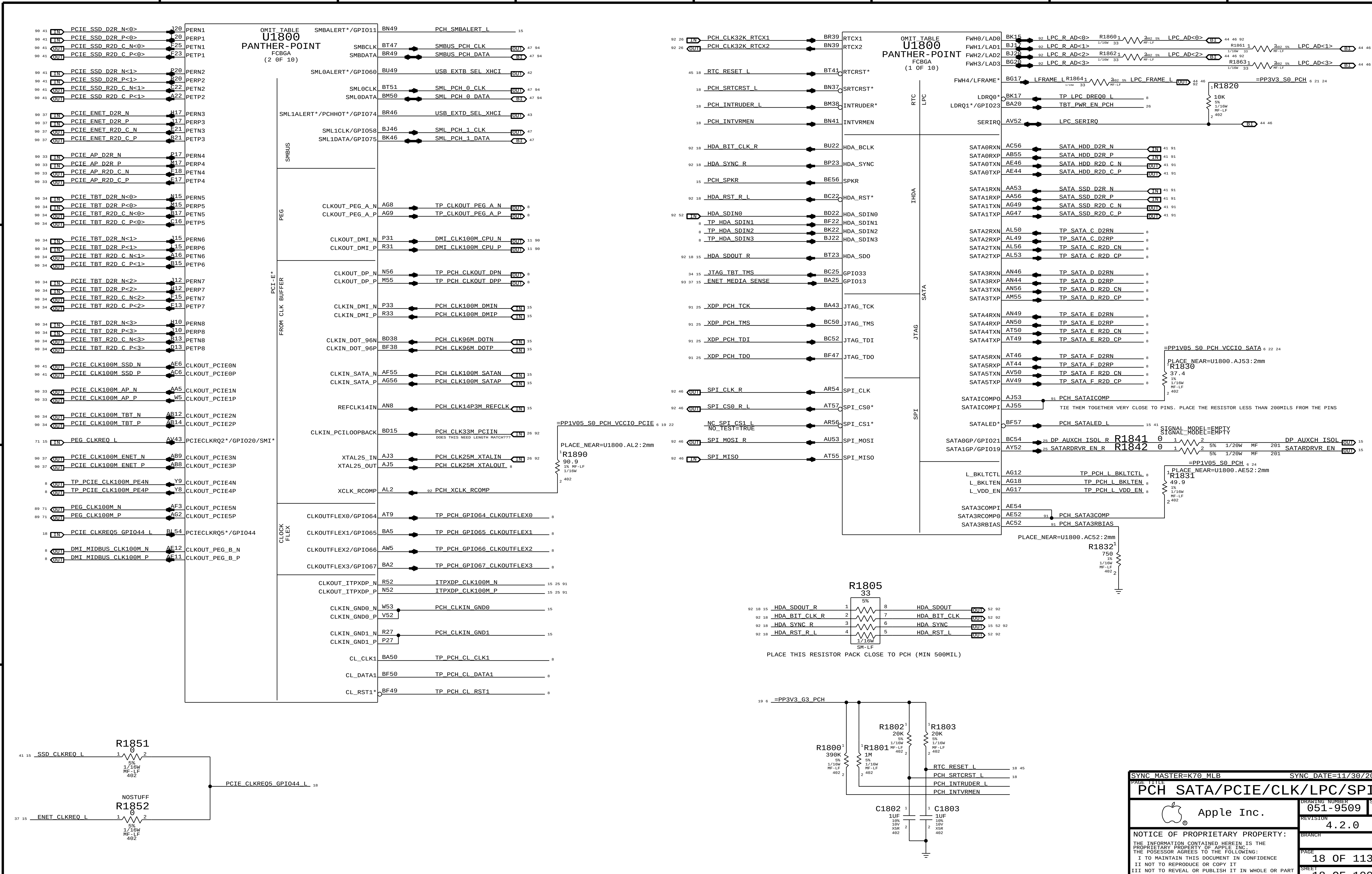

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH SATA/PCIE/CLK/LPC/SPI

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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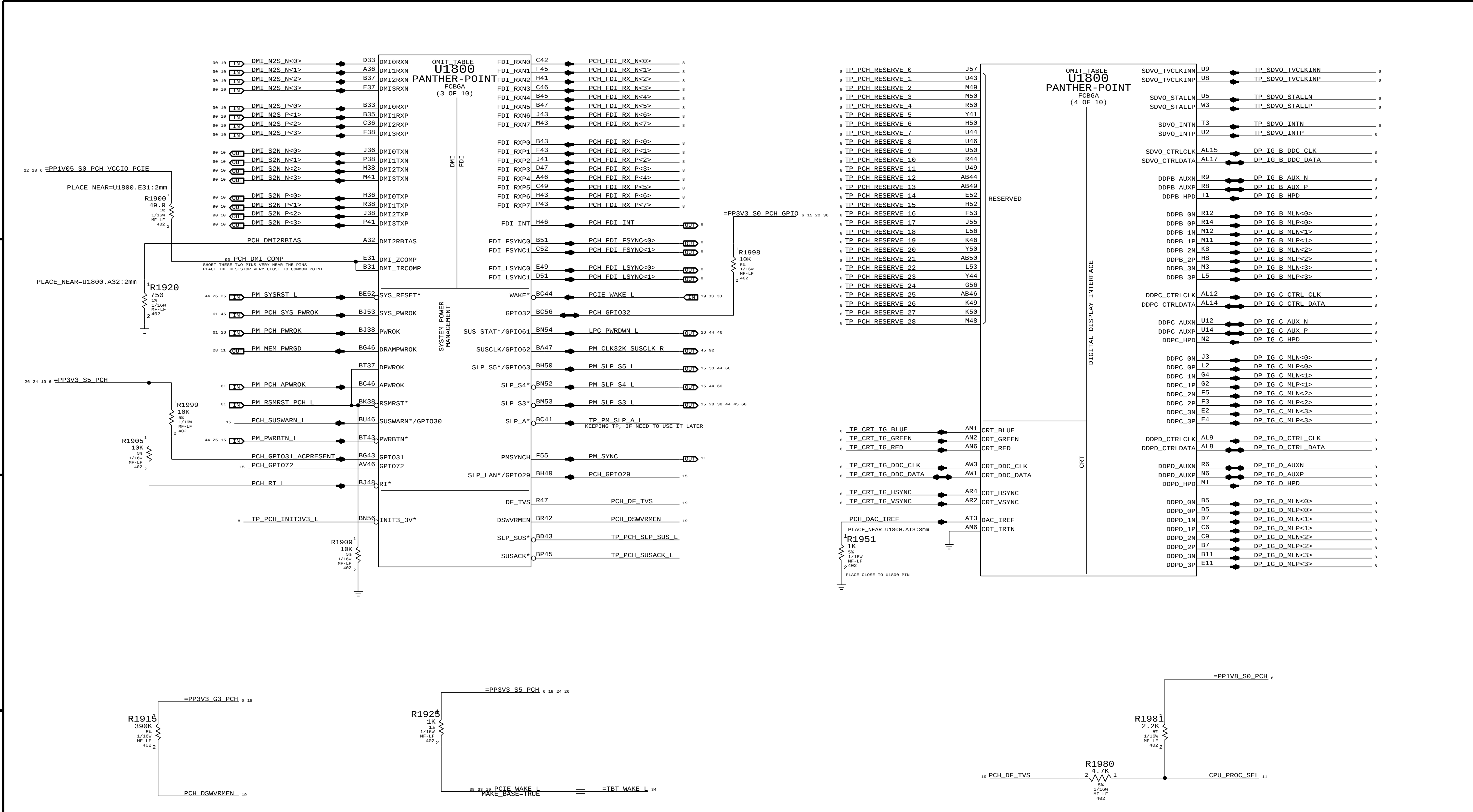

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH DMI/FDI/GRAPHICS

DRAWING NUMBER 051-9509 SIZE D

Apple Inc.

REVISION 4.2.0

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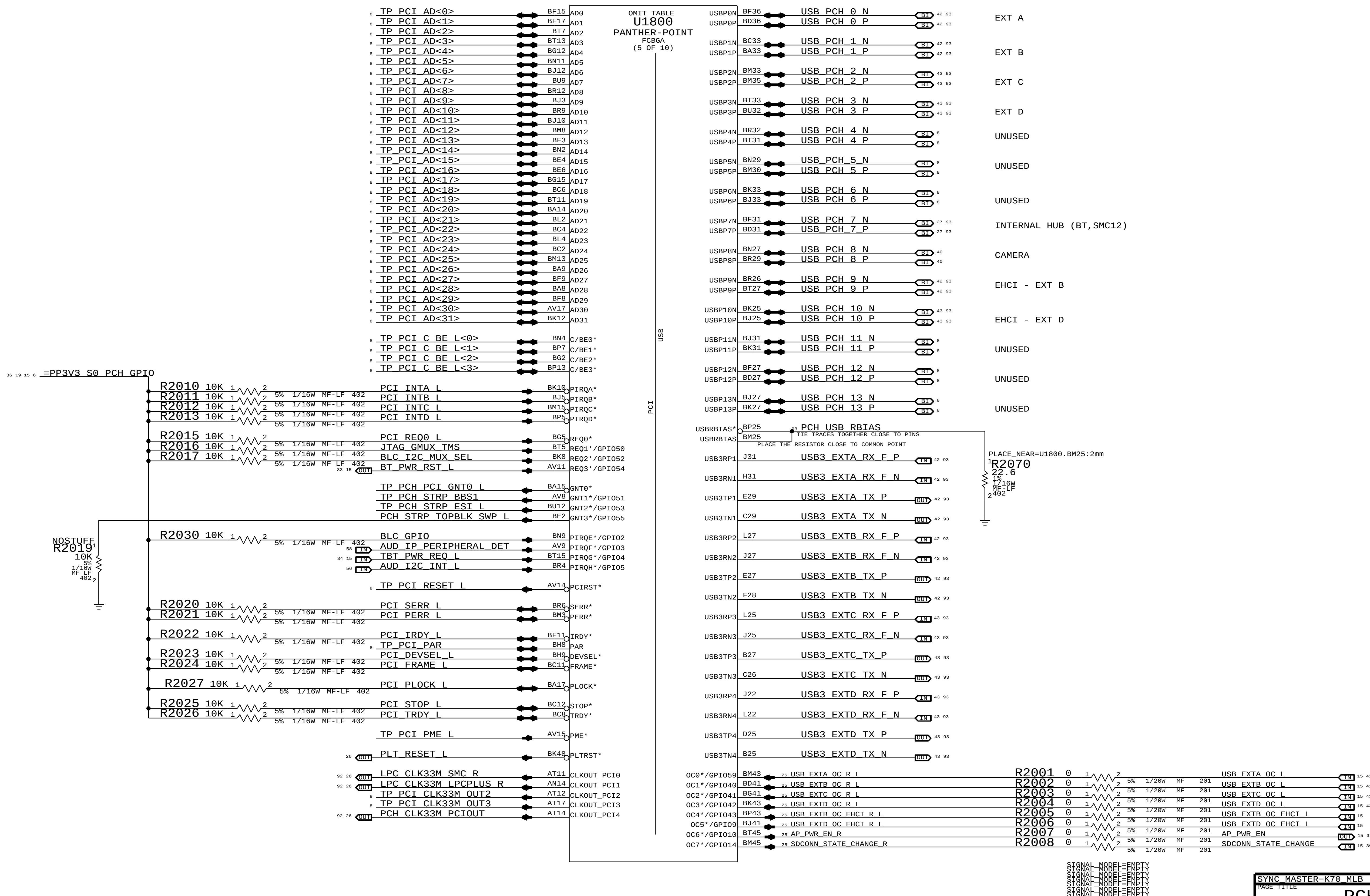

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH PCI/USB

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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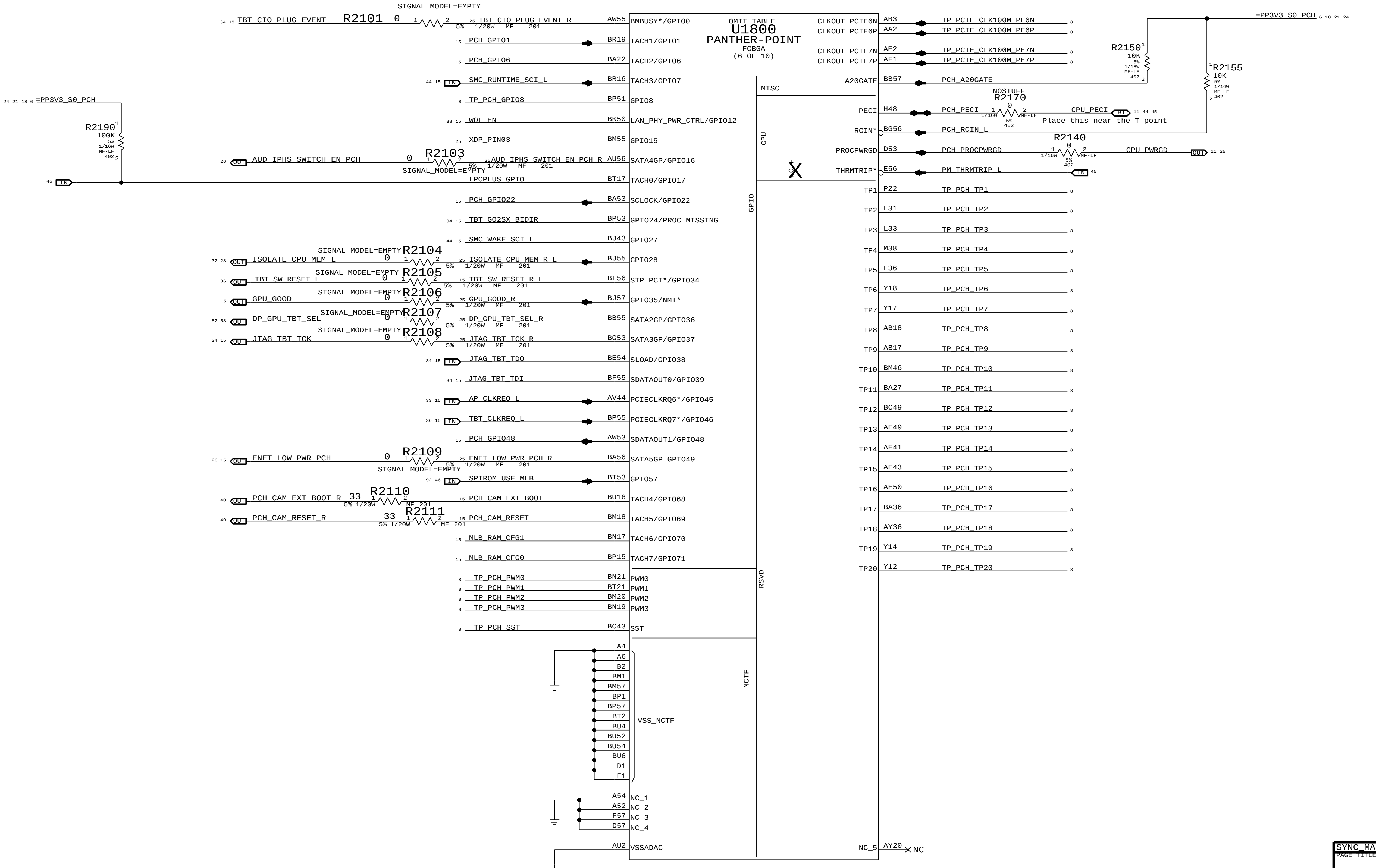

SYNC_MASTER=D7_TONY SYNC_DATE=01/11/2012

PAGE TITLE

PCH MISC

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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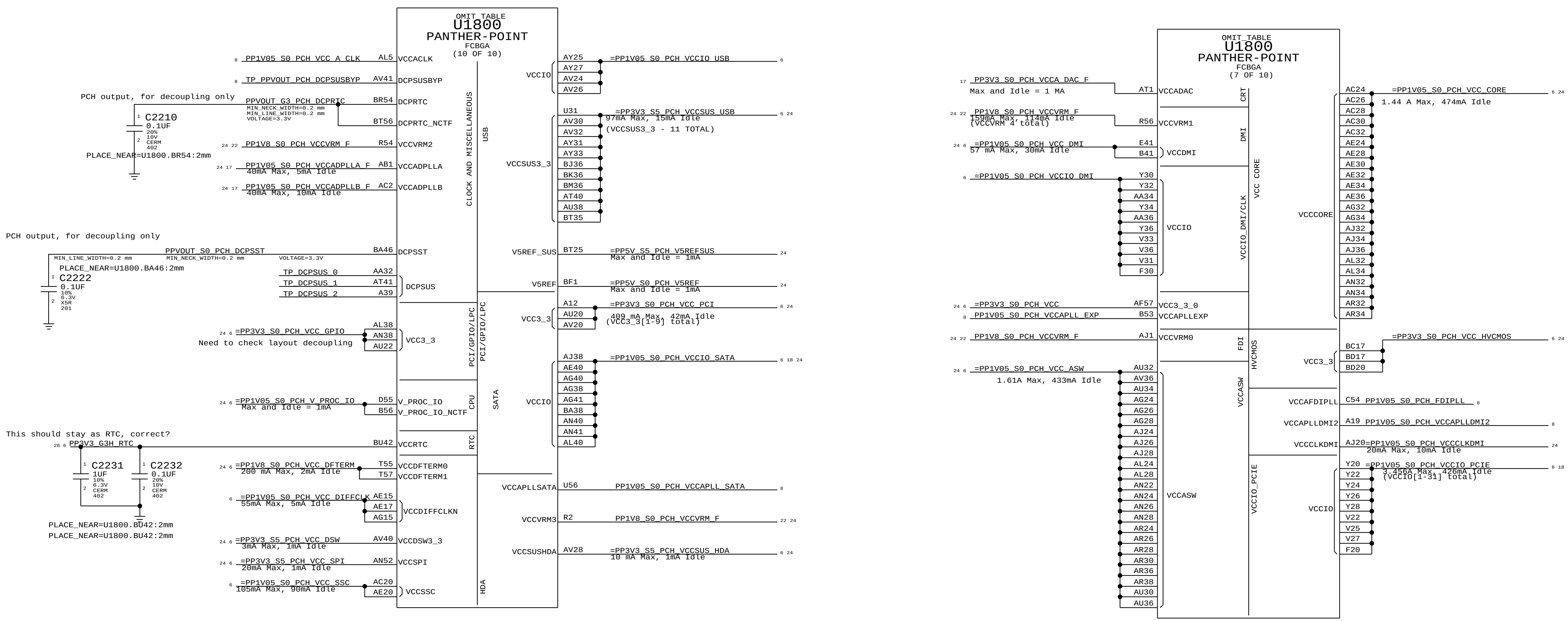
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH POWER

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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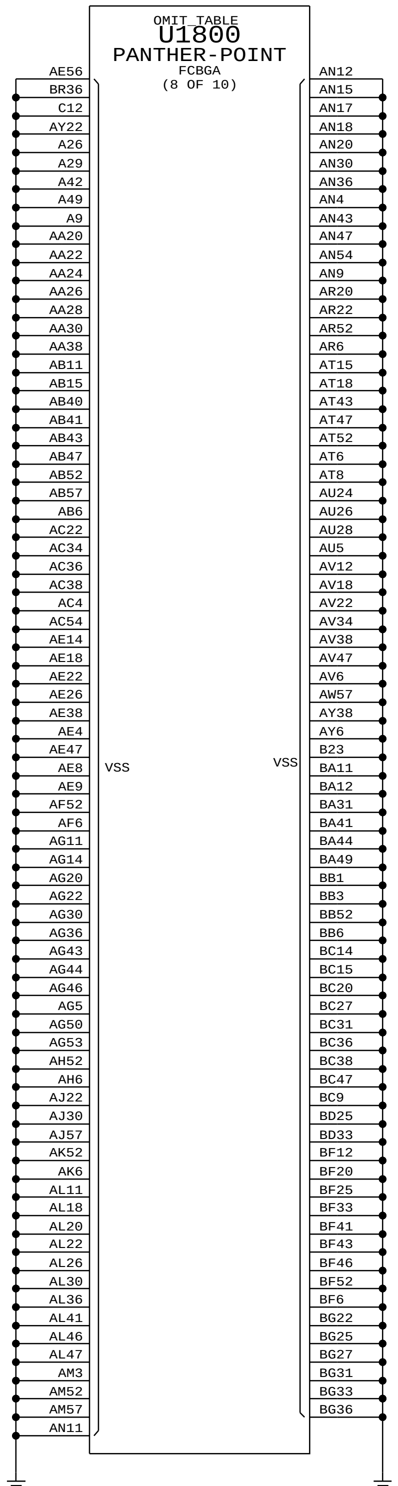
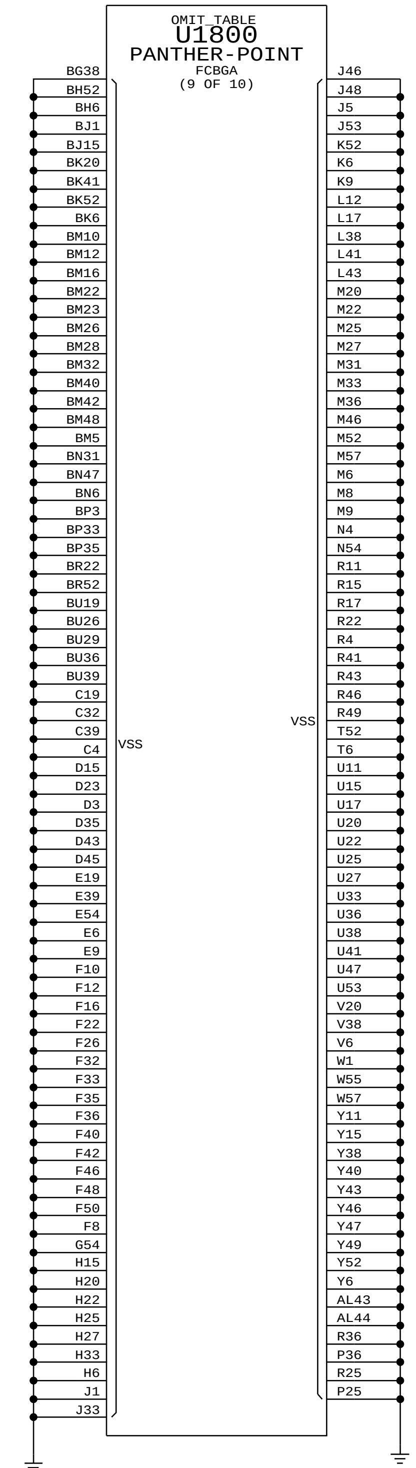
8 7 6 5 4 3 2 1





SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH GROUNDS

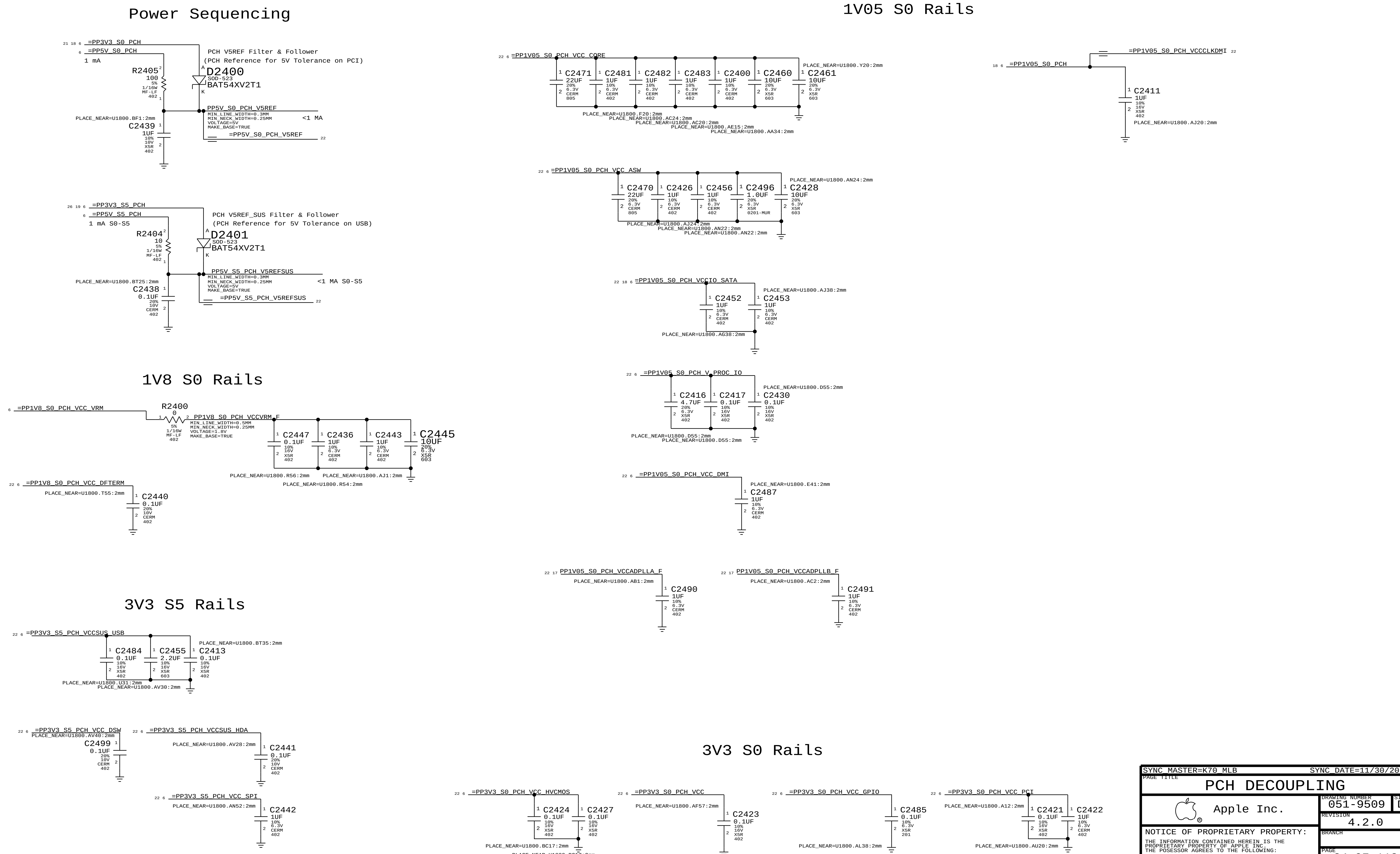
Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

8 7 6 5 4 3 2 1



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

PCH DECOUPLING

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REVISION 4.2.0

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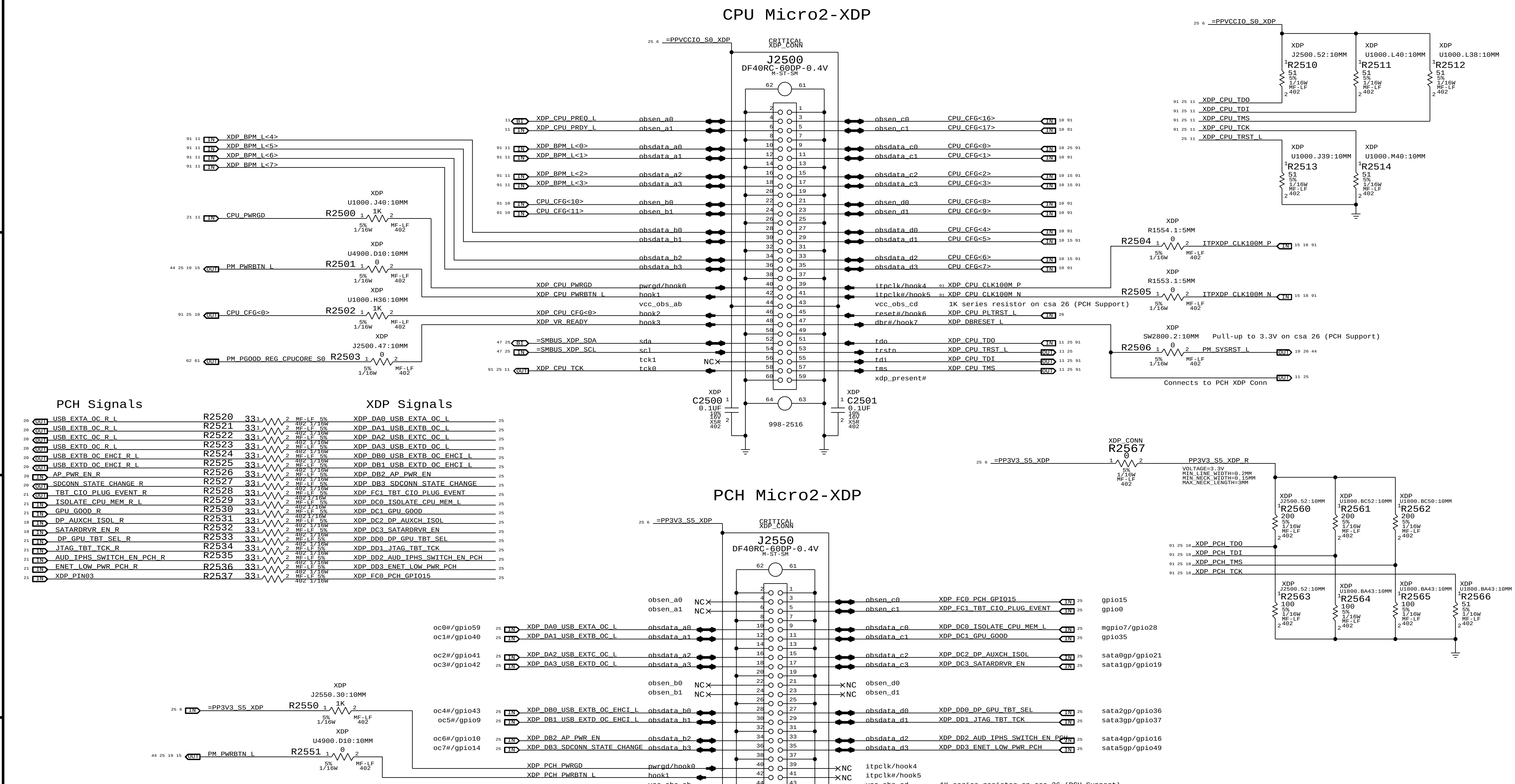


PCH/XDP Signal Isolation Notes:

- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
  - 'Output' non-XDP signals require pulls.
  - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

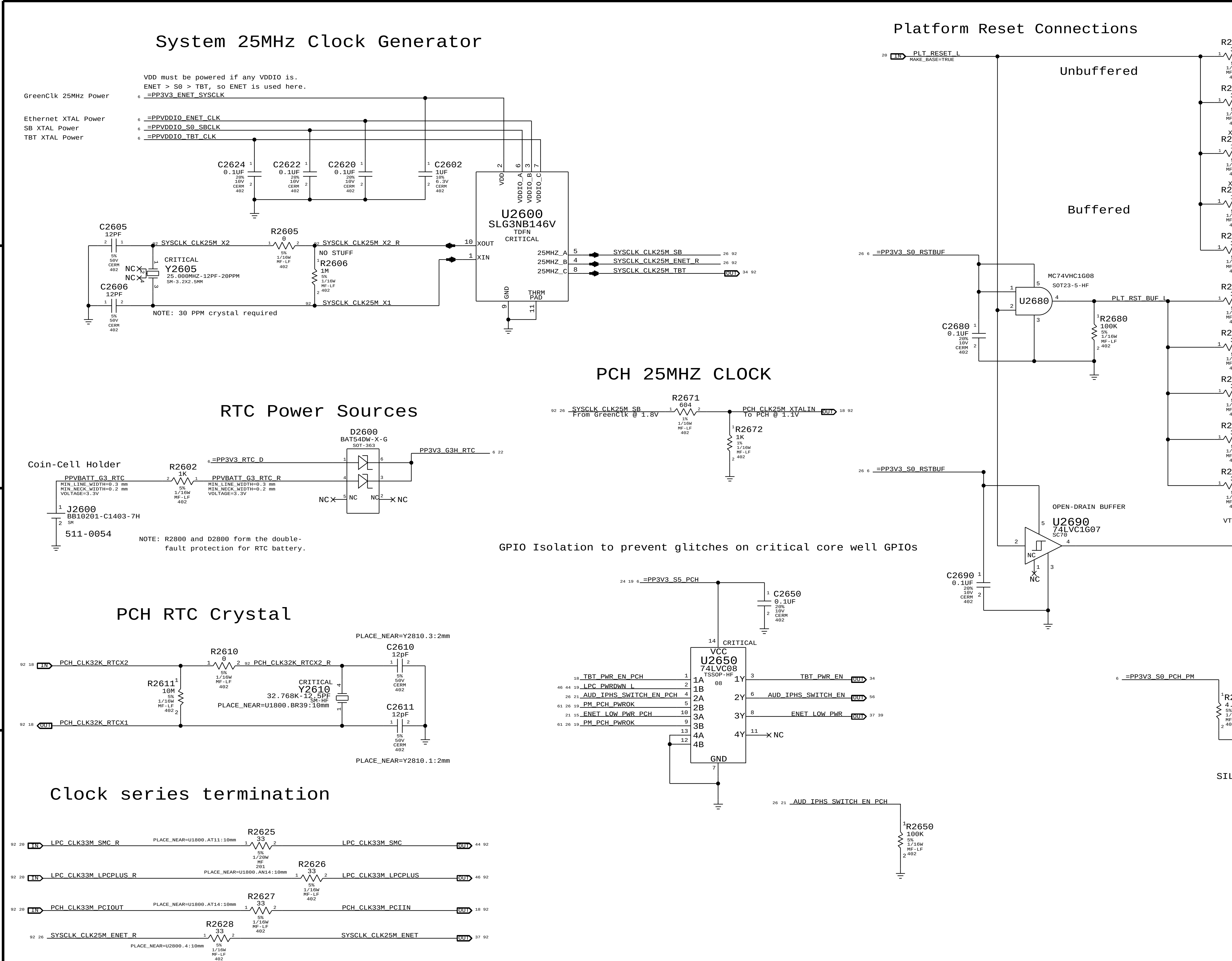
PAGE TITLE: CPU and PCH XDP

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

BRANCH



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

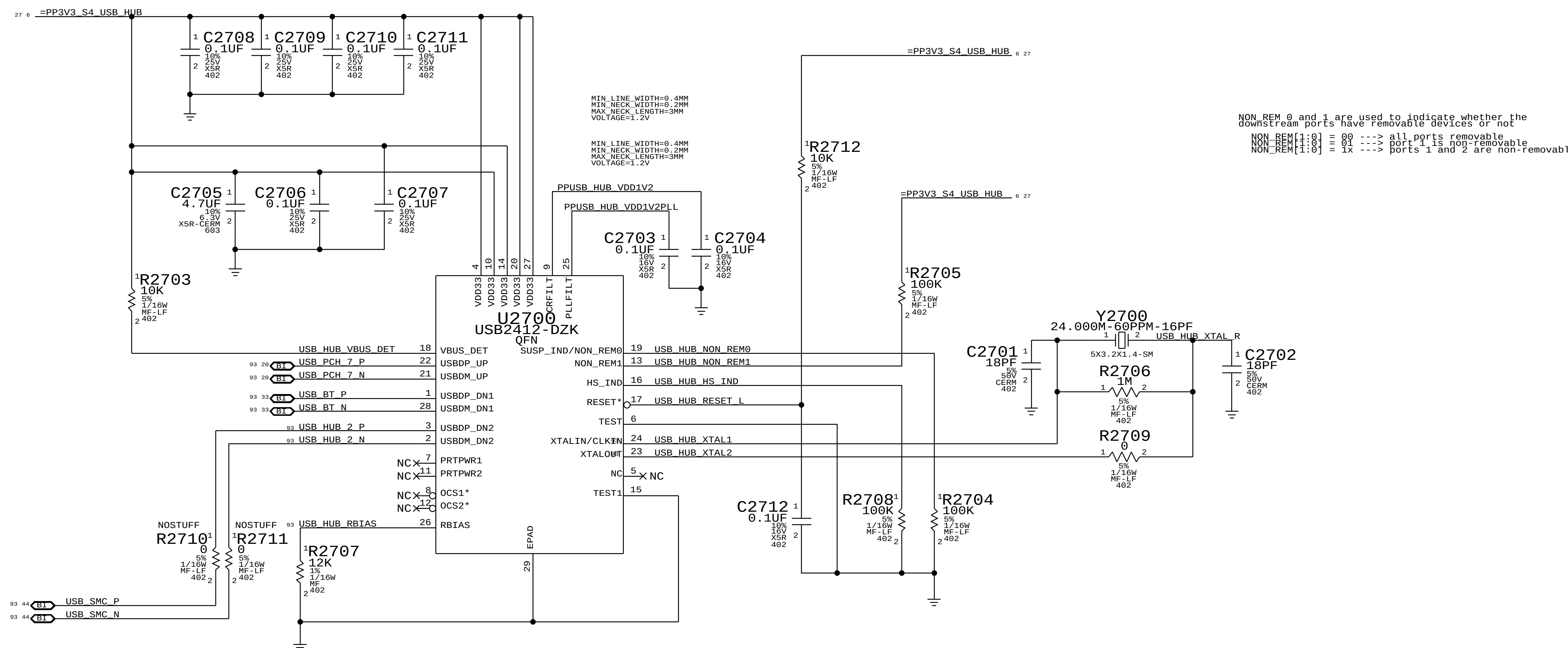
PAGE TITLE: CHIPSET SUPPORT

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

BRANCH



SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE

USB HUB

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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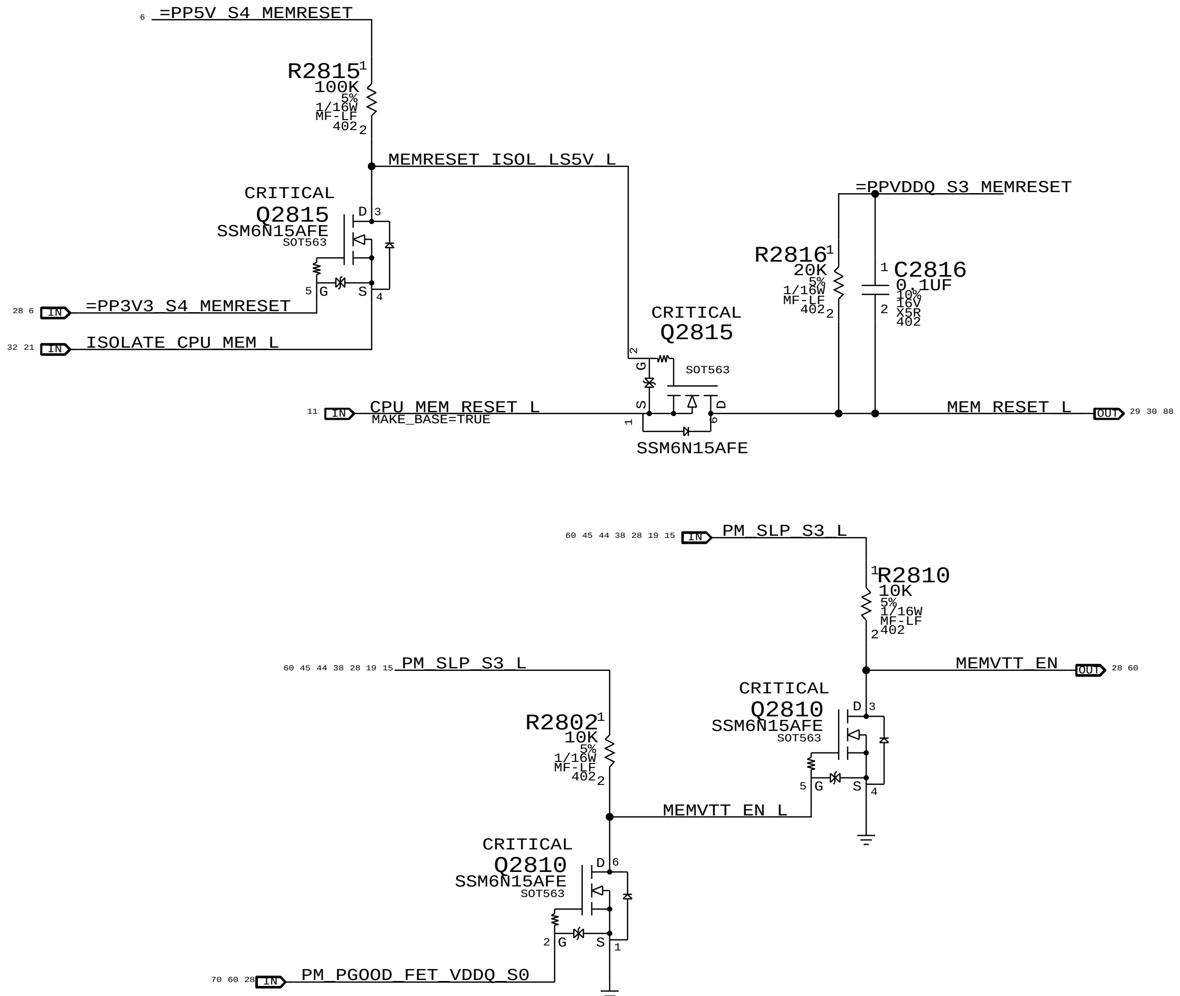
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well
as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

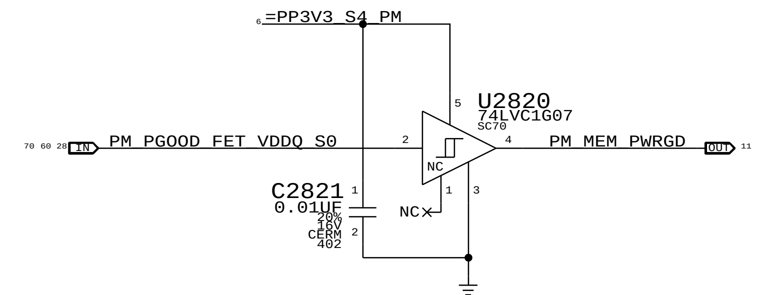
ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN   = PM_PGOOD_FET_VDDQ_S0 * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



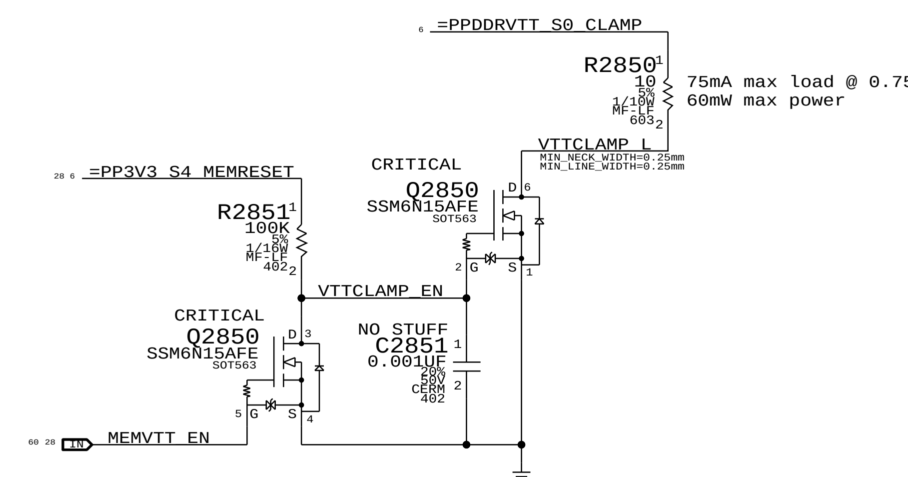
1V5 S0 "PGOOD" for CPU

PM_MEM_PWRGD pull-up to CPU VTT rail is on CPU page



MEMVTT Clamp

Ensures CKE signals are held low in S3



	Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1
to	1	0	1	1	1	1	1
	2	0	0	1	1	1	0
S3	3	0	0	0	X	1	0
	4	0	0	1	X	1	0
to	5	0	1	1	0 (*)	1	1
	6	0	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE
CPU Memory S3 Support

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D
REVISION 4.2.0
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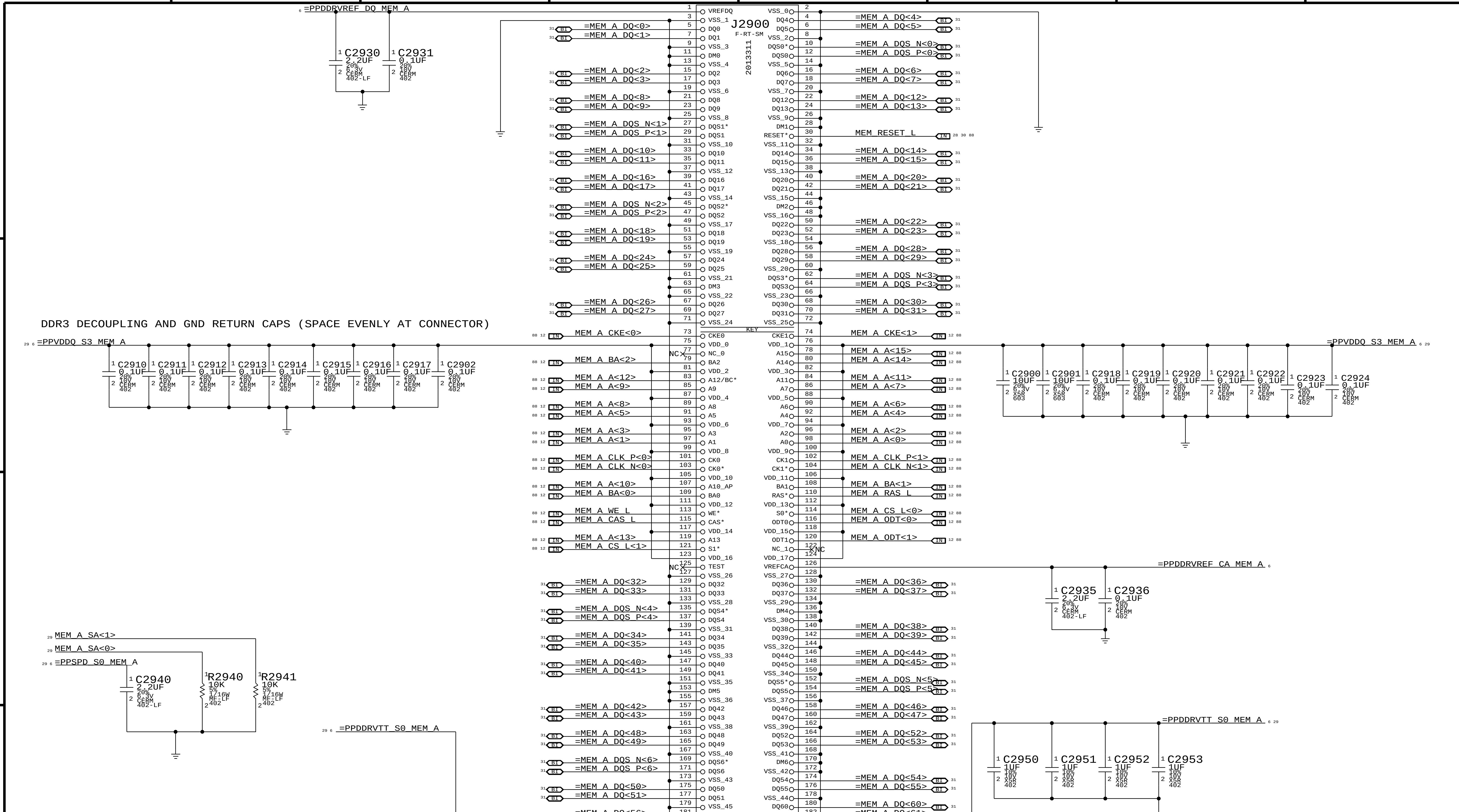


SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

DDR3 SO-DIMM Connector A

Apple Inc.

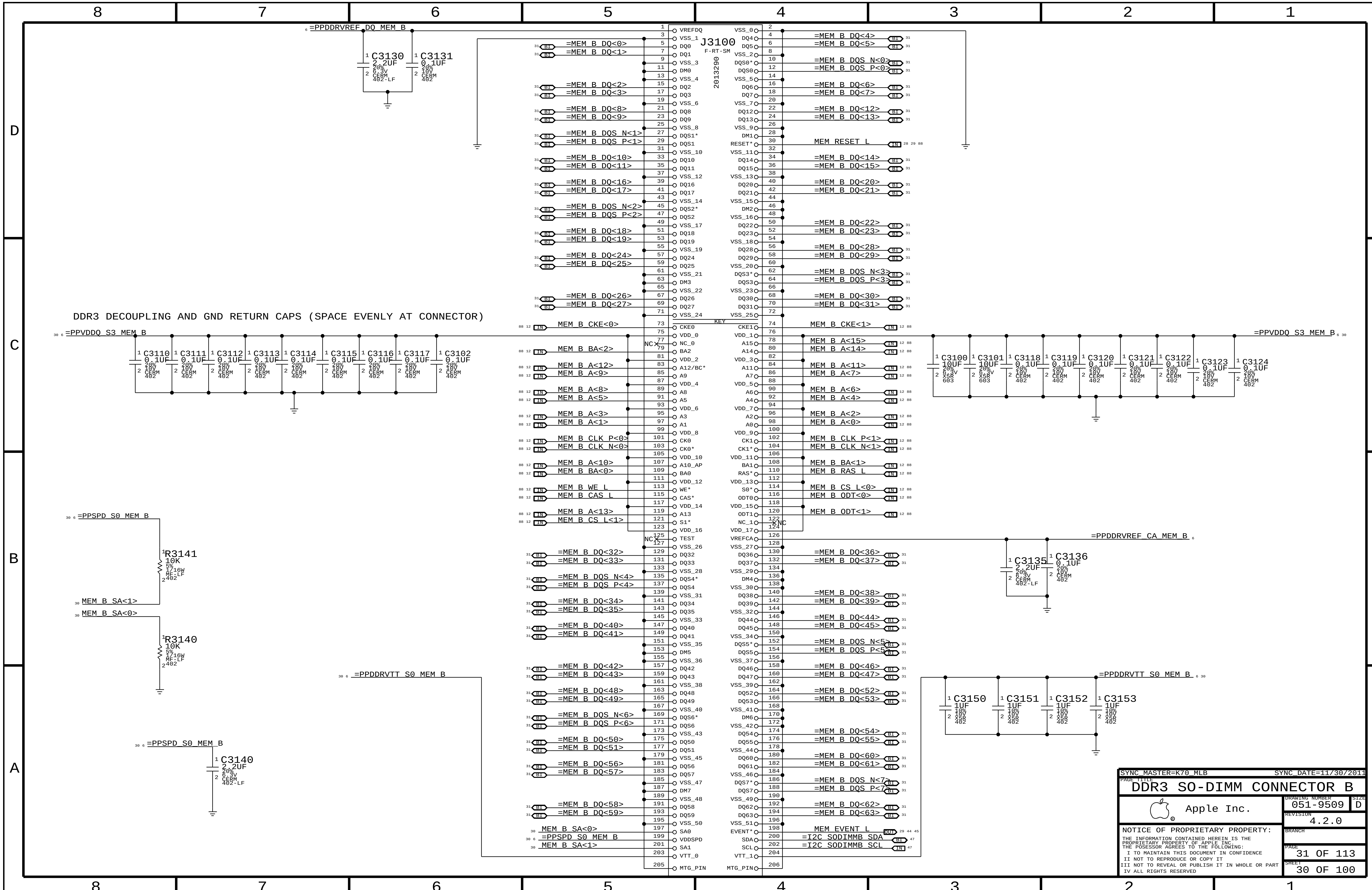
DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

PAGE 29 OF 113

SHEET 29 OF 100



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B

DRAWING NUMBER 051-9509 SIZE D

Apple Inc.

REVISION 4.2.0

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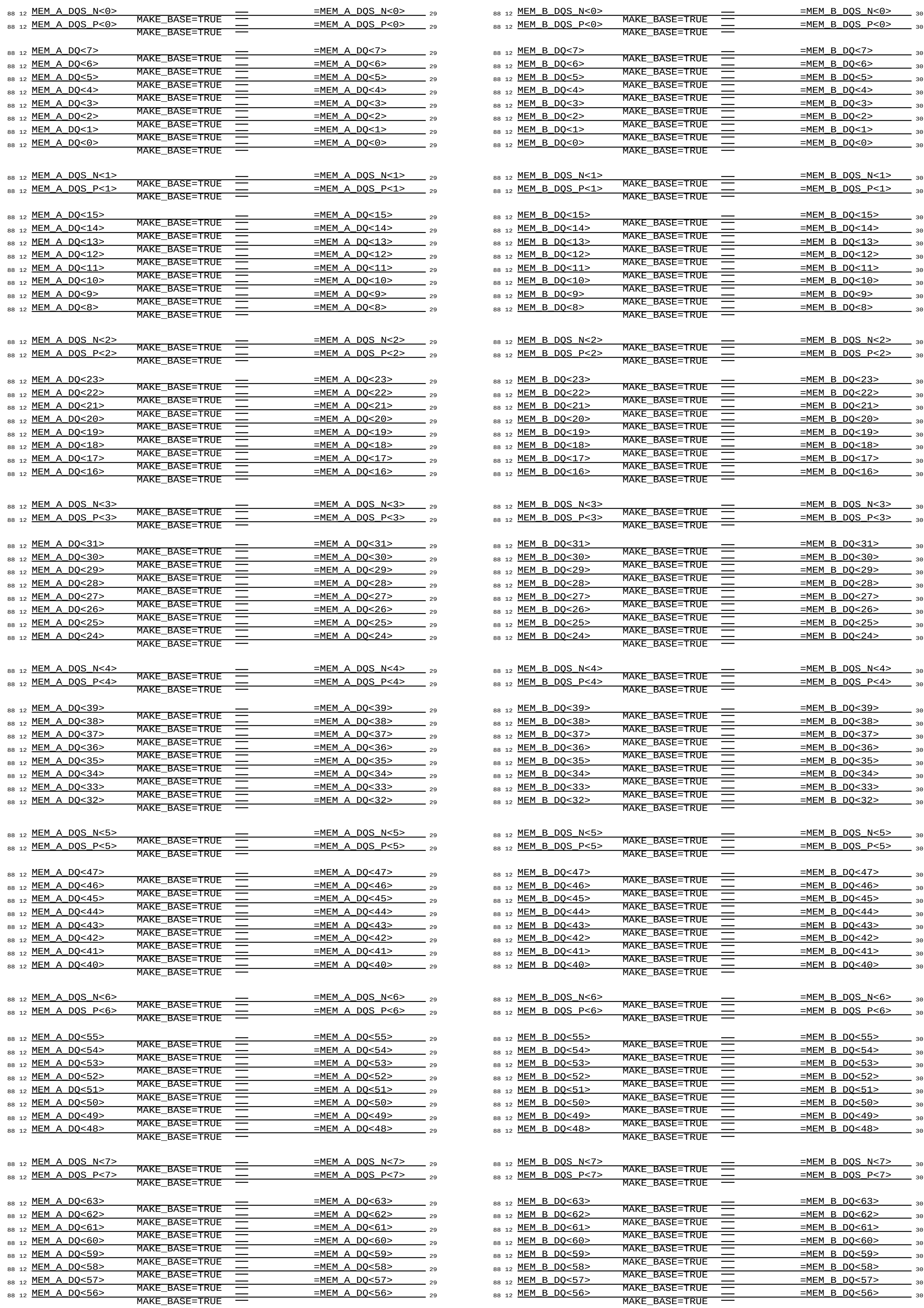


SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE

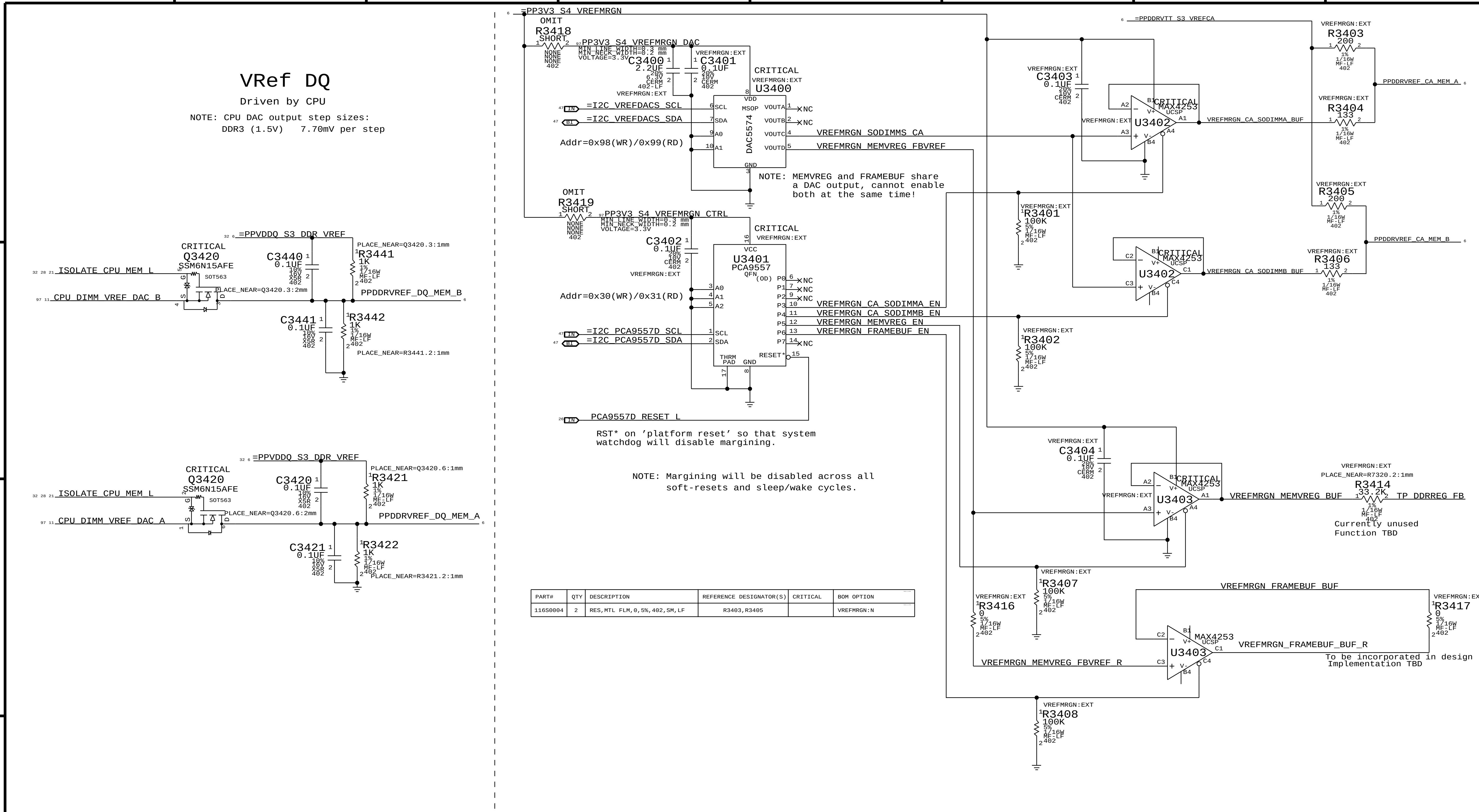
DDR3 ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH



	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	C	C	D	D
PCA9557D Pin:	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE
DDR3/FRAMEBUF VREF MARGINING

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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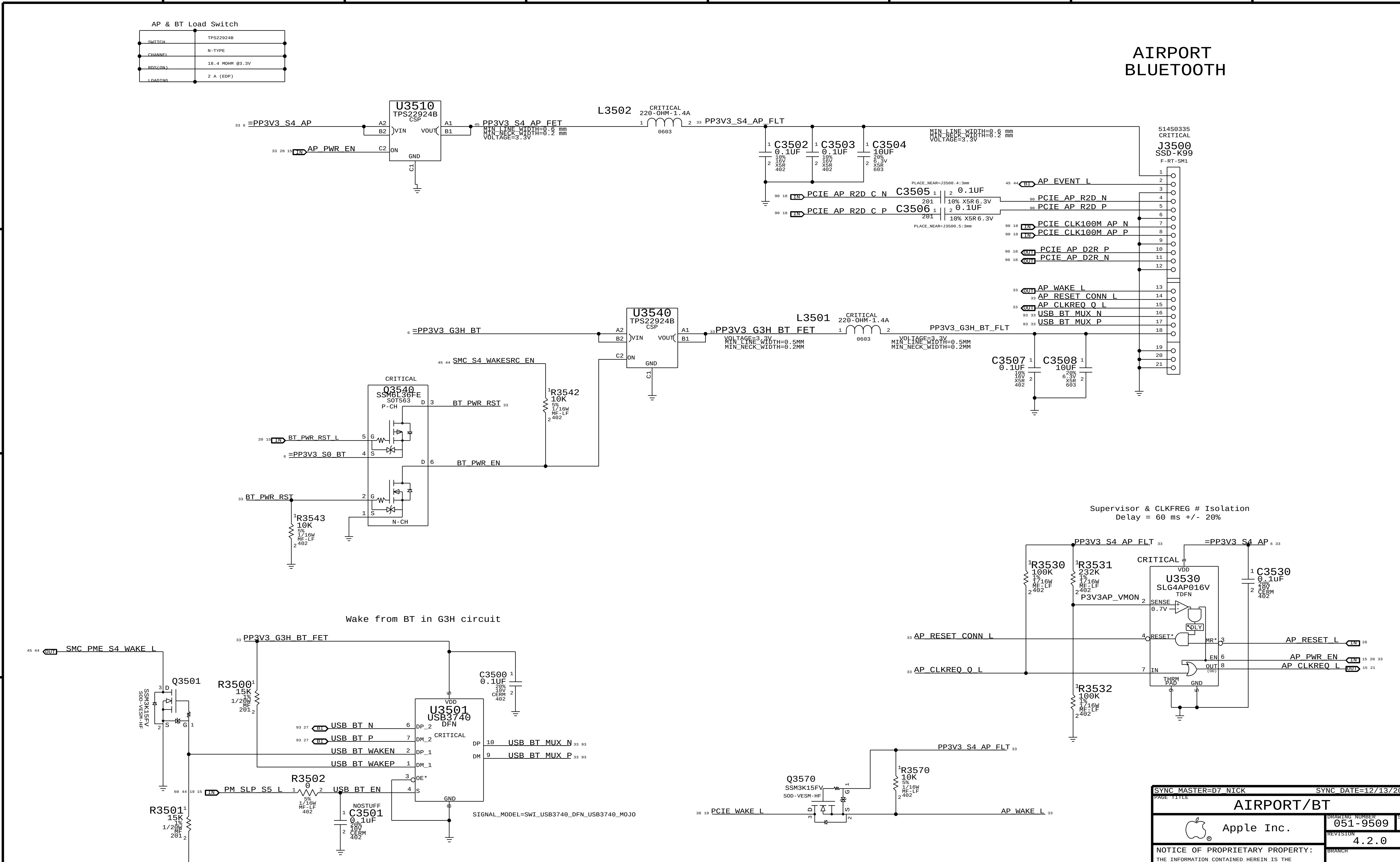

SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE

AIRPORT/BT

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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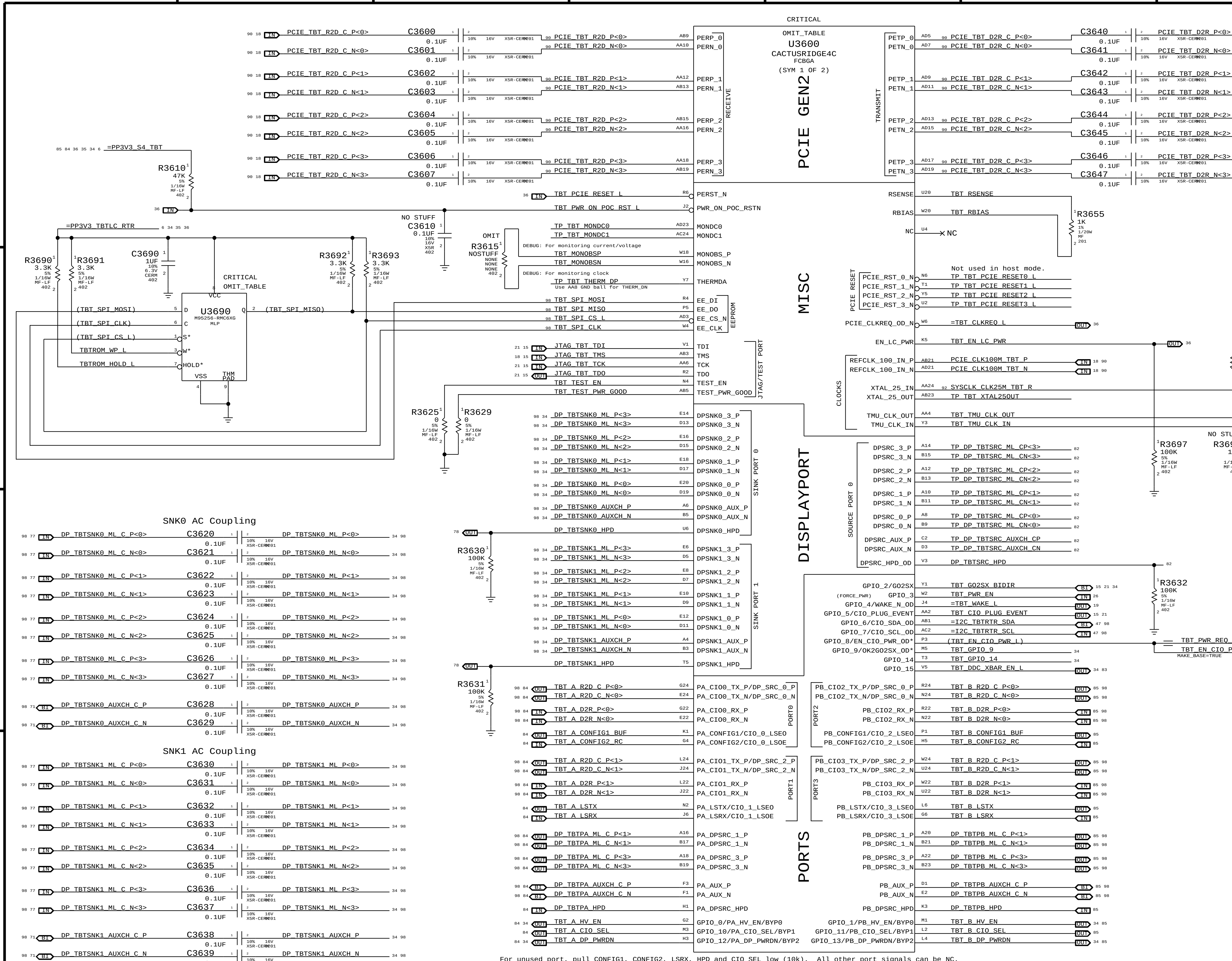

SYNC_MASTER=D7_DOUG SYNC_DATE=01/11/2012

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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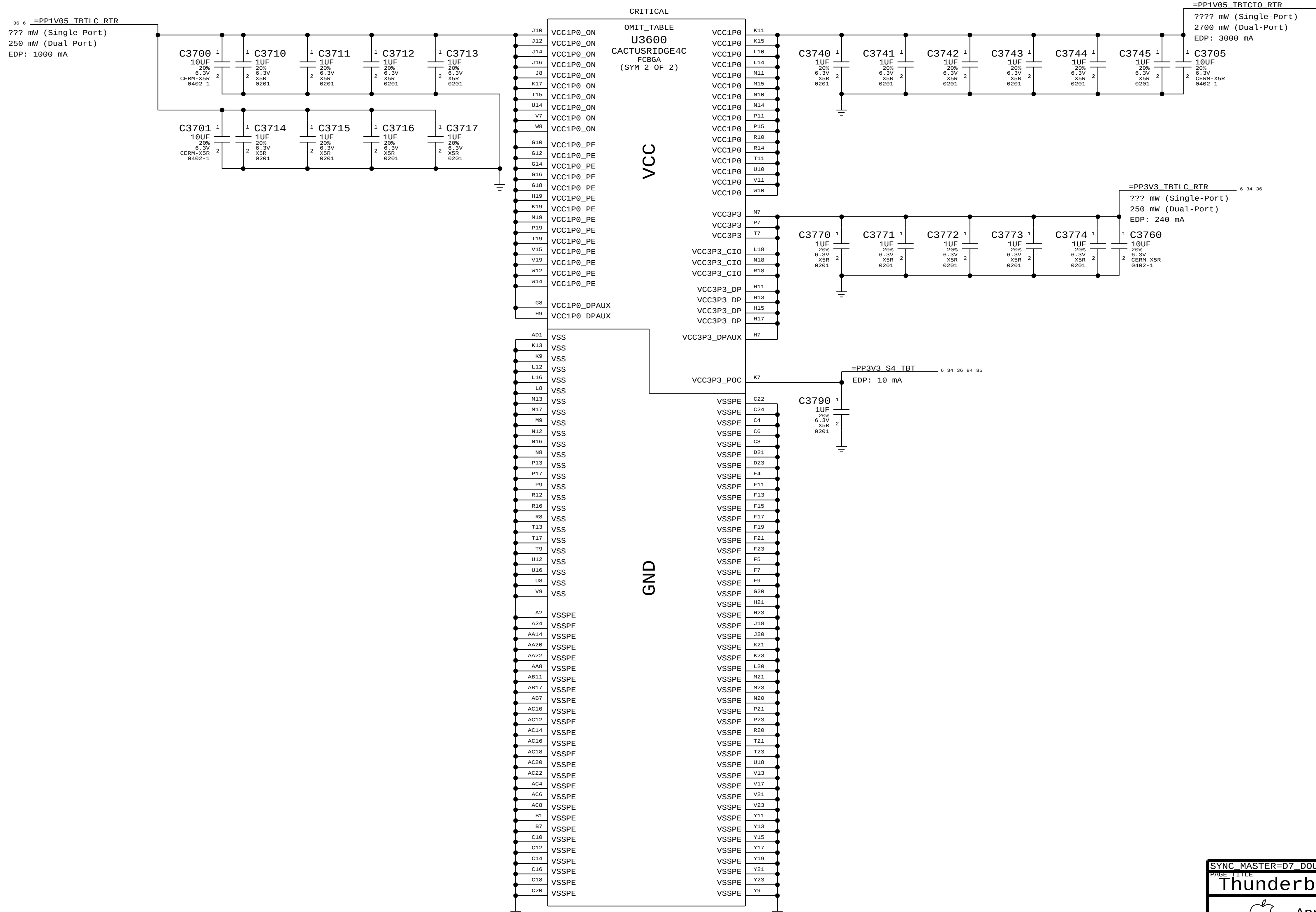
PAGE 36 OF 113

SHEET 34 OF 100



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC_MASTER=D7_DOUG SYNC_DATE=01/11/2012

PAGE TITLE
Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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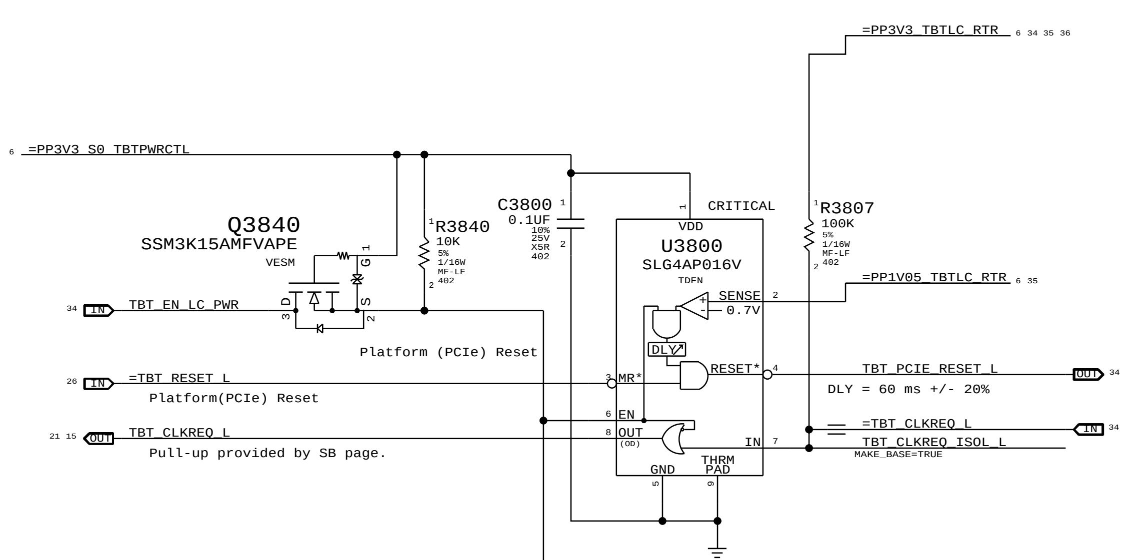
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

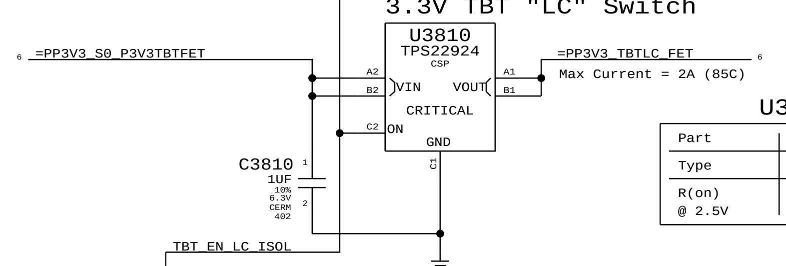
Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

Supervisor & CLKREQ# Isolation



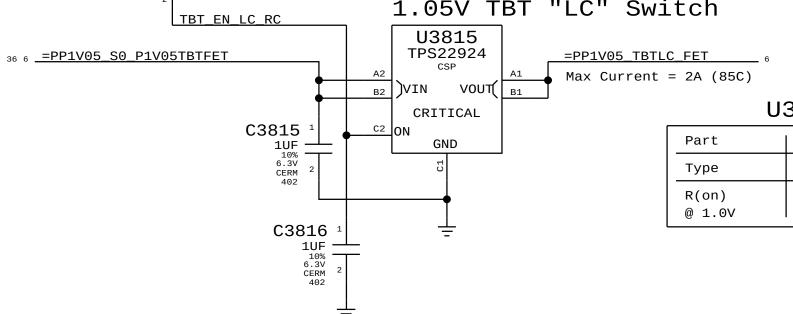
3.3V TBT "LC" Switch



U3810

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

1.05V TBT "LC" Switch

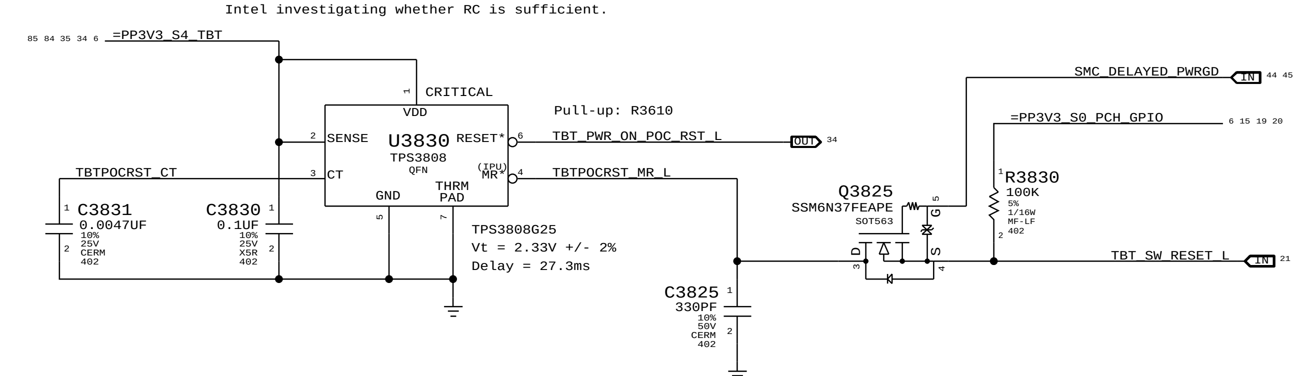


U3815

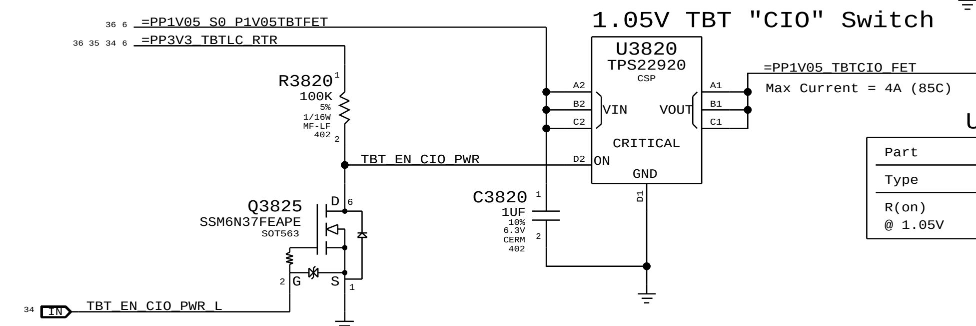
Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



U3820

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

SYNC_MASTER=D7_DOUG SYNC_DATE=01/11/2012

PAGE TITLE: Thunderbolt Power Support

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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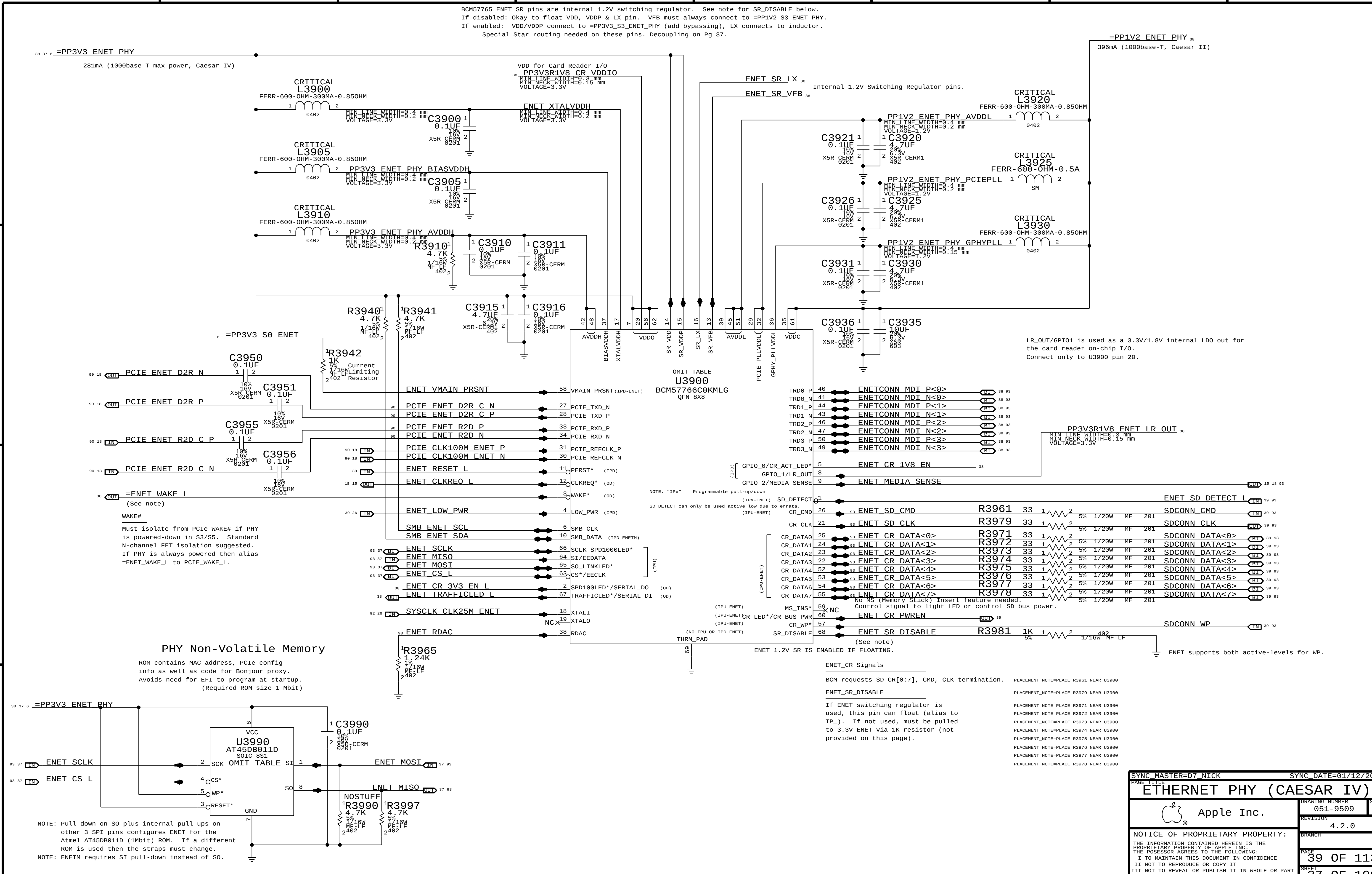

SYNC_MASTER=D7_NICK SYNC_DATE=01/12/2012

PAGE TITLE

ETHERNET PHY (CAESAR IV)

Apple Inc.

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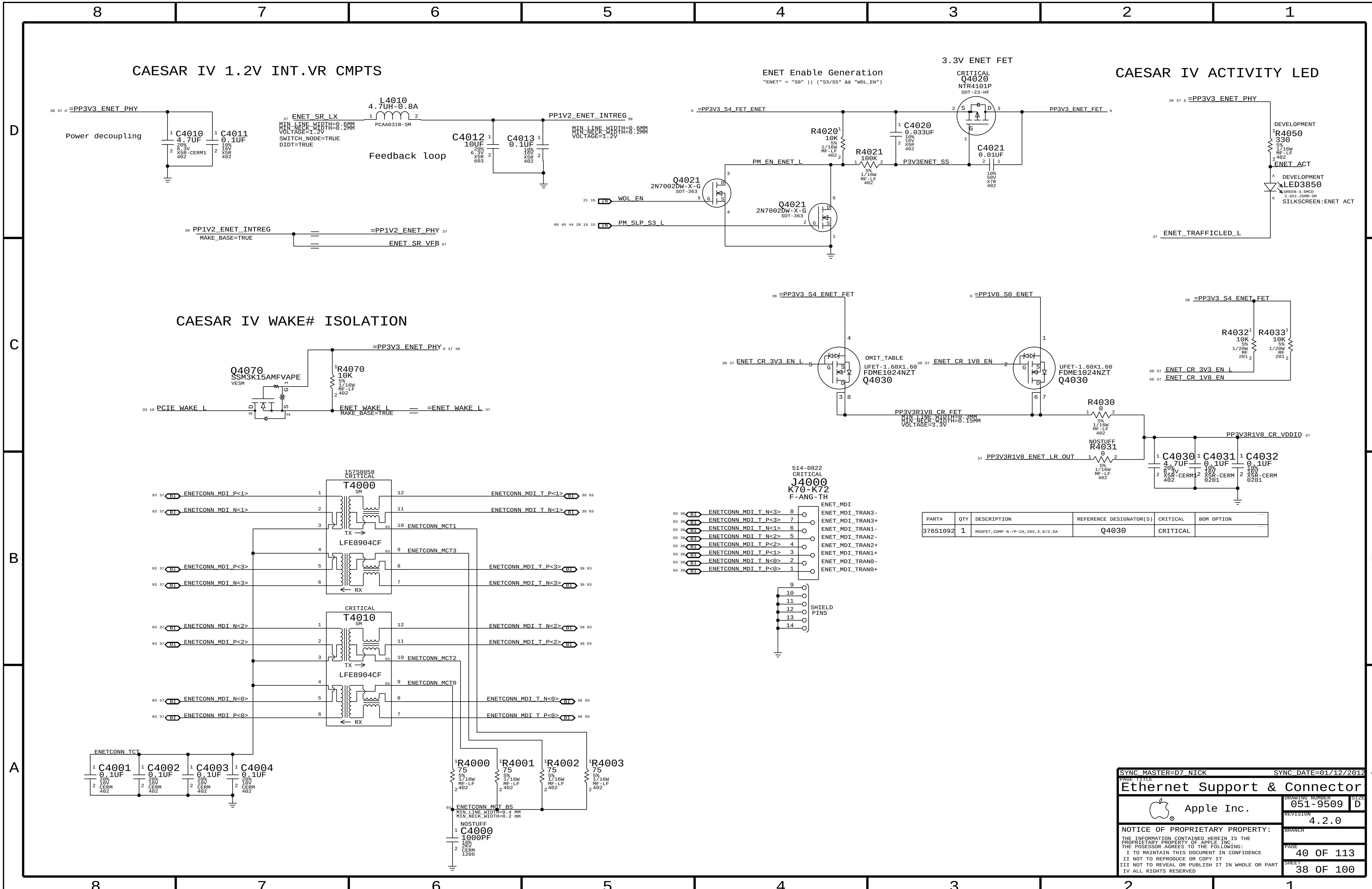

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S1092	1	MOSFET,COMP N-/P-CH,20V,3.8/2.6A	Q4030	CRITICAL	

SYNC_MASTER=D7_NICK SYNC_DATE=01/12/2012

PAGE TITLE
Ethernet Support & Connector

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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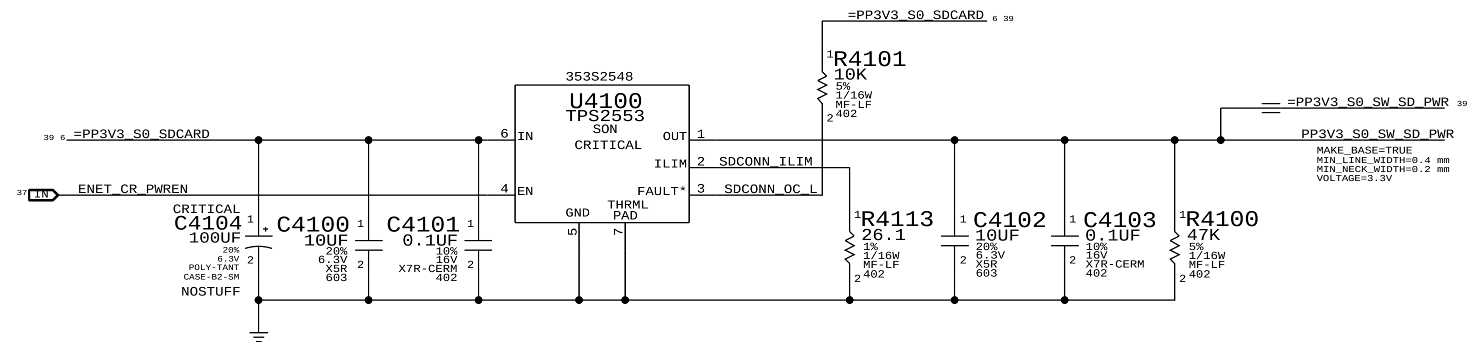
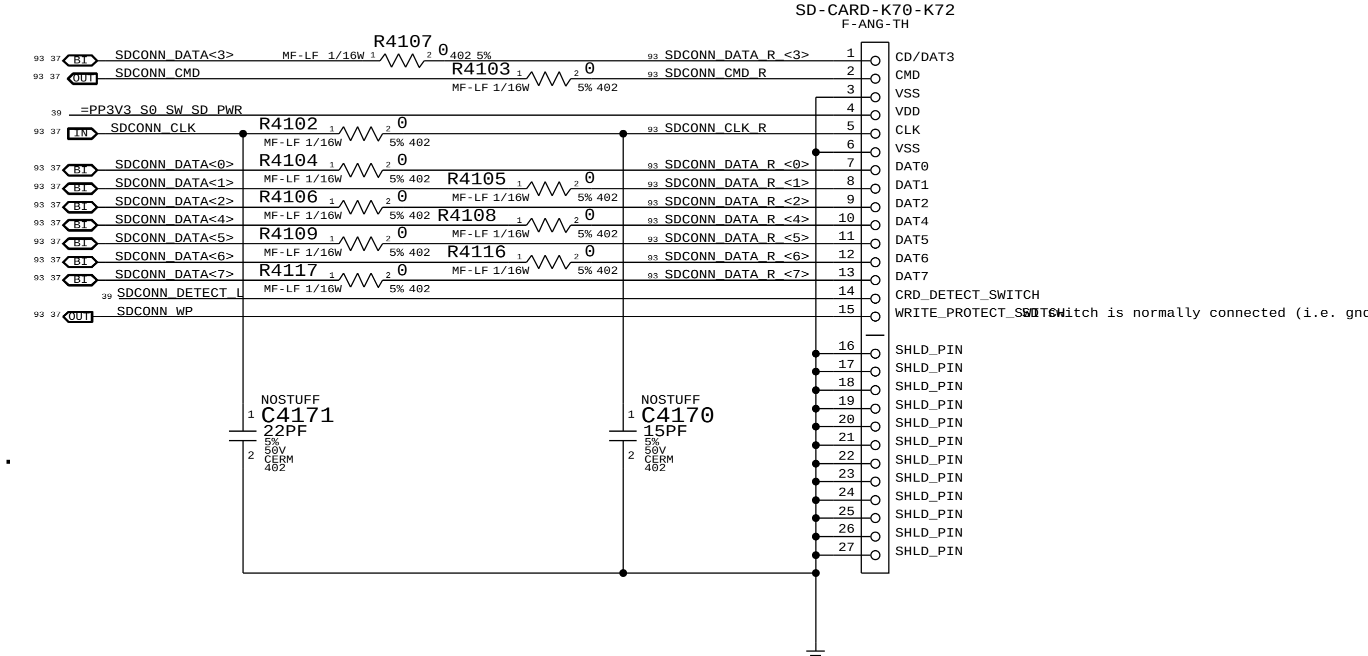
SD CARD 3.3V OVERCURRENT PROTECTION CHIP



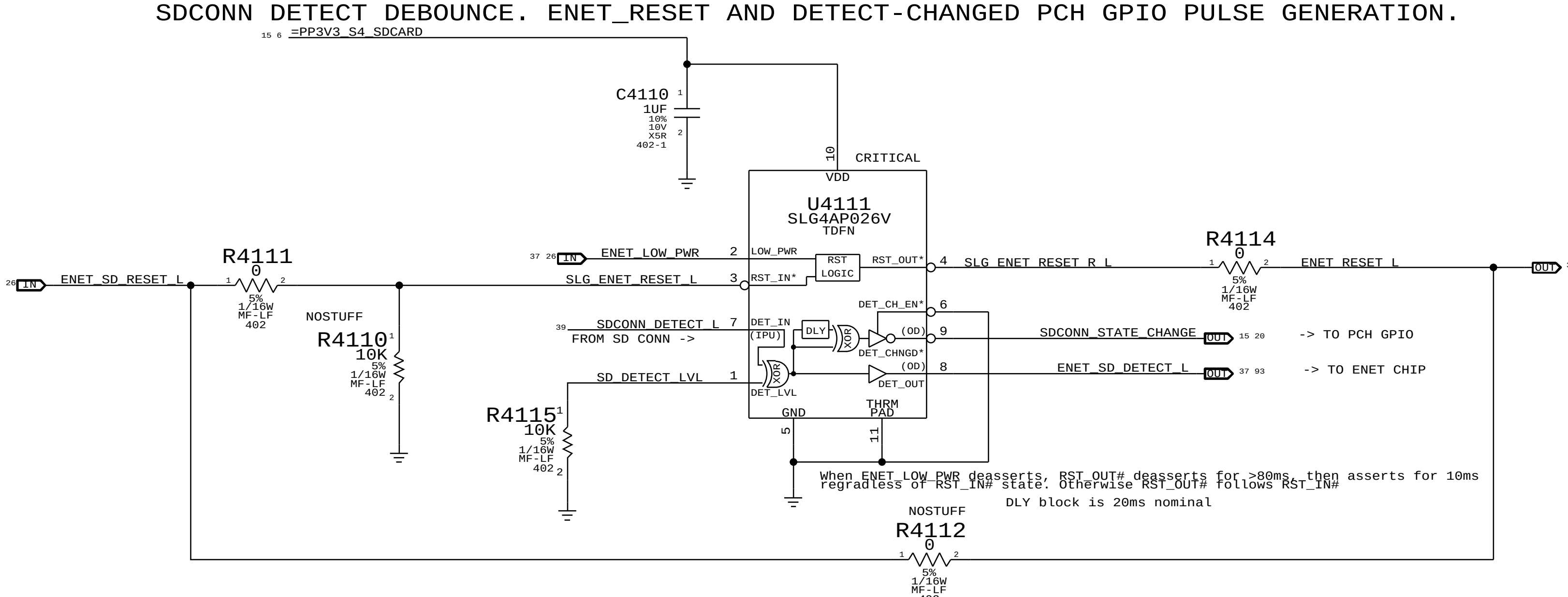
516-0249

SD CARD CONNECTOR

J4100
SD-CARD-K70-K72
F-ANG-TH



SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



SYNC_MASTER=D7_NICK SYNC_DATE=01/12/2012

PAGE TITLE

SD READER CONNECTOR

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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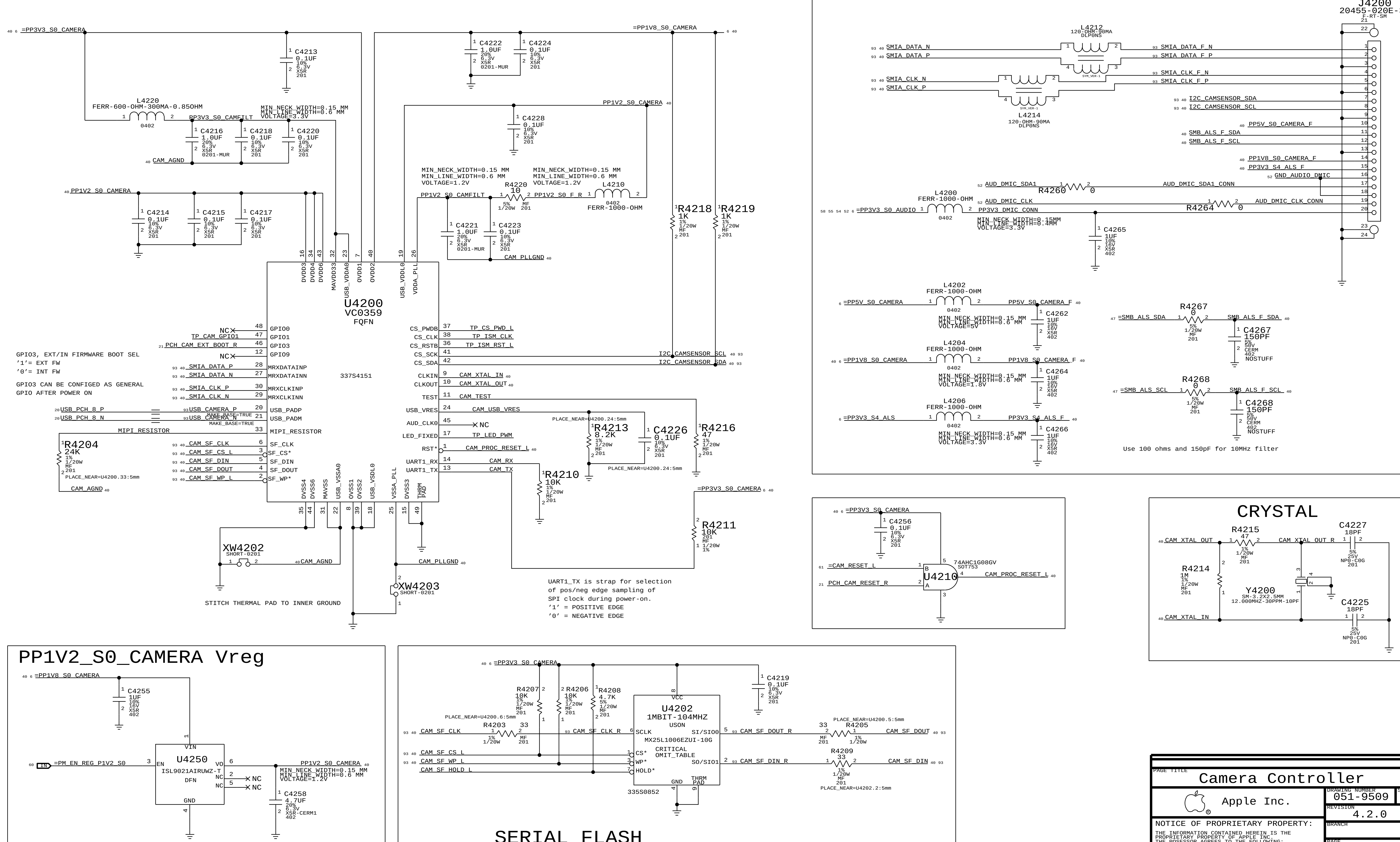
USB CAMERA CONTROLLER



PAGE TITLE
Camera Controller

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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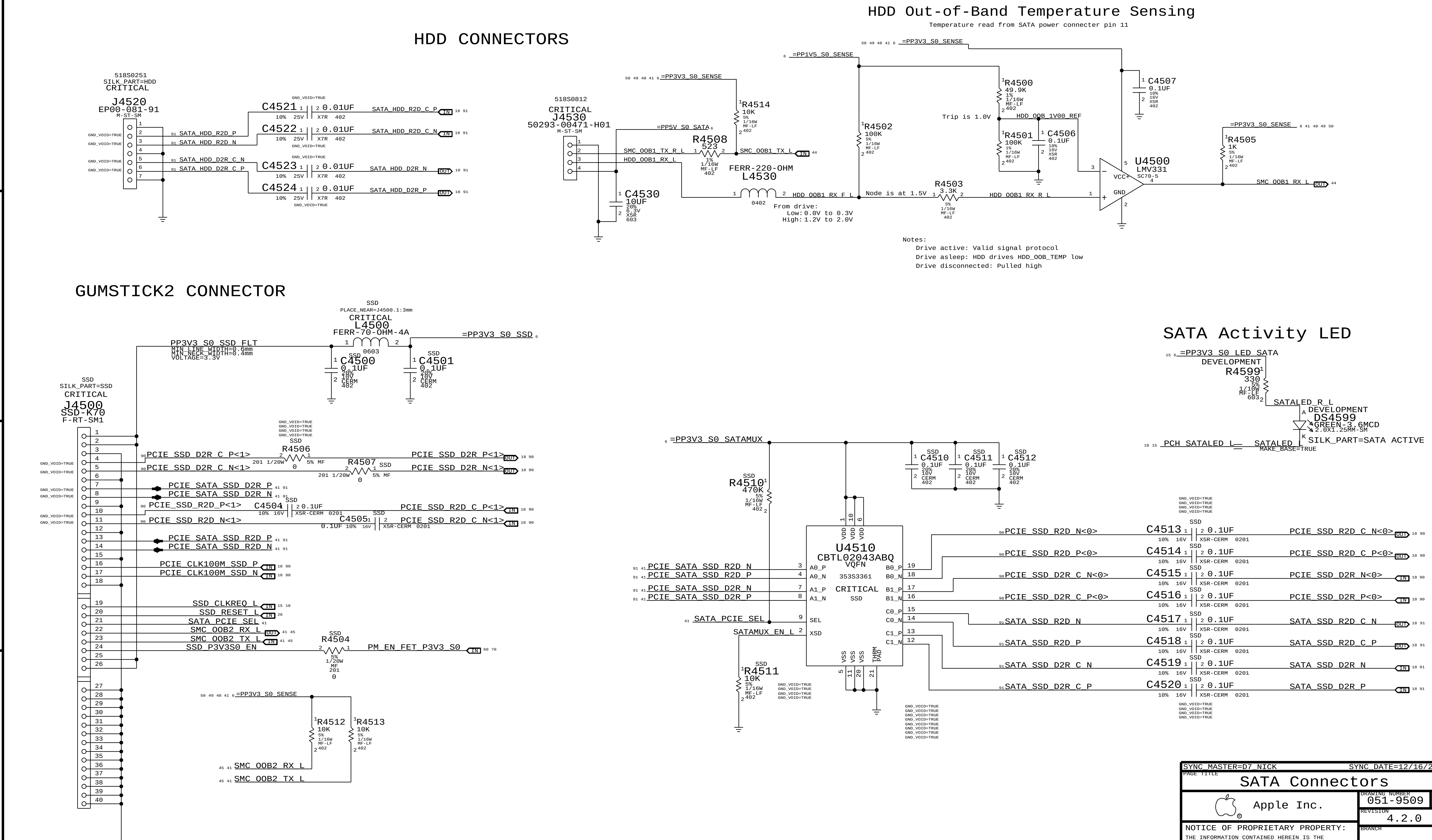

SYNC_MASTER=D7_NICK SYNC_DATE=12/16/2011

PAGE TITLE

SATA Connectors

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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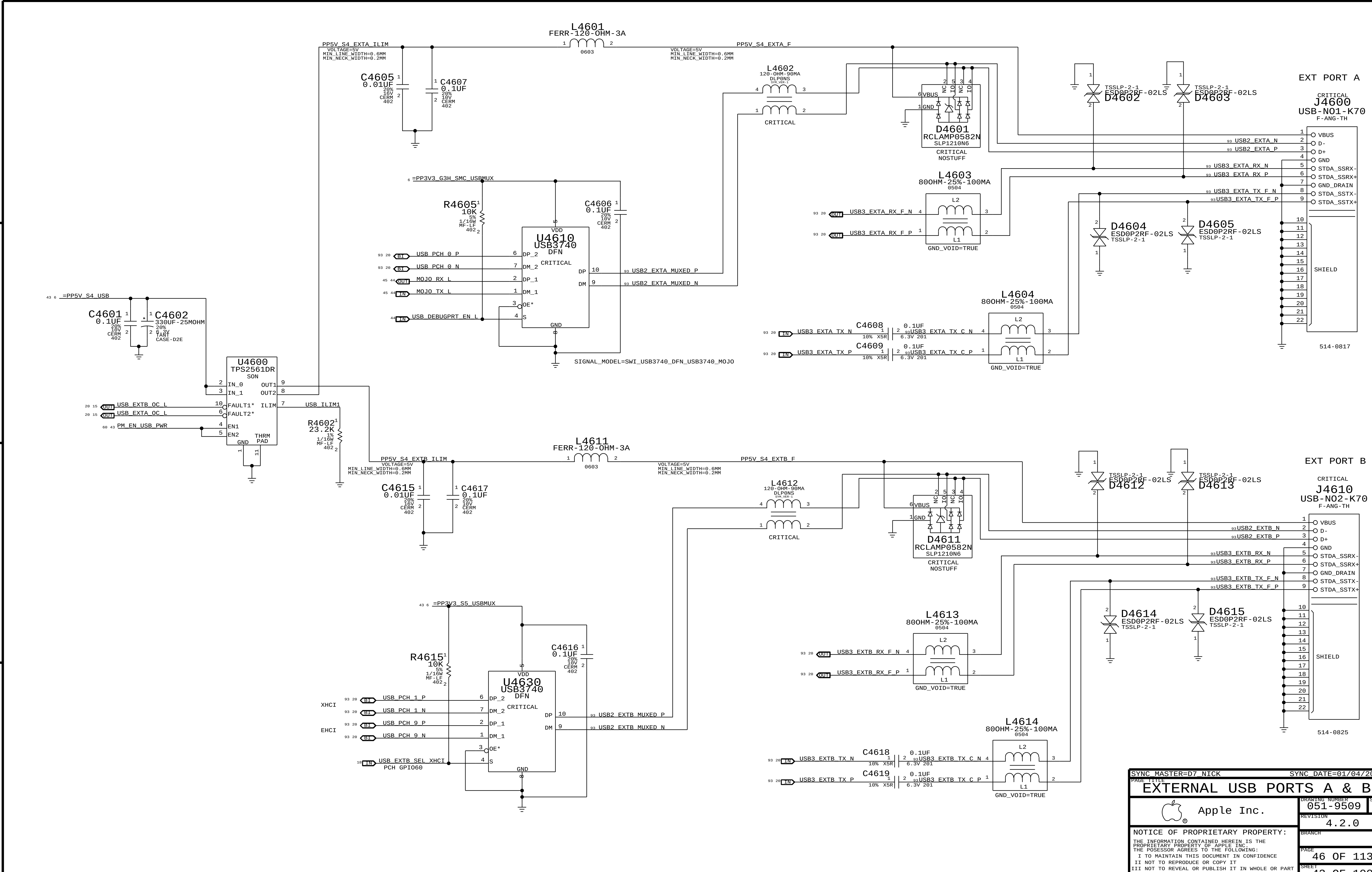

SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE

EXTERNAL USB PORTS A & B

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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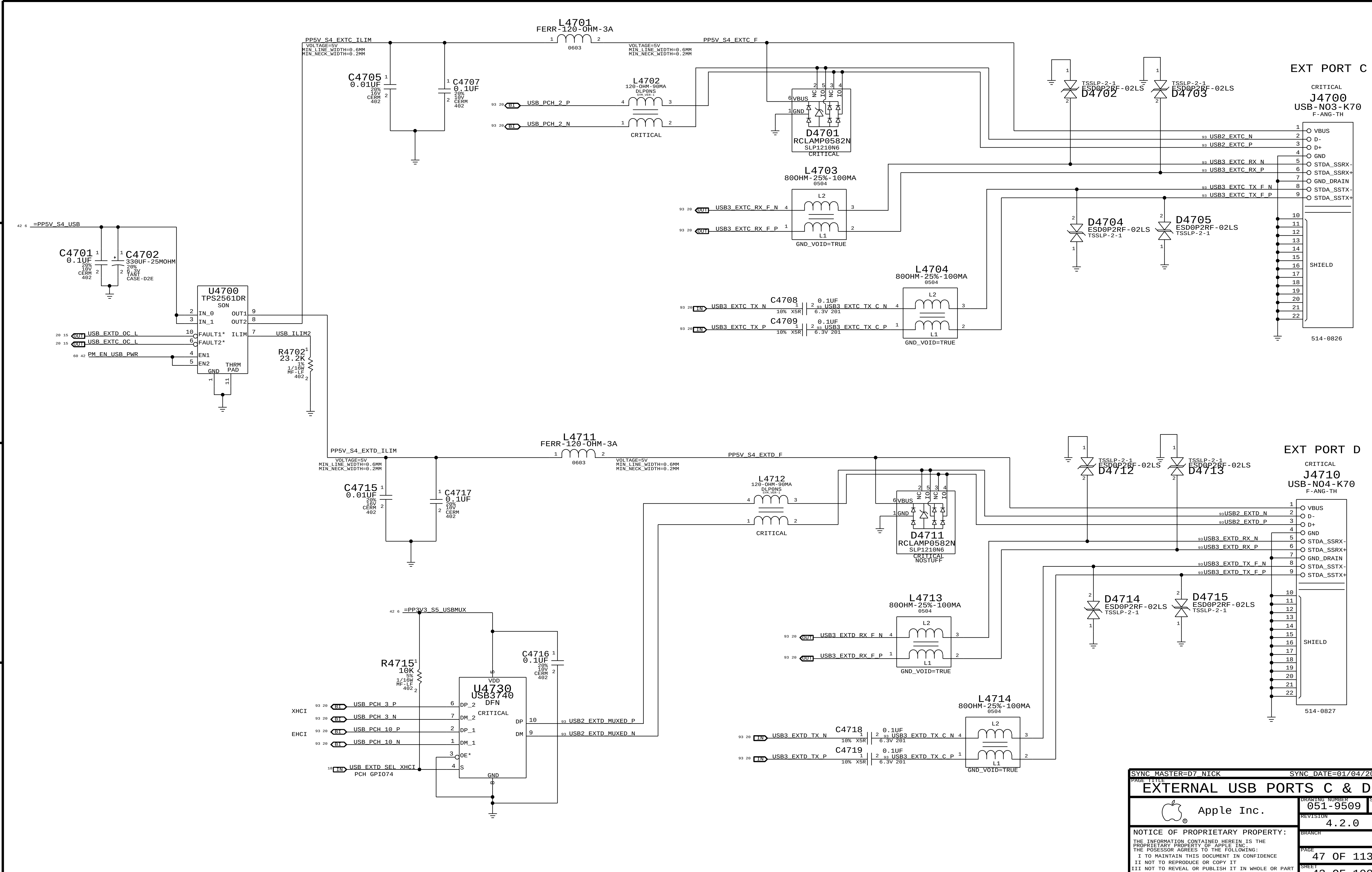
PAGE 46 OF 113

SHEET 42 OF 100



SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE

EXTERNAL USB PORTS C & D

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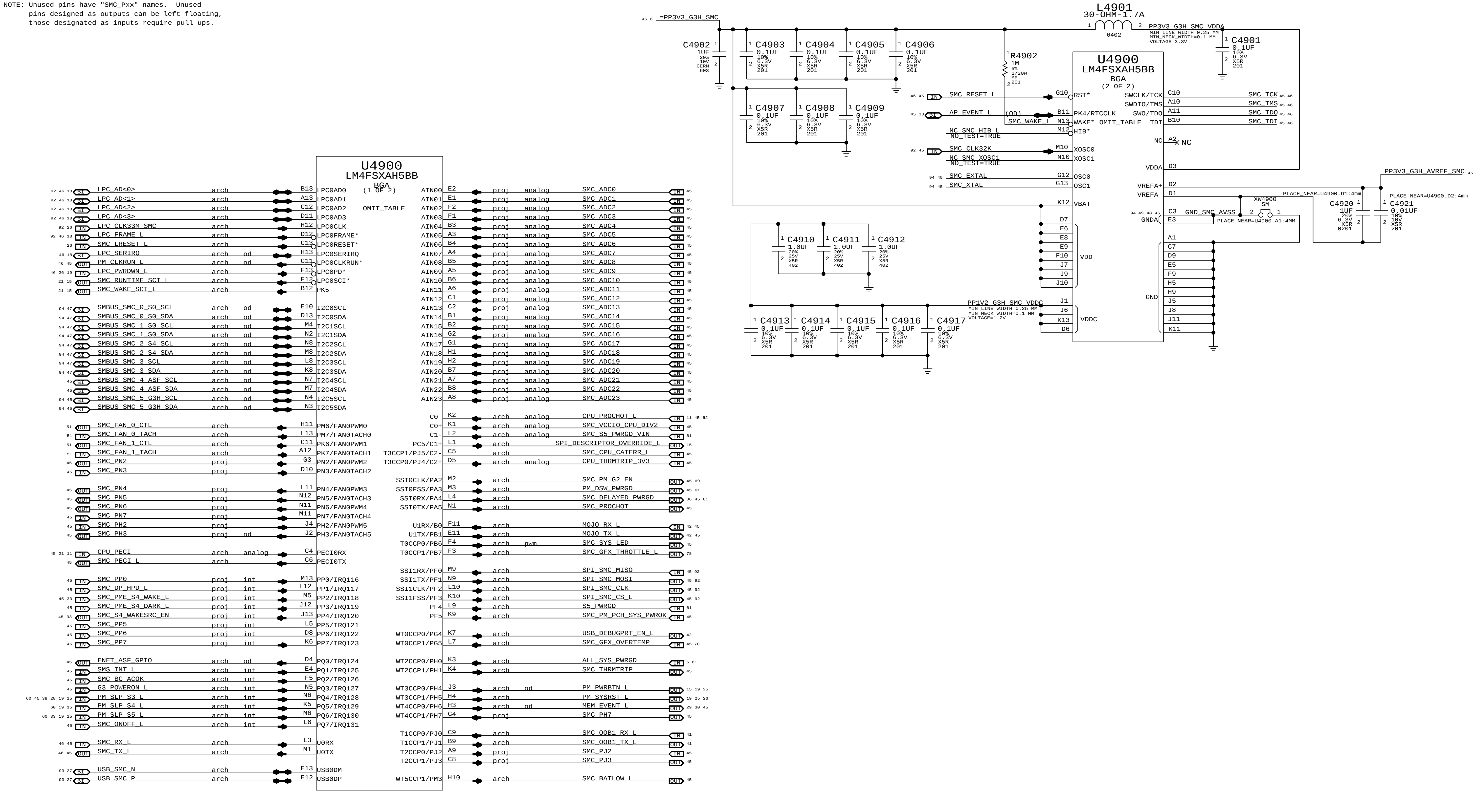
NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.



SYNC_MASTER=D7_DOUG SYNC_DATE=01/11/2012

PAGE TITLE

SMC

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

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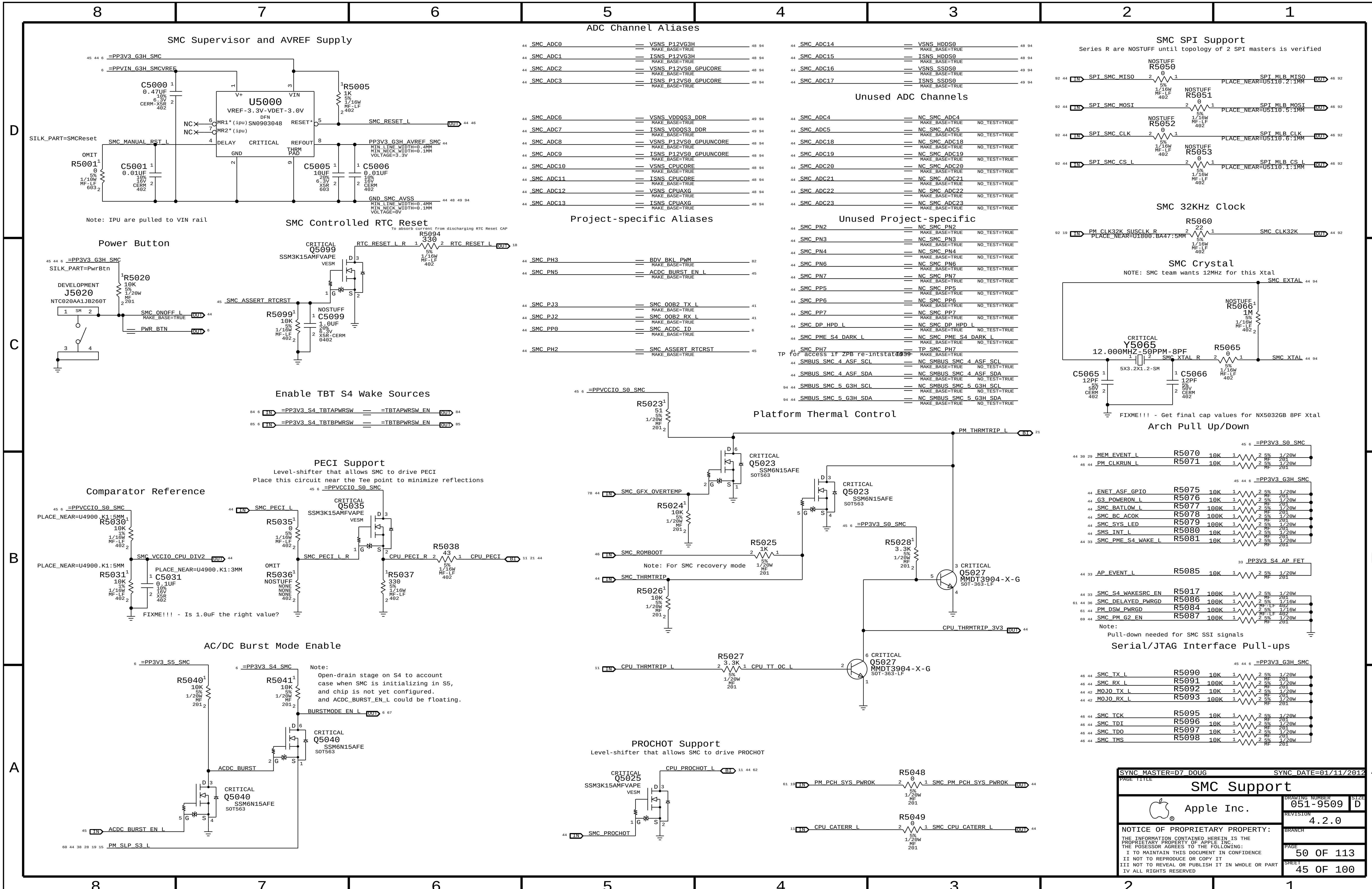

SYNC_MASTER=D7_DOUG SYNC_DATE=01/11/2012

PAGE TITLE

SMC Support

DRAWING NUMBER 051-9509 SIZE D

Apple Inc.

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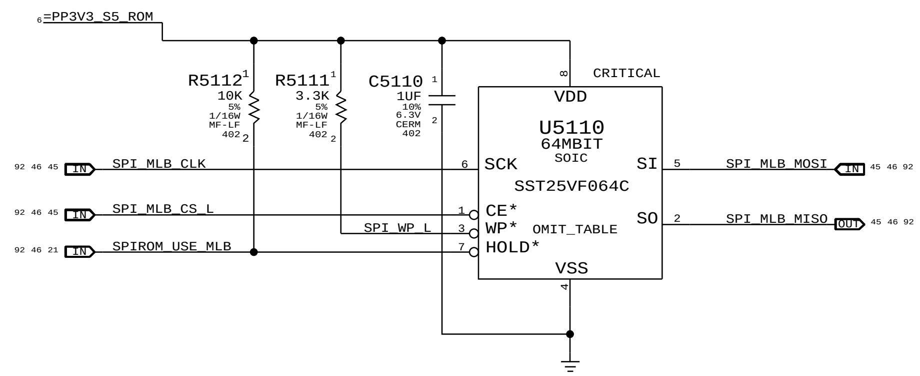
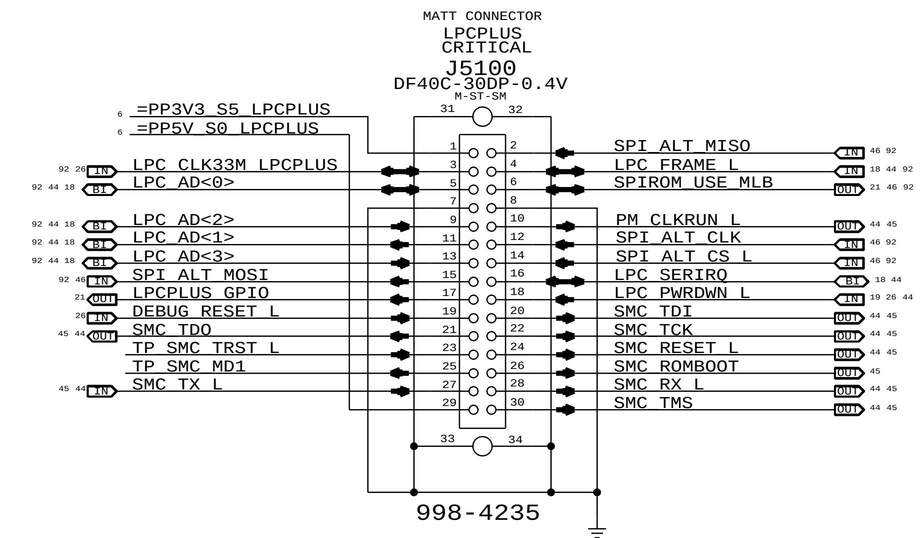
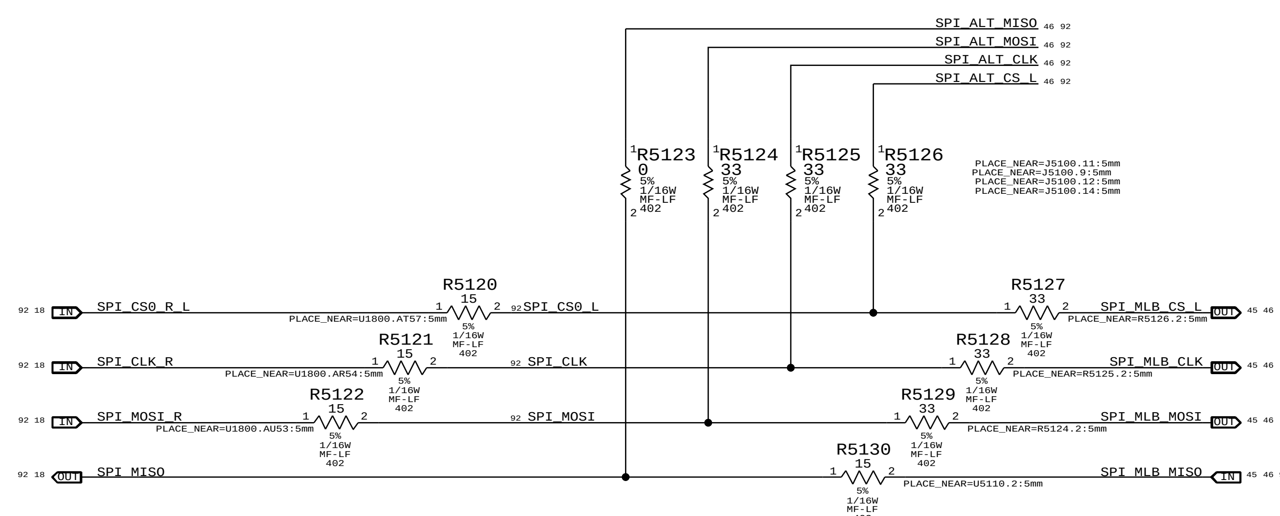
SPI BootROM



LPC+SPI Connector



SPI Series Termination



SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE

SPI and Debug Connector

Apple Inc.

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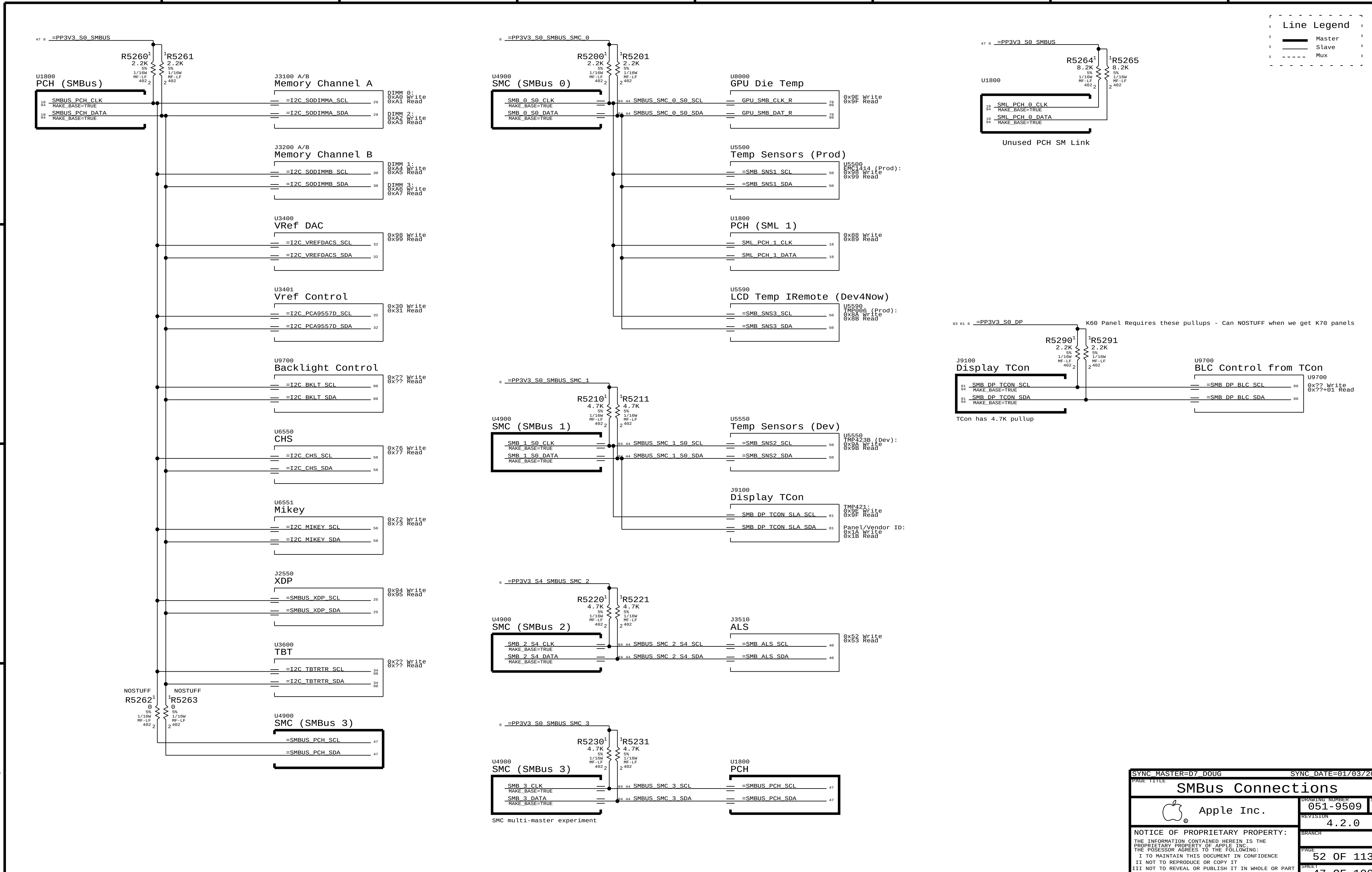

SYNC_MASTER=D7_DOUG SYNC_DATE=01/03/2012

PAGE TITLE

SMBus Connections

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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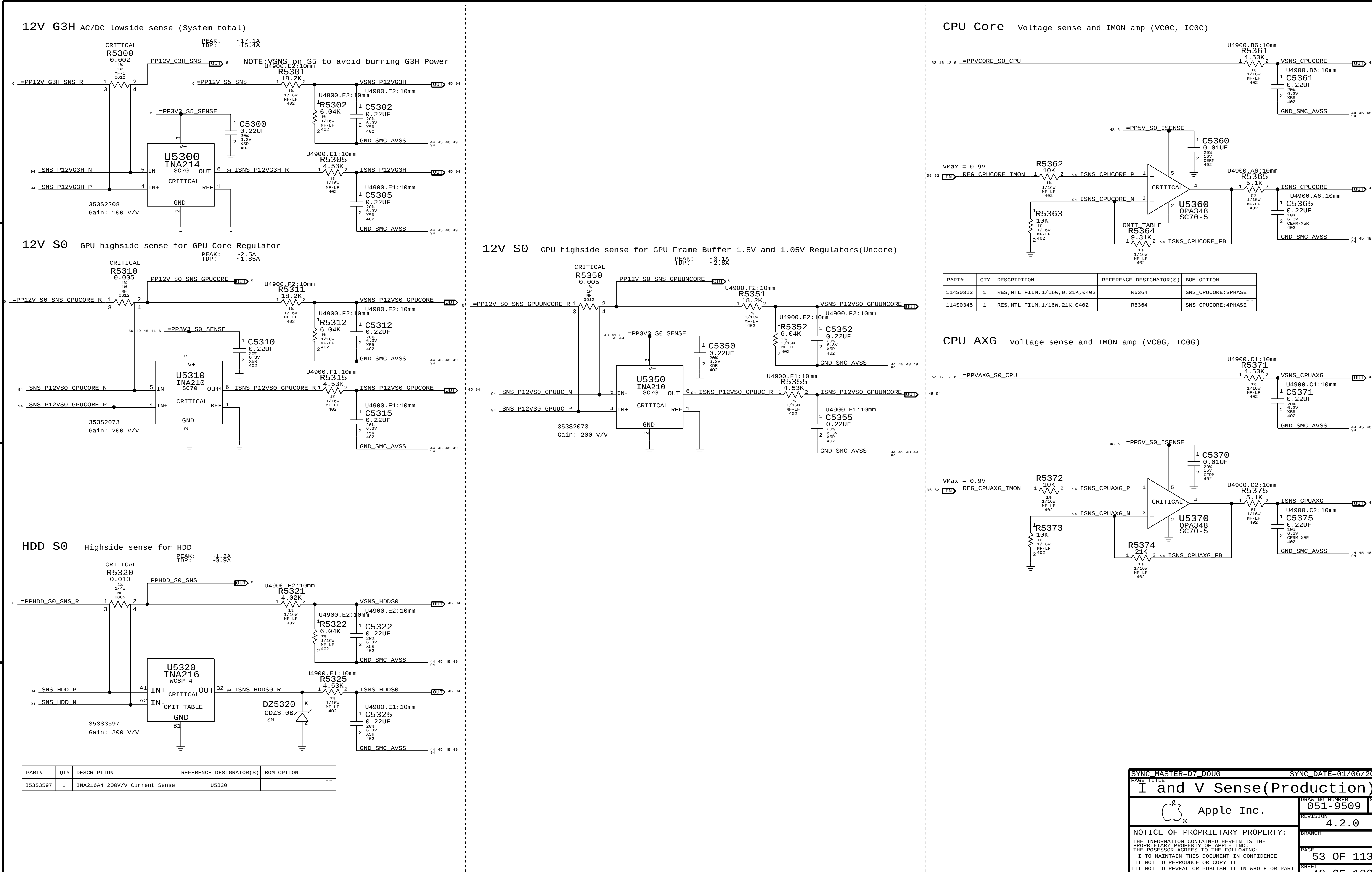
PAGE 52 OF 113

SHEET 47 OF 100



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0312	1	RES,MTL FILM,1/16W,9.31K,0402	R5364	SNS_CPUCORE:3PHASE
114S0345	1	RES,MTL FILM,1/16W,21K,0402	R5364	SNS_CPUCORE:4PHASE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S3597	1	INA216A4 200V/V Current Sense	U5320	

SYNC_MASTER=D7_DOUG SYNC_DATE=01/06/2012

PAGE TITLE

I and V Sense(Production)

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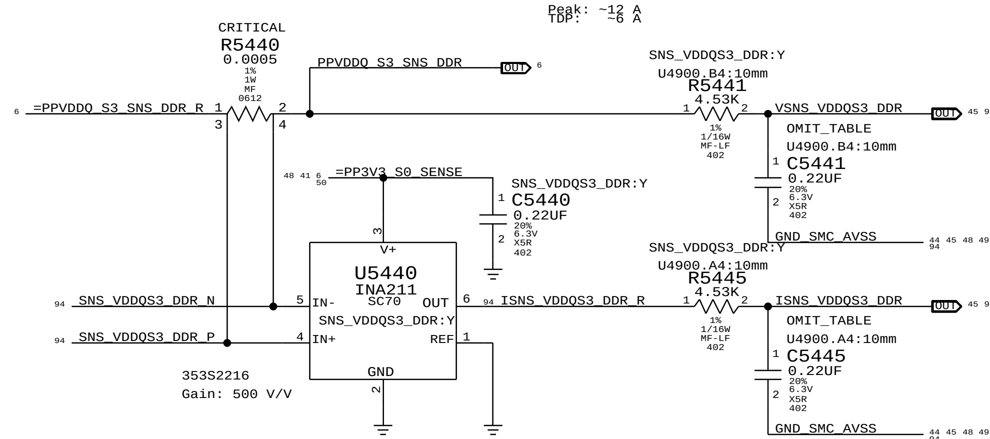
VDDQ S3 VDDQ lowside sense for SO-DIMM modules

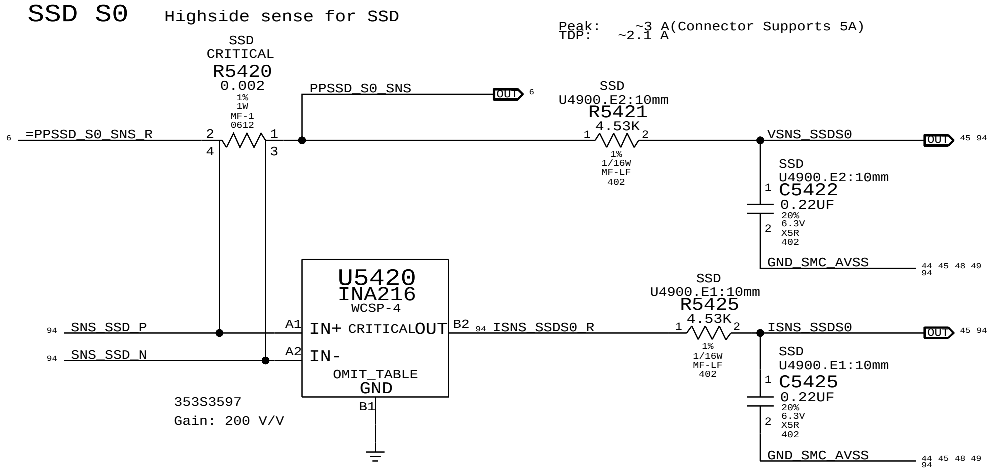
Peak: ~12 A
TDP: ~6 A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	2	CAP,0.22UF,402	C5441,C5445	SNS_VDDQS3_DDR:Y
116S0004	2	RES,0 OHM,402	C5441,C5445	SNS_VDDQS3_DDR:N

SSD S0 Highside sense for SSD

Peak: ~3 A(Connector Supports 5A)
TDP: ~2.1 A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S3597	1	INA216A4 200V/V Current Sense	U5420	SSD

SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE
I and V Sense(Development)

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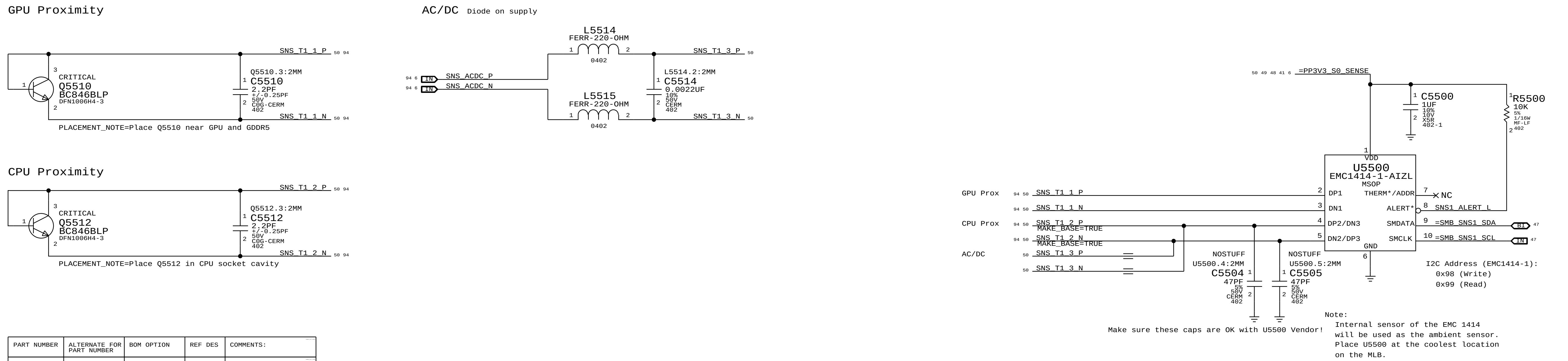
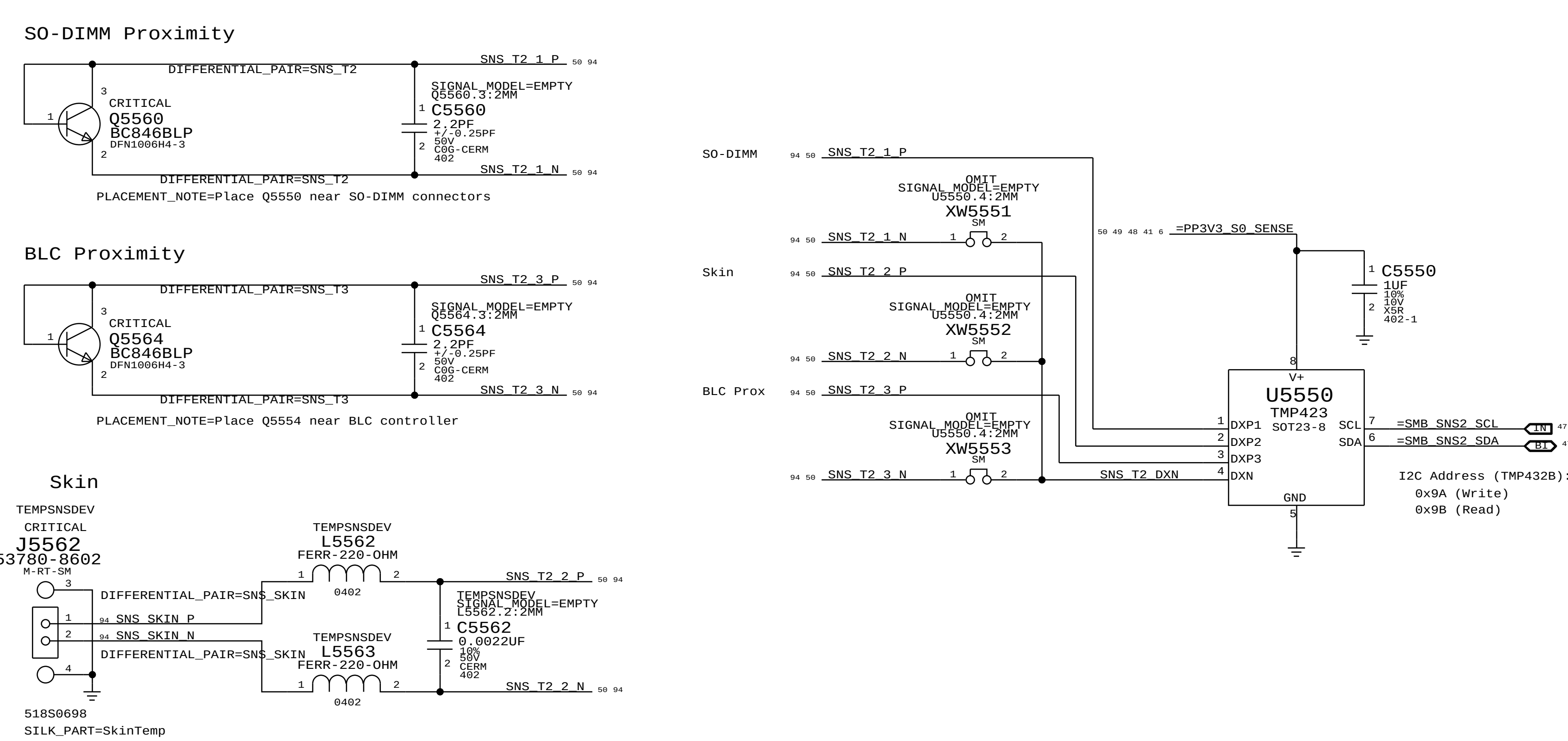
Temperature Sensor T1: Production Bound

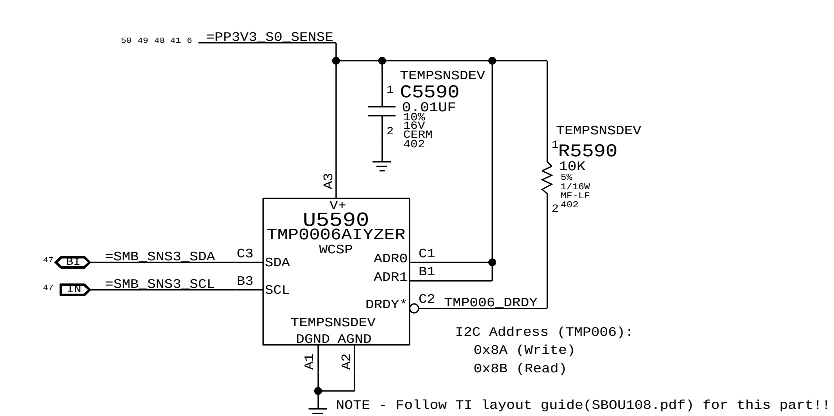


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

Temperature Sensor T2: Development Only



Temperature Sensor T3: LCD Remote Sensor(Dev4Now)



SYNC_MASTER=D7_DOUG SYNC_DATE=12/13/2011

PAGE TITLE: Temperature Sensors

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DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

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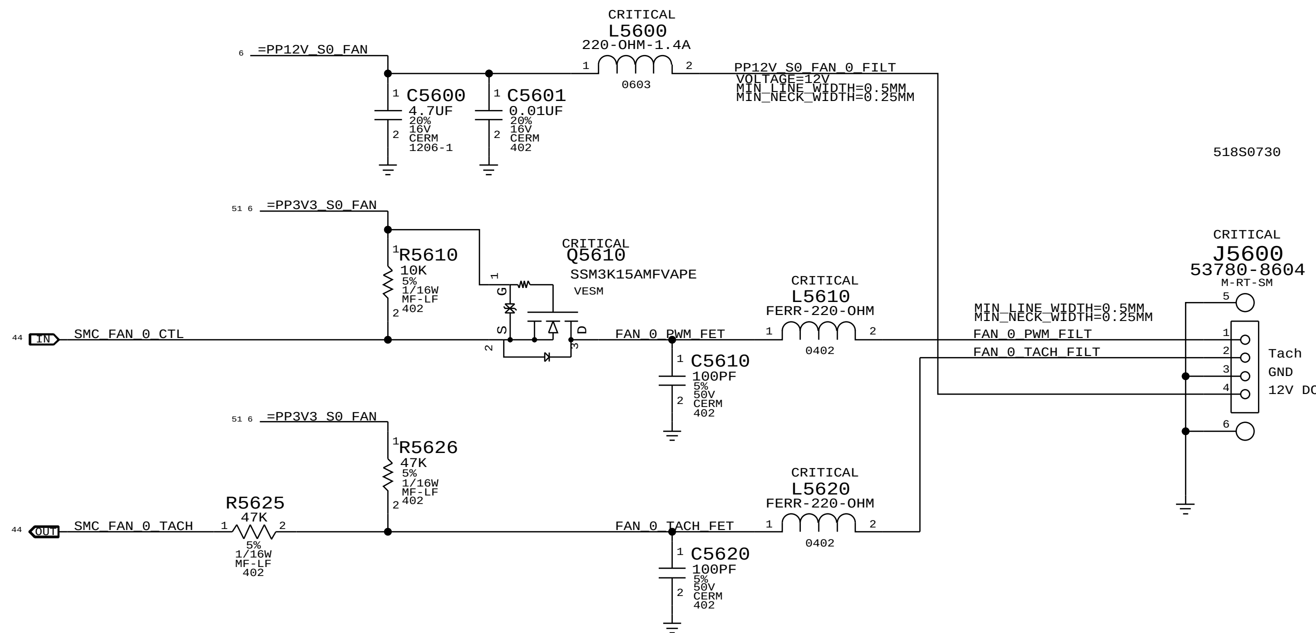
SMC Fan 0 (System)

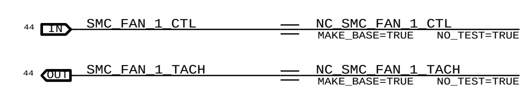
Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.



SMC Fan 1 (Unused)



SYNC_MASTER=K70_MLB SYNC_DATE=11/30/2011

PAGE TITLE: System Fan

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

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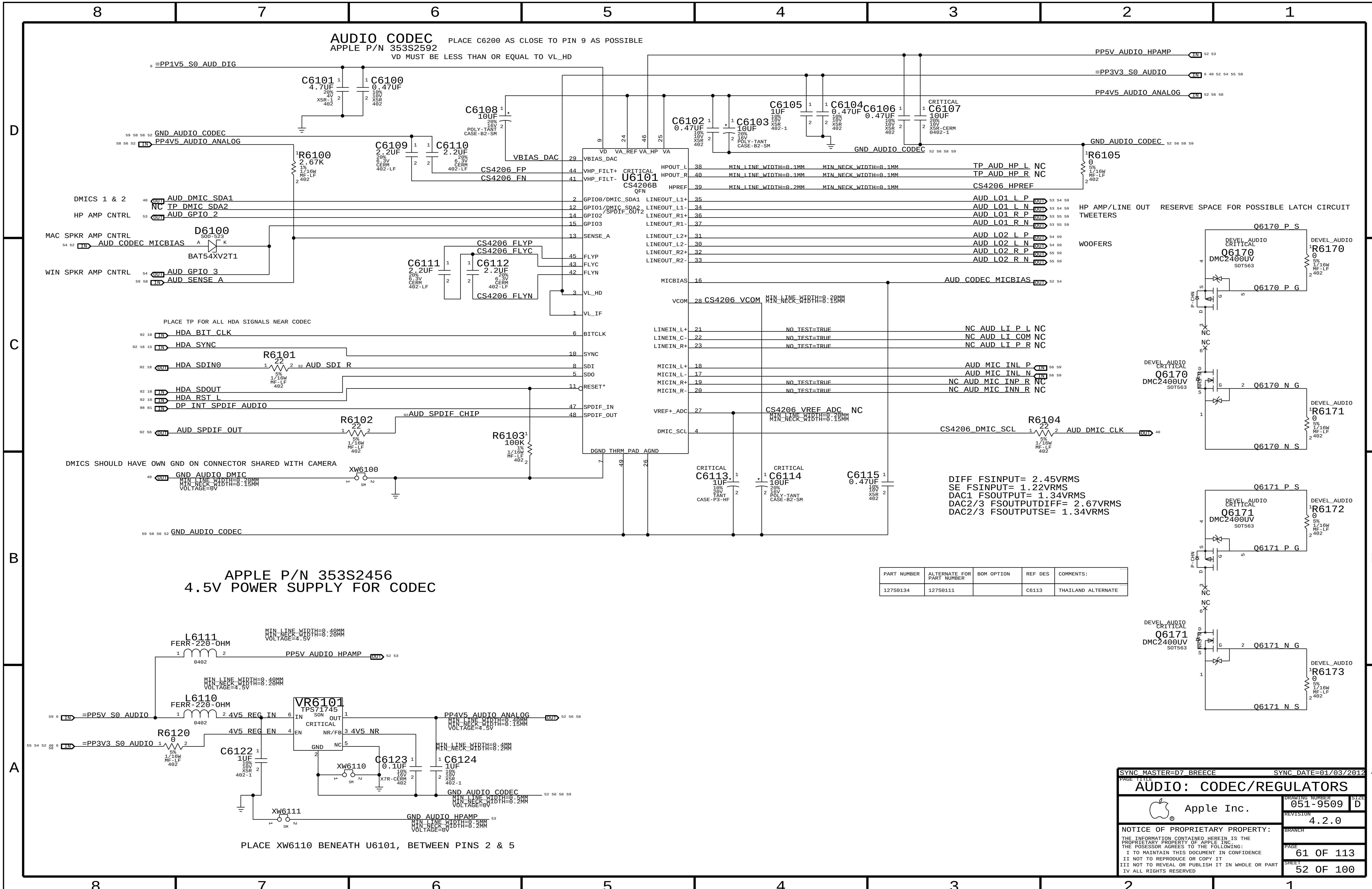

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE
AUDIO: CODEC/REGULATORS

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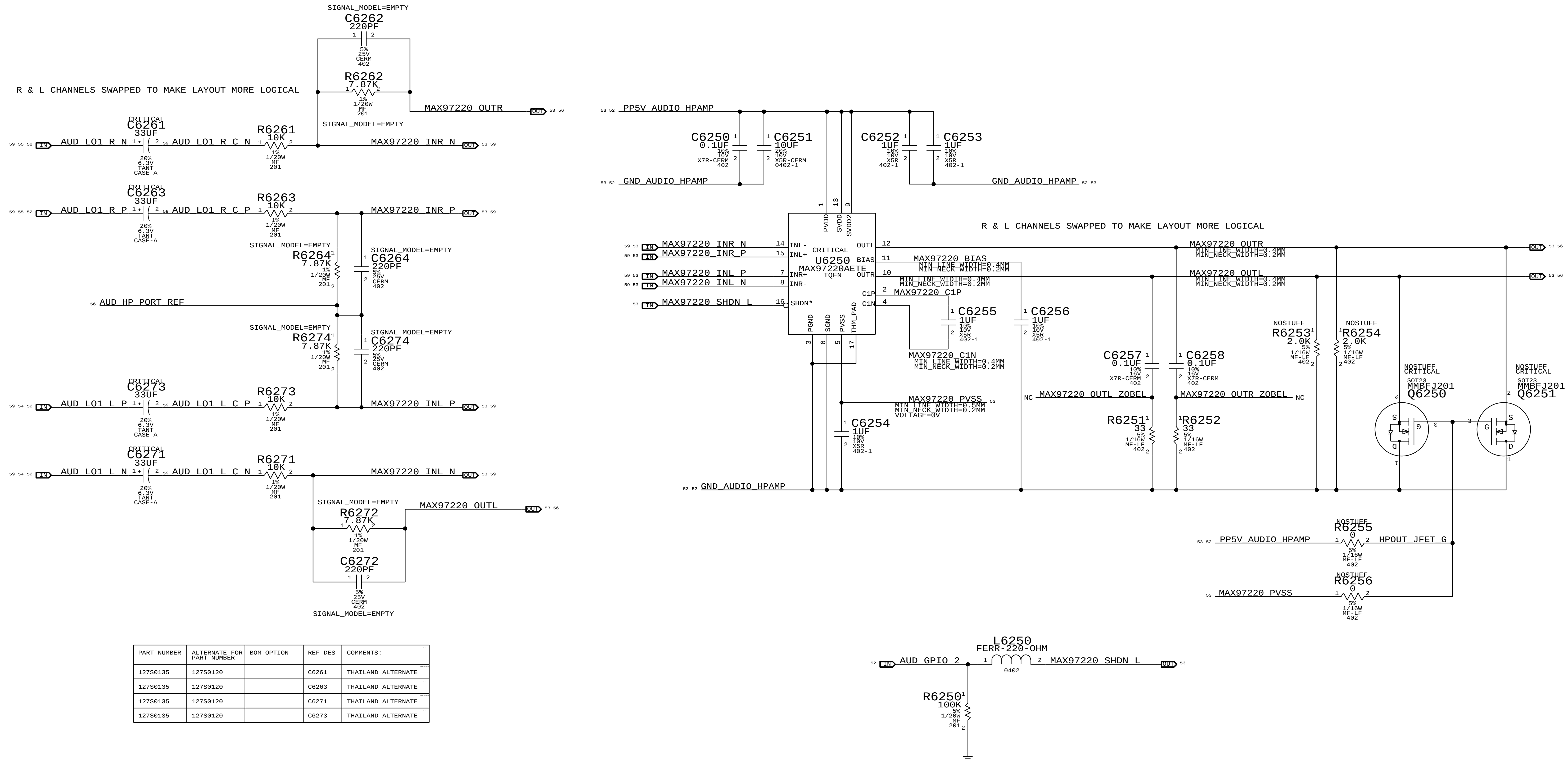

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0135	127S0120		C6261	THAILAND ALTERNATE
127S0135	127S0120		C6263	THAILAND ALTERNATE
127S0135	127S0120		C6271	THAILAND ALTERNATE
127S0135	127S0120		C6273	THAILAND ALTERNATE

SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE

AUDIO: HEADPHONE AMP

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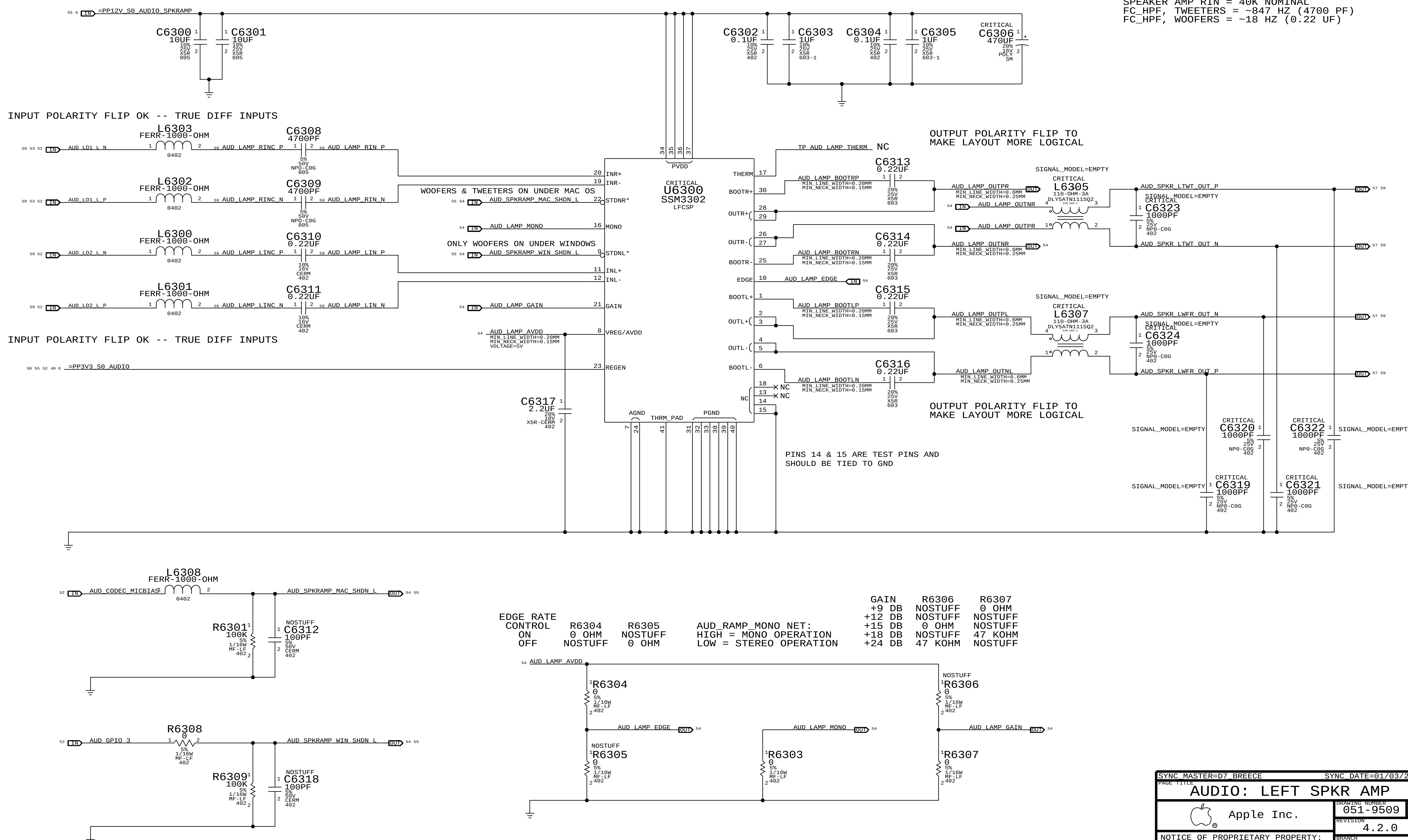
LEFT CH SPEAKER AMP

APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~18 HZ (0.22 UF)



EDGE RATE CONTROL	R6304	R6305
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6306	R6307
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	NOSTUFF
+15 DB	0 OHM	NOSTUFF
+18 DB	NOSTUFF	47 KOHM
+24 DB	47 KOHM	NOSTUFF

SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE: AUDIO: LEFT SPKR AMP

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DRAWING NUMBER 051-9509 SIZE D

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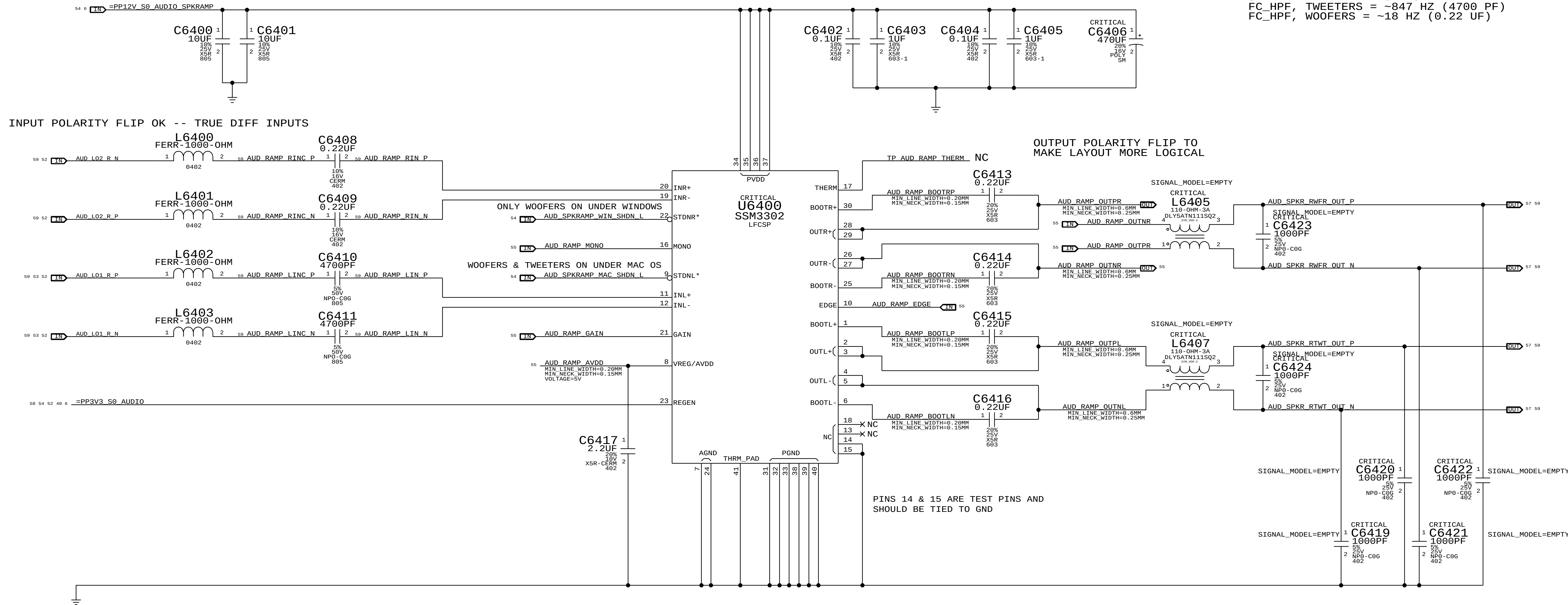
RIGHT CH SPEAKER AMP

APPLE P/N 353S3163

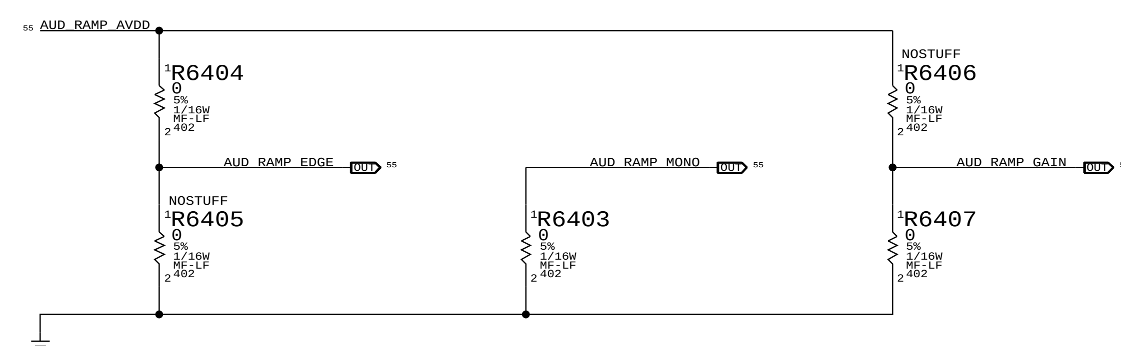
SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~18 HZ (0.22 UF)



EDGE RATE CONTROL	R6404	R6405
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6406	R6407
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	NOSTUFF
+15 DB	0 OHM	NOSTUFF
+18 DB	NOSTUFF	47 KOHM
+24 DB	47 KOHM	NOSTUFF



SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE: AUDIO: RIGHT SPKR AMP

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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640

PORT B LEFT(HEADSET MIC)
HP=80HZ, LP=10.63KHZ

MIKEY 1A
APN:353S2640
MIKEY ADDRESS: WRITE=72H, READ=73H

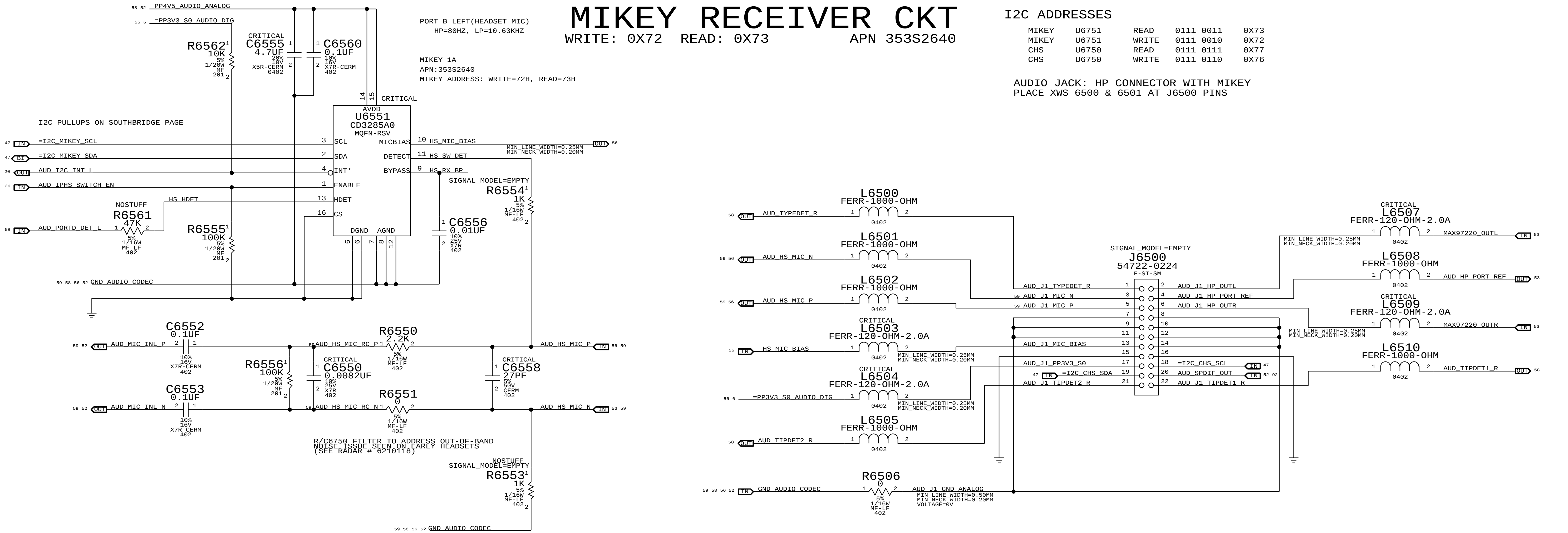
I2C ADDRESSES

MIKEY	U6751	READ	0111 0011	0X73
MIKEY	U6751	WRITE	0111 0010	0X72
CHS	U6750	READ	0111 0111	0X77
CHS	U6750	WRITE	0111 0110	0X76

AUDIO JACK: HP CONNECTOR WITH MIKEY
PLACE XWS 6500 & 6501 AT J6500 PINS

I2C PULLUPS ON SOUTHBRIDGE PAGE

R/C6750 FILTER TO ADDRESS OUT-OF-BAND NOISE ISSUE SEEN ON EARLY HEADSETS (SEE RADAR # 6210118)



SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE: AUDIO: Jack, Mikey, CHS Switch

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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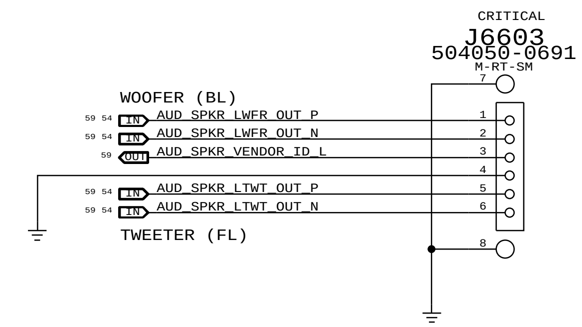
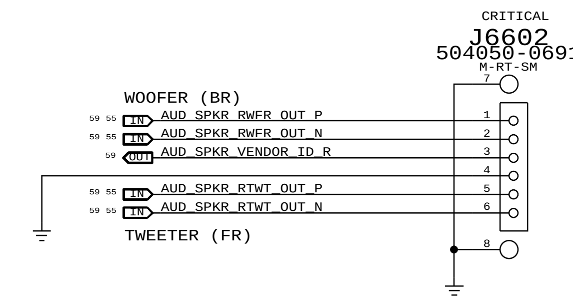
SHEET 56 OF 100

SPEAKER CABLE CONNECTORS





SYNC_MASTER=D7_BREECE	SYNC_DATE=01/03/2012
PAGE TITLE	Audio: Spkr/Mic Conn.
Apple Inc.	DRAWING NUMBER 051-9509 SIZE D
	REVISION 4.2.0
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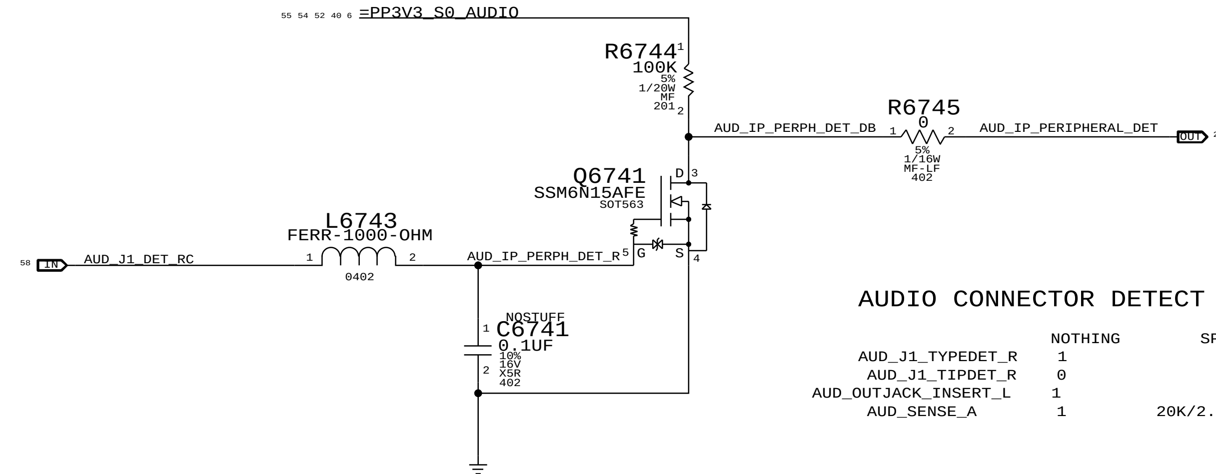
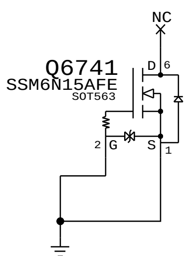
	SHEET 57 OF 100

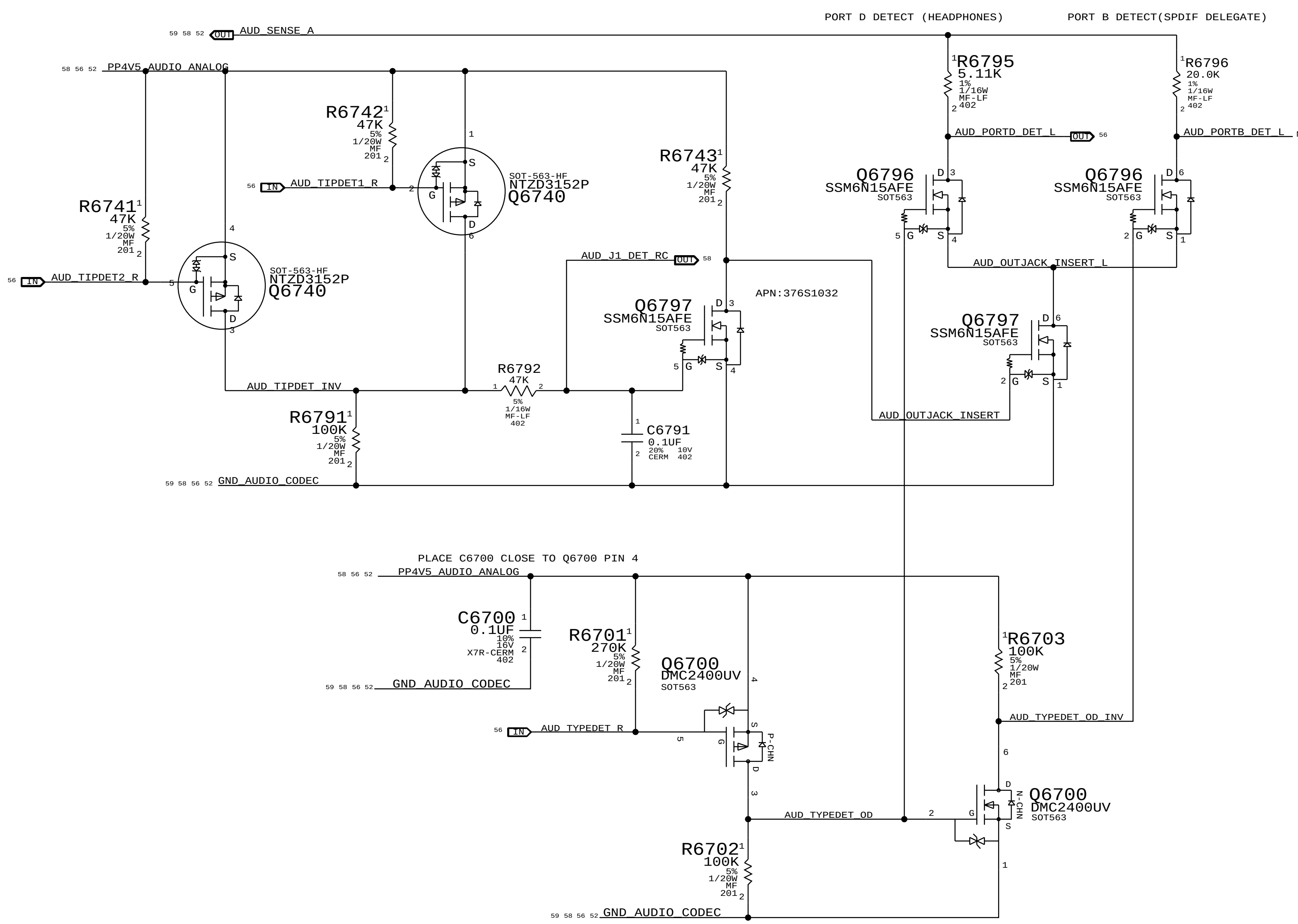
IPHS HS Detect Debounce CKT





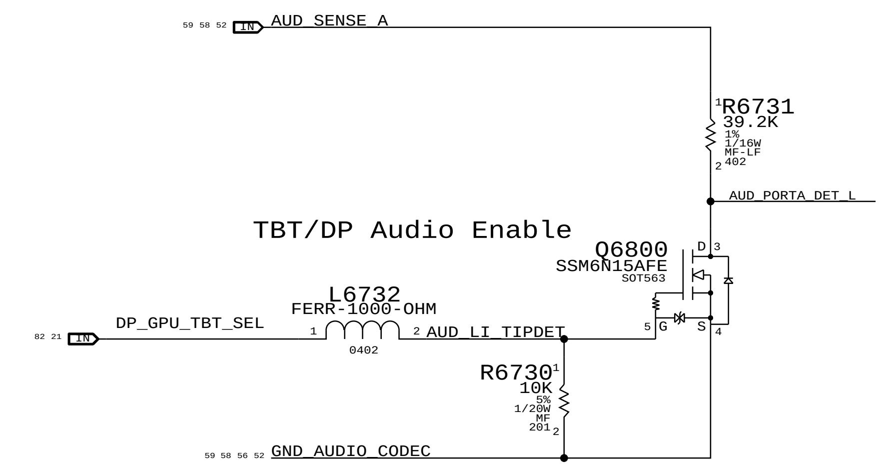
AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_J1_TYPEDET_R	1	1	0
AUD_J1_TIPDET_R	0	1	1
AUD_OUTJACK_INSERT_L	1	0	0
AUD_SENSE_A	1	20K/2.67K RDIV	39.2K/2.67K RDIV



LI Insert Detect (DETECT A)

TBT/DP Audio Enable



SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE: AUDIO: Detects/Grounding

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DRAWING NUMBER 051-9509 SIZE D

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0x0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	PANTHER POINT GPIO 16	PANTHER POINT GPIO 5 (RCVR_INT) PANTHER POINT GPIO 3 (PERIPH DET)

OTHER DETECT

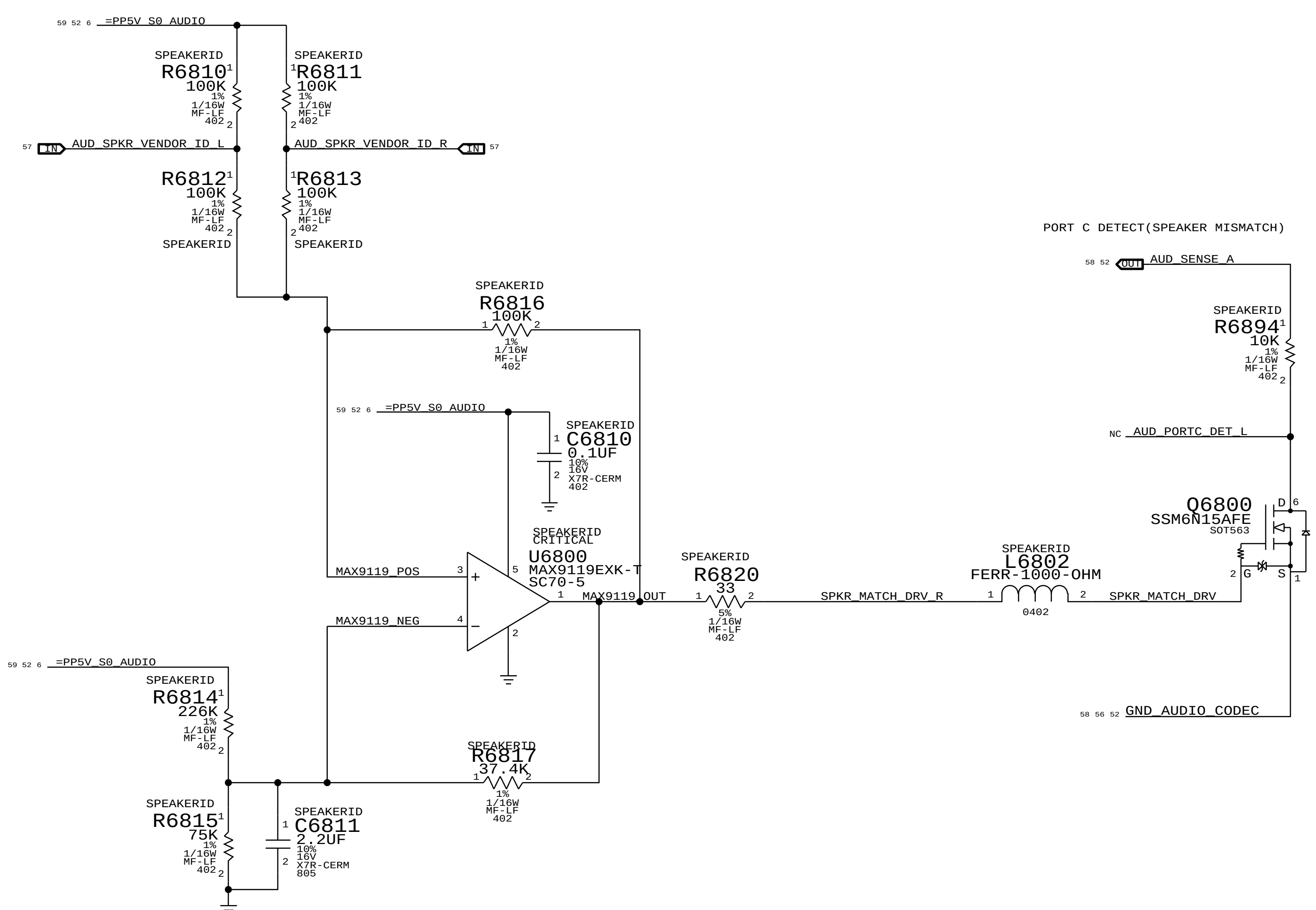
FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	52 53 54
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	52 53 54
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	53
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	53
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	52 53 55
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	52 53 55
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	53
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	53
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	52 54
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	52 54
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	52 55
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	52 55
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	55
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	55
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	55
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	55
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	55
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	55
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	55
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	55
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	54
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	54
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	54
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	54
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	54
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	54
I264	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	54
I263	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	54
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	53
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	53
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	53
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	53
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	55 57
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	55 57
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	55 57
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	55 57
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	54 57
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	54 57
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	54 57
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	54 57
I326	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	52 56
I327	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	52 56
I328	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_P	56
I329	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_N	56
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_P	56
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_N	56
I324	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_P	56
I325	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_N	56



SYNC_MASTER=D7_BREECE SYNC_DATE=01/03/2012

PAGE TITLE

AUDIO: Speaker ID

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

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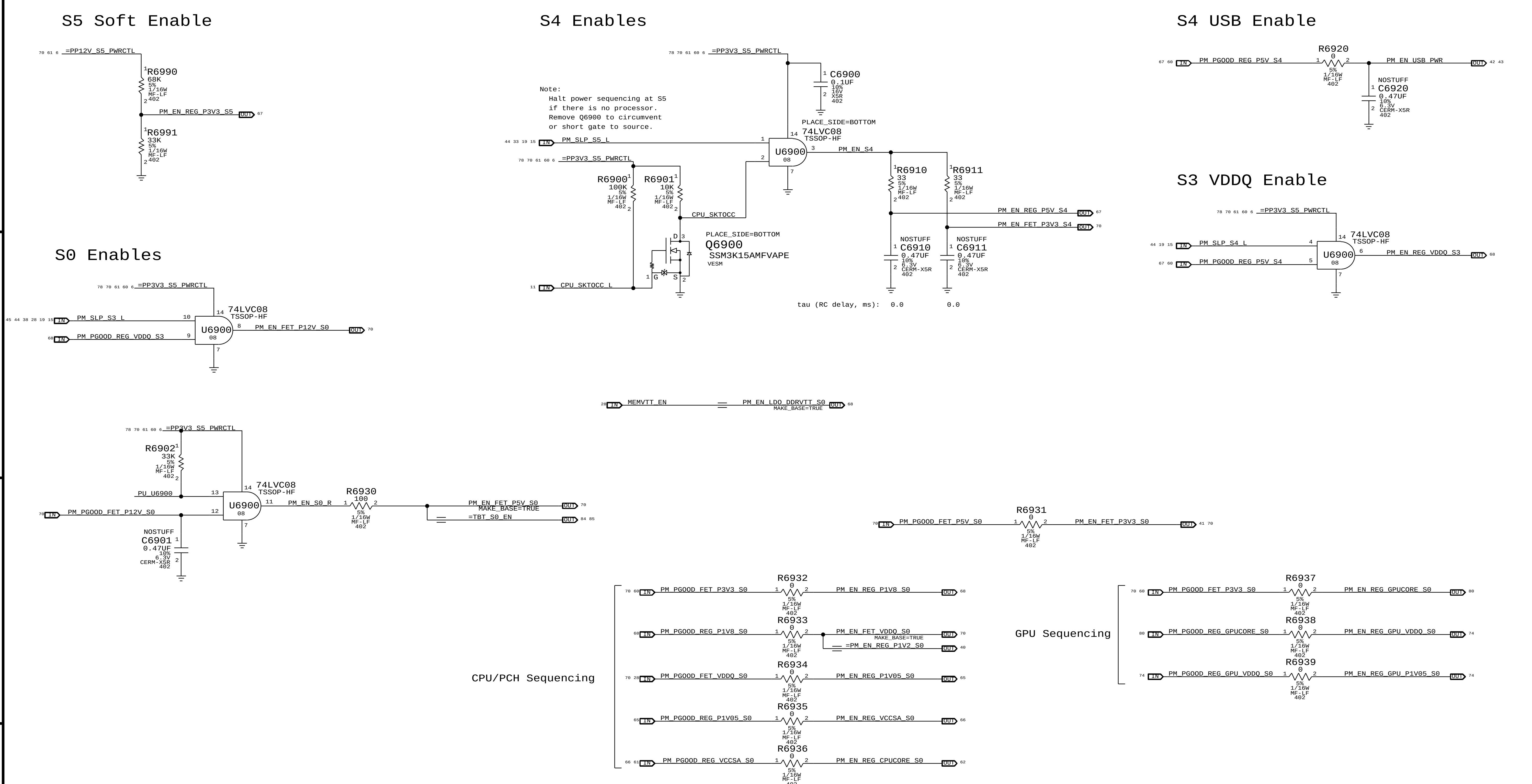

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.8 V 1.5 V, PCH Core/PLL/VRM)
Uncore: VccSA, VDDQ, VccA (1.8 V), VccIO (VccSA, VccA, and VccIO must ramp within 50 ms of each other)

Notes on sequencing requirements

Intel:
1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK

NVIDIA:
1. 3V3_S0 must ramp first
2. IFPA/B_IOVDD (1.8 V) can ramp simultaneously or after 3V3_S0 (unused)
3. NVVDD (GPU_CORE) must ramp after IFPA/B_IOVDD
4. VDDQ must ramp after CPU_CORE
5. PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
6. All rails must reach their target voltages in more than 40 uS

SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE
PM Regulator Enables

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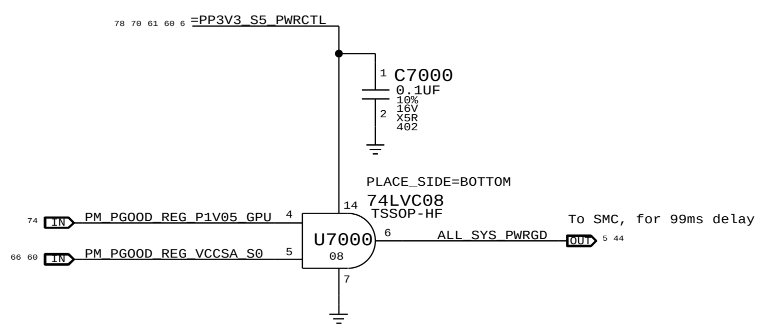
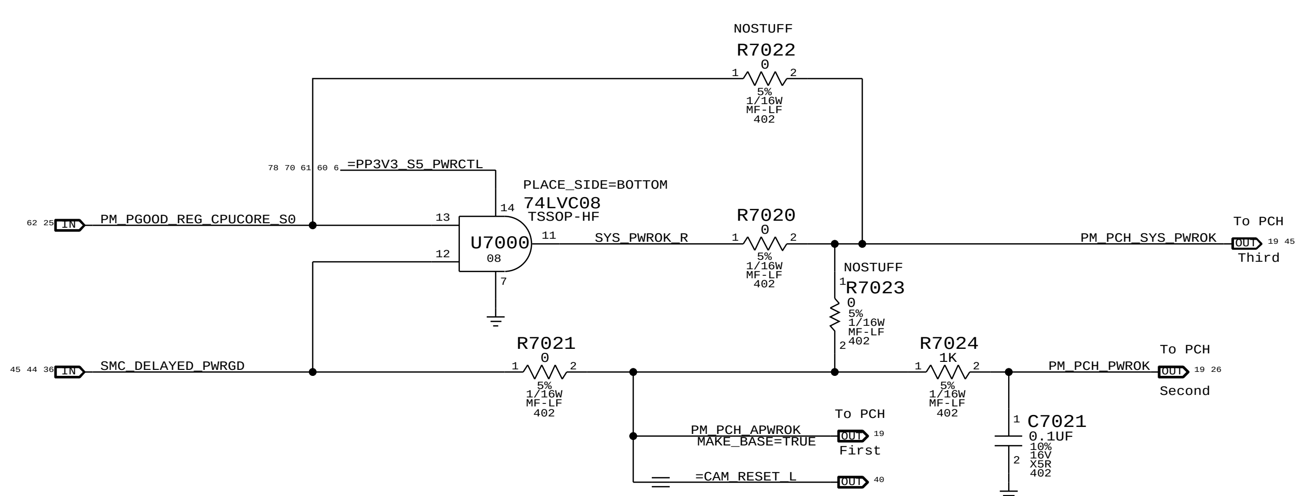
Platform and UnCore Power Good

Derive SMC ALL_SYS_PWRGD

Note: GPU power goods are implicitly included because the power goods for VDDQ, DPVDDC, and GPU Core are wired-or together



PCH Power Goods



Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

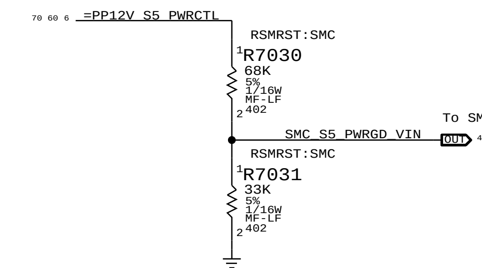
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

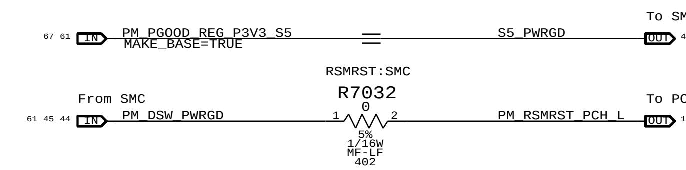
Requirements:
- Power on:
  - Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
  - Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.

SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.

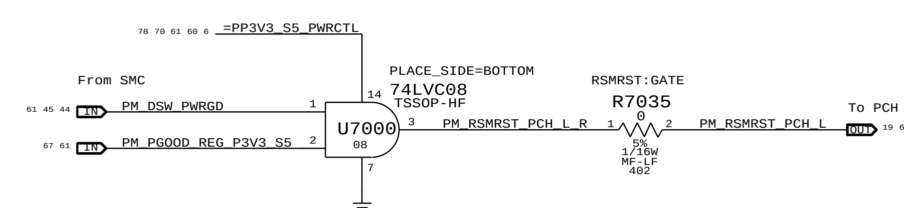
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.





Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=D7_NICK SYNC_DATE=12/13/2011

PAGE TITLE: PM Power Good

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DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

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PAGE: 70 OF 113

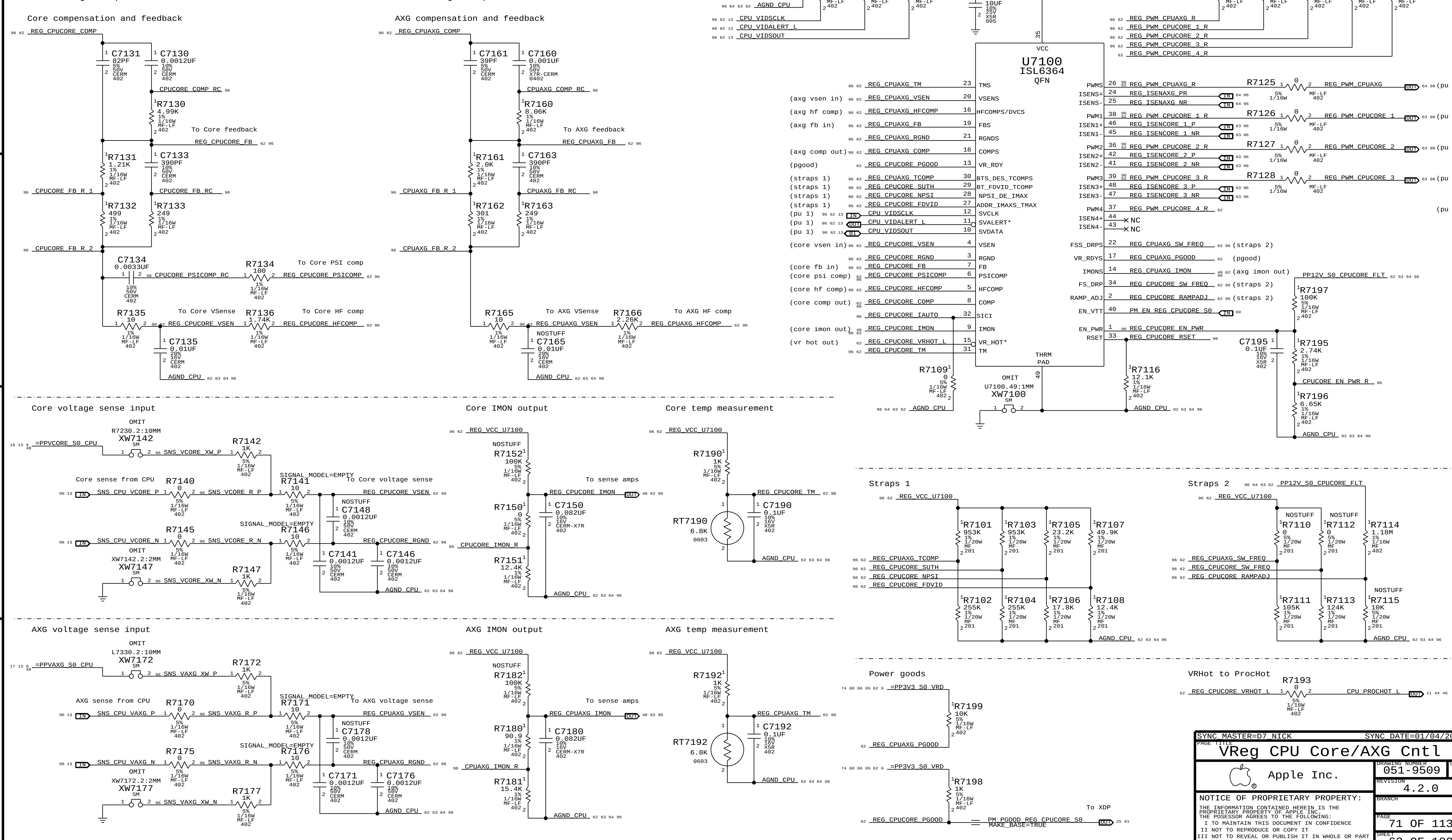
SHEET: 61 OF 100

CPU Core S0 Regulator

Max avg current: ? A (design)/ 41.05 A (budget)
Max peak current: ? A (design)/75.05 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

CPU AXG S0 Regulator

Max avg current: ? A (design)/ 10 A (budget)
Max peak current: ? A (design)/ 30 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz



SYNC_MASTER=D7_NICK	SYNC_DATE=01/04/2012
PAGE TITLE	VReg CPU Core/AXG Cntl
Apple Inc.	DRAWING NUMBER 051-9509 SIZE D
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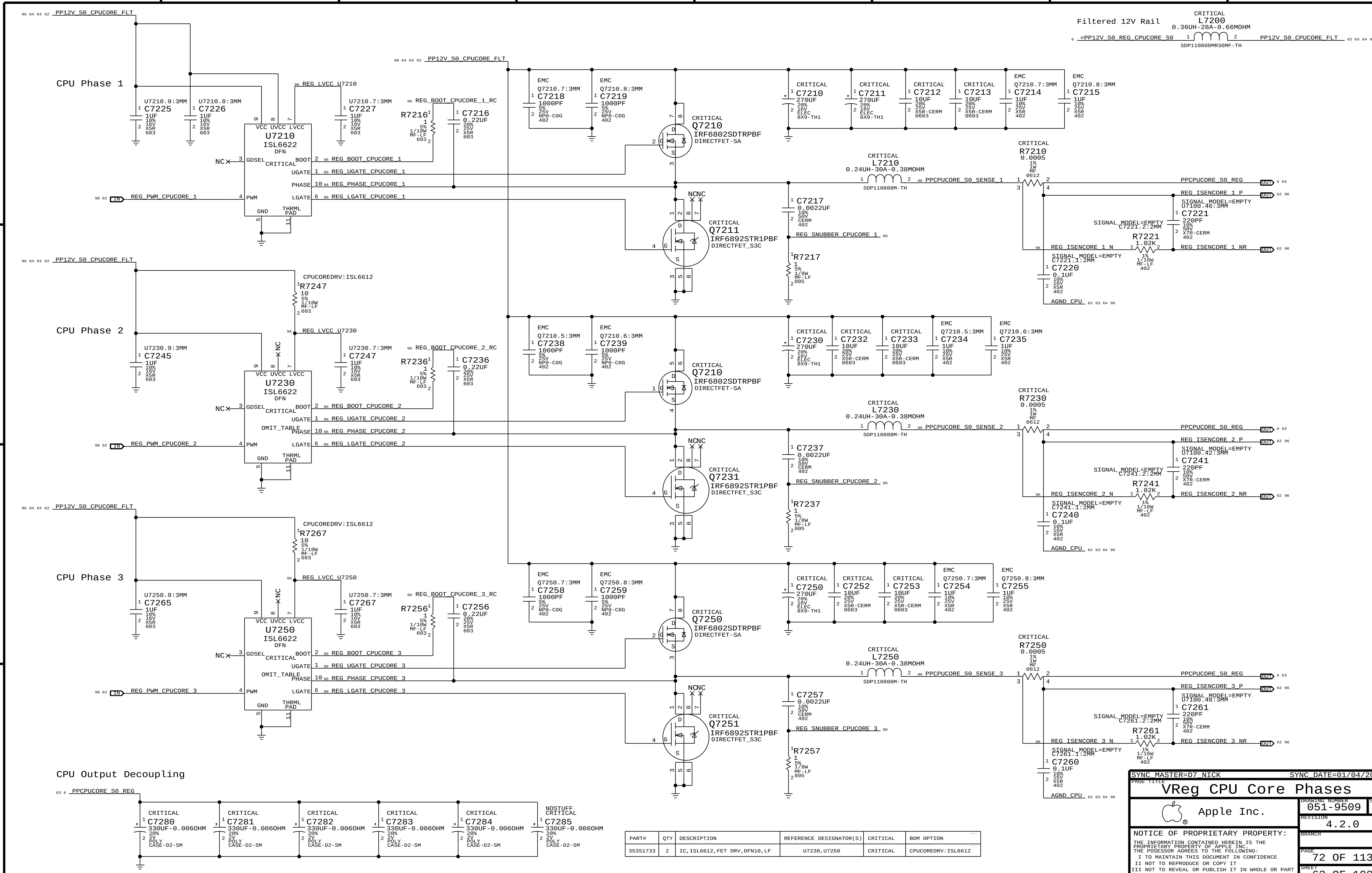

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S1733	2	IC,ISL6612,FET DRV,DFN10,LF	U7230,U7250	CRITICAL	CPUCOREDRV:ISL6612

SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE

VReg CPU Core Phases

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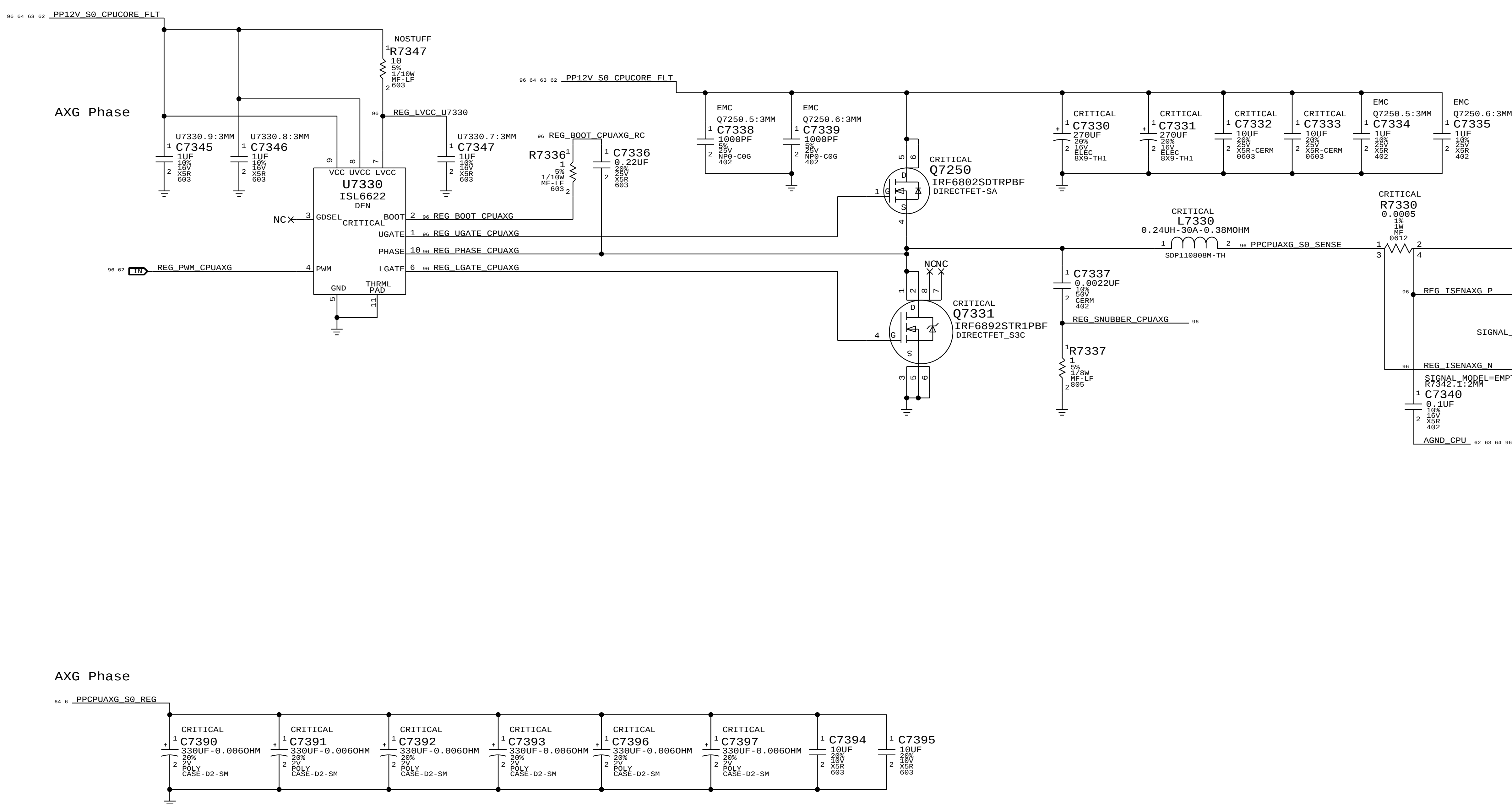

SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE

VReg CPU AXG Phases

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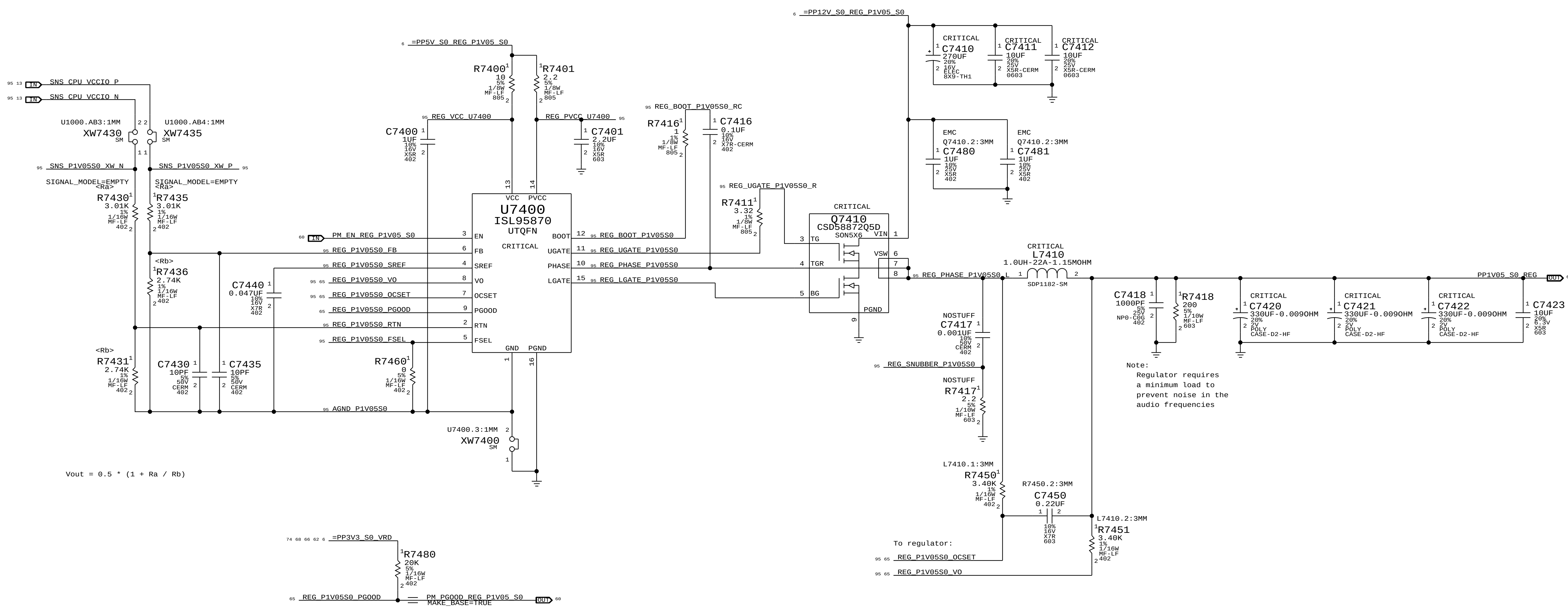
CPU VccIO/PCH (1.05V) S0 Regulator

Max avg current: ? A (design)/ 14.38 A (budget)
Max peak current: ? A (design)/ 18.38 A (budget)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE: VReg CPU/PCH 1.05V S0

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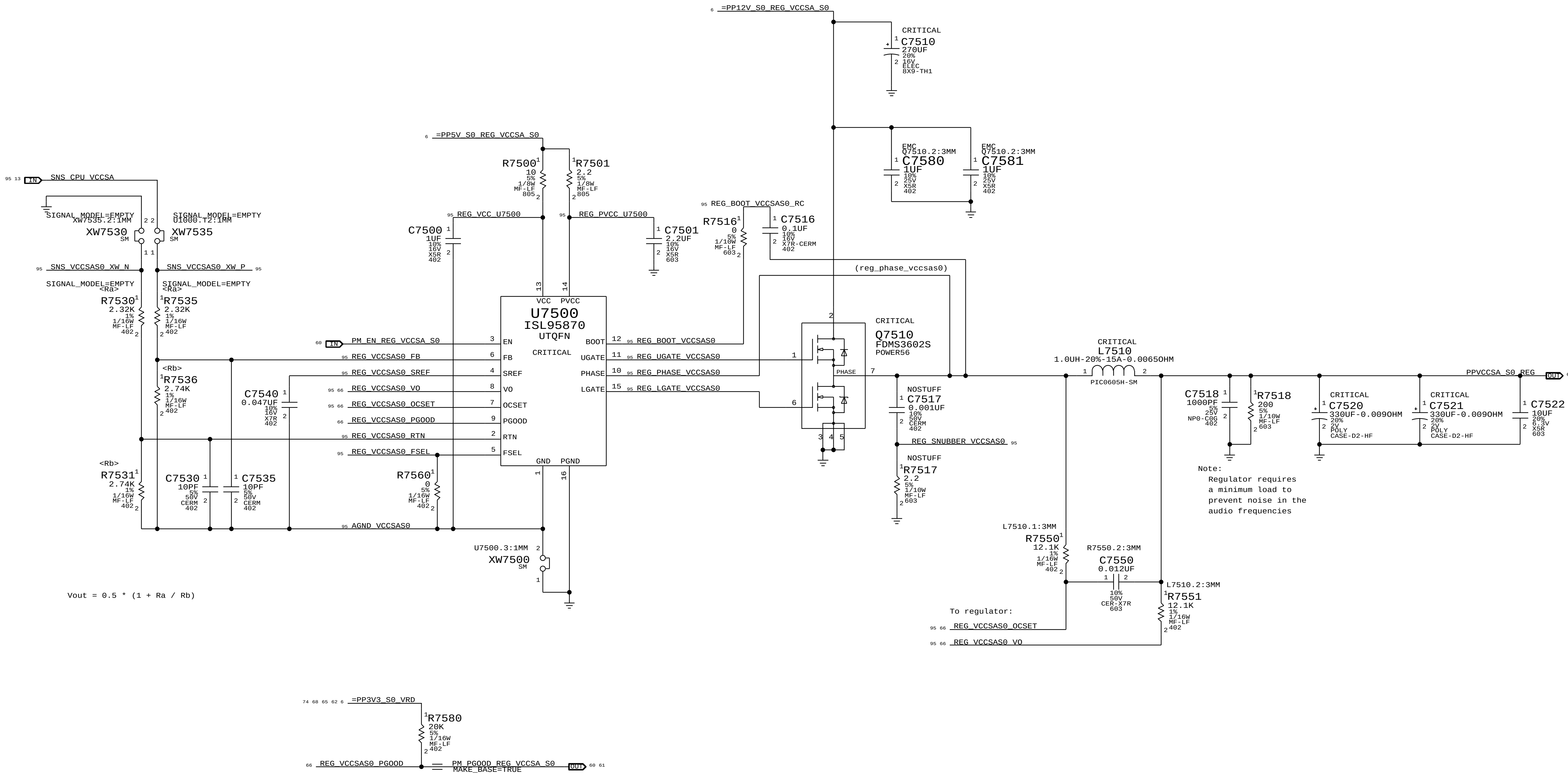
CPU VccSA (0.925V) S0 Regulator

Max avg current: ? A (design)/ 1.51 A (budget)
Max peak current: ? A (design)/ 8.76 A (budget)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE

VReg CPU VccSA S0

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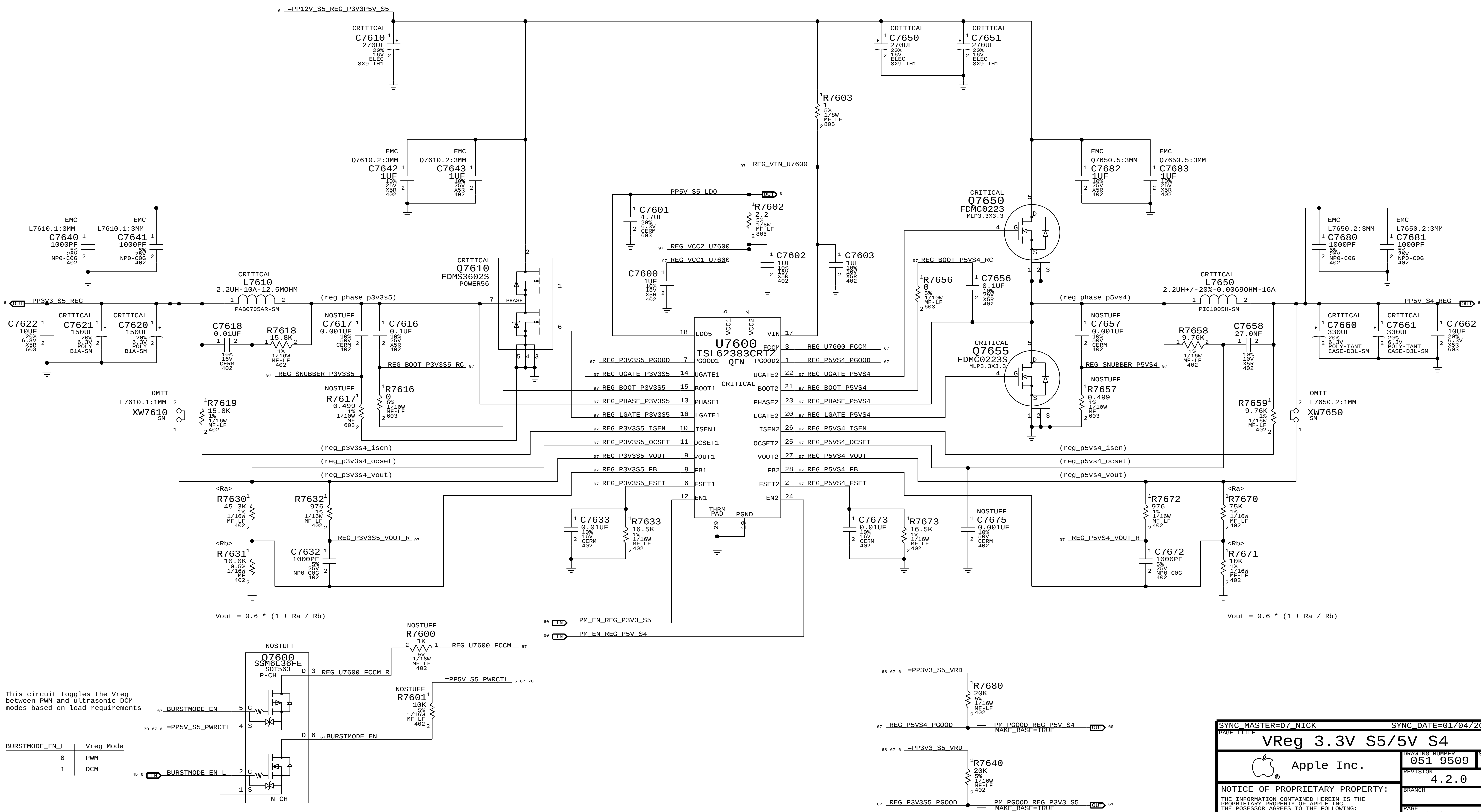
3.3V S5 Regulator

Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



This circuit toggles the Vreg between PWM and ultrasonic DCM modes based on load requirements

BURSTMODE_EN_L	Vreg Mode
0	PWM
1	DCM

SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE: VReg 3.3V S5/5V S4

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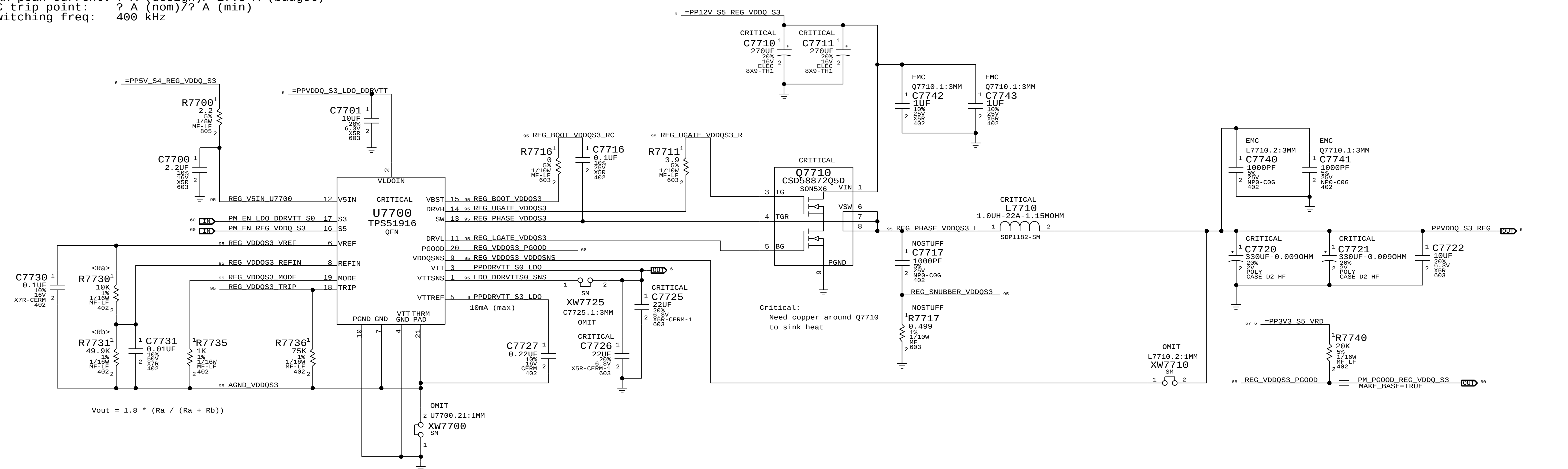
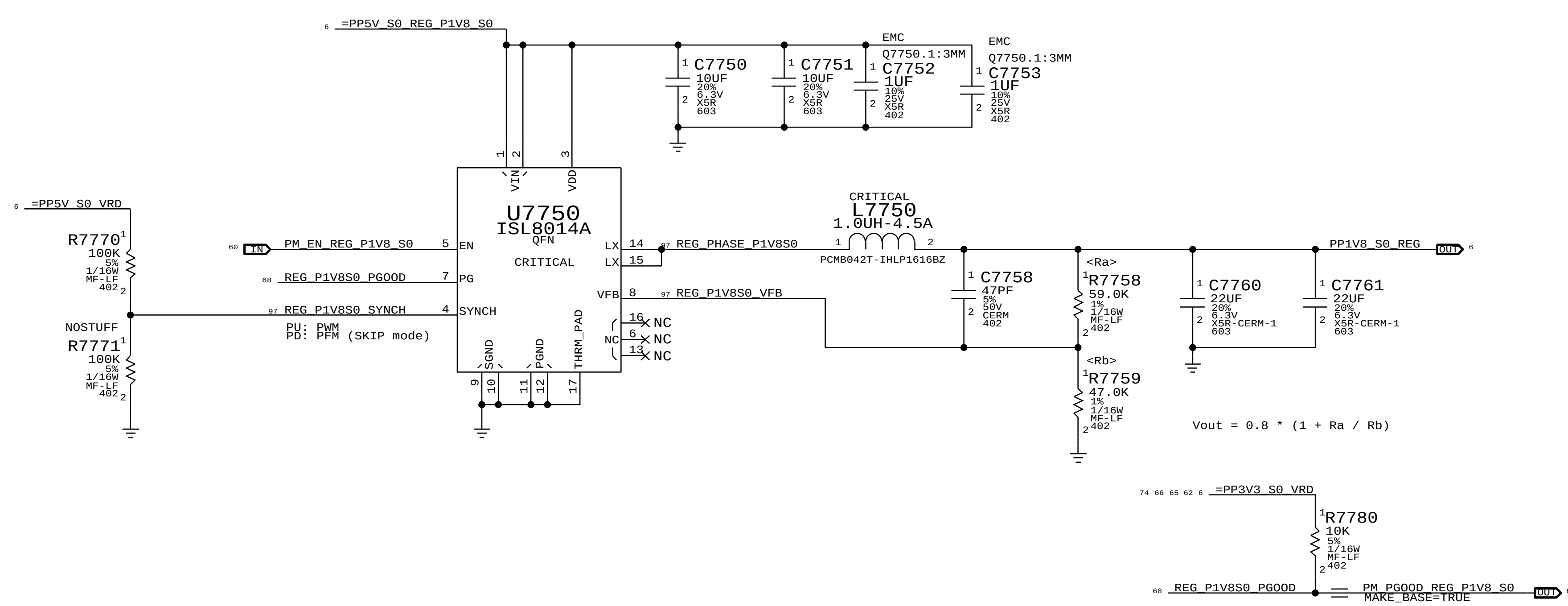
VDDQ (1.5V) S3 Regulator

Max avg current: ? A (design)/ 8 A (budget)
Max peak current: ? A (design)/ 17.8 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 400 kHz



1.8V S0 Regulator

Max avg current: 3 A (design)/ 0.61 A (budget)
Max peak current: 3 A (design)/ 1.83 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



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PAGE TITLE
VReg VDDQ and 1.8V S0

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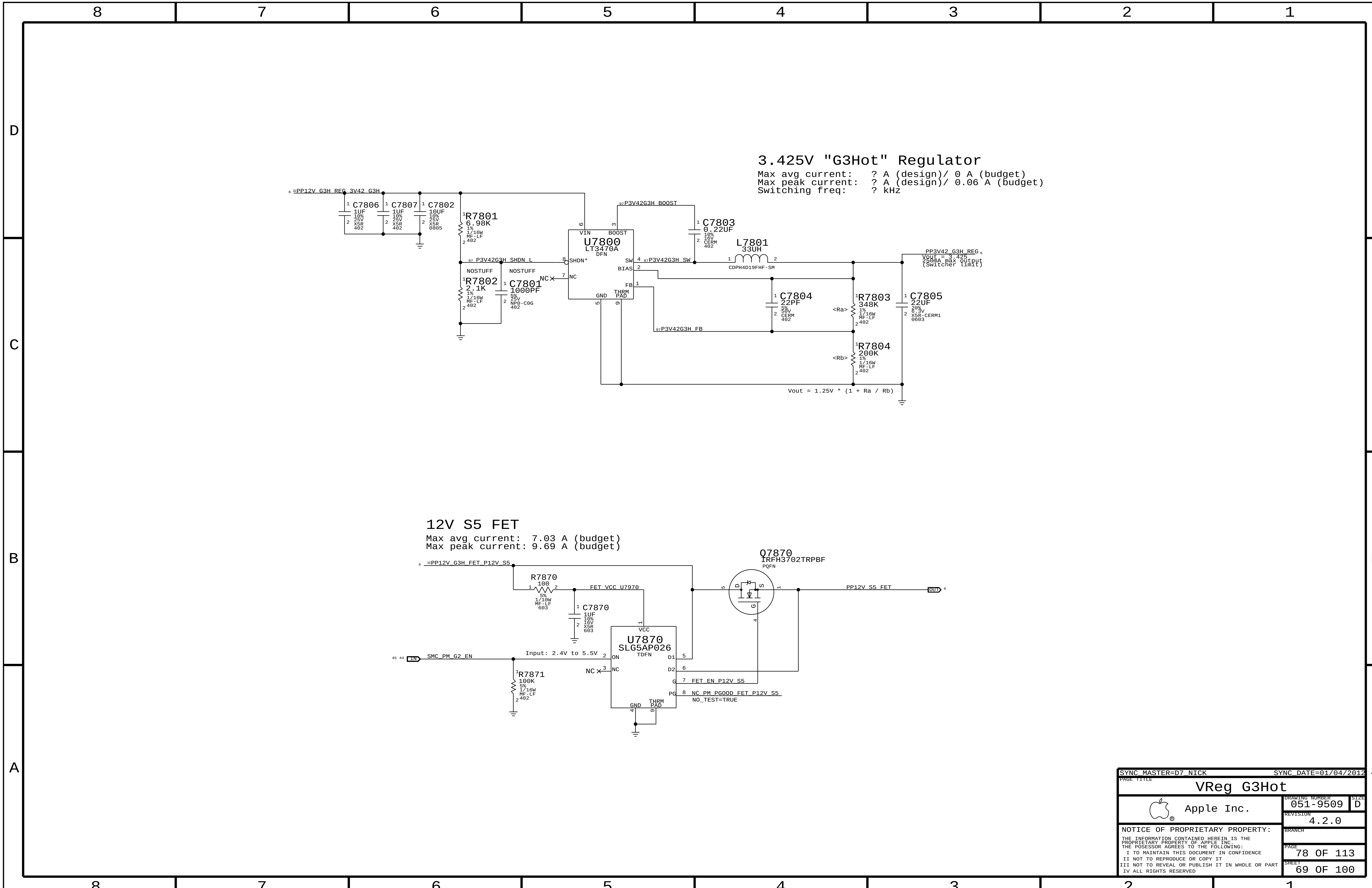
3.425V "G3Hot" Regulator
Max avg current: ? A (design)/ 0 A (budget)
Max peak current: ? A (design)/ 0.06 A (budget)
Switching freq: ? kHz
=PP12V_G3H_REG_3V42_G3H
C7806 1UF 10% 25V X5R 402
C7807 1UF 10% 25V X5R 402
C7802 10UF 10% 25V X5R 0805
R7801 6.98K 1% 1/16W MF-LF 402
P3V42G3H_BOOST
C7803 0.22UF 10% 16V CERM 402
U7800 LT3470A DFN
VIN BOOST SHDN* SW BIAS NC FB GND THRM PAD
P3V42G3H_SHDN_L
P3V42G3H_SW
L7801 33UH CDPH4D19FHF-SM
PP3V42_G3H_REG
Vout = 3.425
250mA max output
(Switcher limit)
NOSTUFF
R7802 2.1K 1% 1/16W MF-LF 402
NOSTUFF
C7801 1000PF 5% 25V NP0-C0G 402
C7804 22PF 5% 50V CERM 402
R7803 348K 1% 1/16W MF-LF 402
<Ra>
C7805 22UF 20% 6.3V X5R-CERM1 0603
P3V42G3H_FB
R7804 200K 1% 1/16W MF-LF 402
<Rb>
Vout = 1.25V * (1 + Ra / Rb)
12V S5 FET
Max avg current: 7.03 A (budget)
Max peak current: 9.69 A (budget)
=PP12V_G3H_FET_P12V_S5
R7870 100 5% 1/10W MF-LF 603
FET_VCC_U7970
C7870 1UF 10% 16V X5R 603
Q7870 IRFH3702TRPBF PQFN
PP12V_S5_FET
OUT
U7870 SLG5AP026 TDFN
VCC ON NC D1 D2 G PG GND THRM PAD
SMC_PM_G2_EN
IN
Input: 2.4V to 5.5V
R7871 100K 5% 1/16W MF-LF 402
FET_EN_P12V_S5
NC_PM_PGOOD_FET_P12V_S5
NO_TEST=TRUE
SYNC_MASTER=D7_NICK
SYNC_DATE=01/04/2012
PAGE TITLE
VReg G3Hot
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SIZE D
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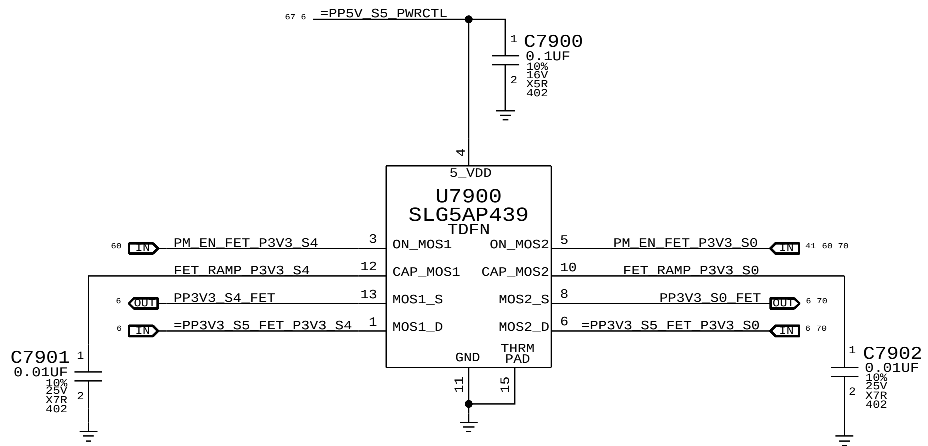
3.3V S4 FET

Max avg current: 0.85 A (budget)
Max peak current: 1.48 A (budget)

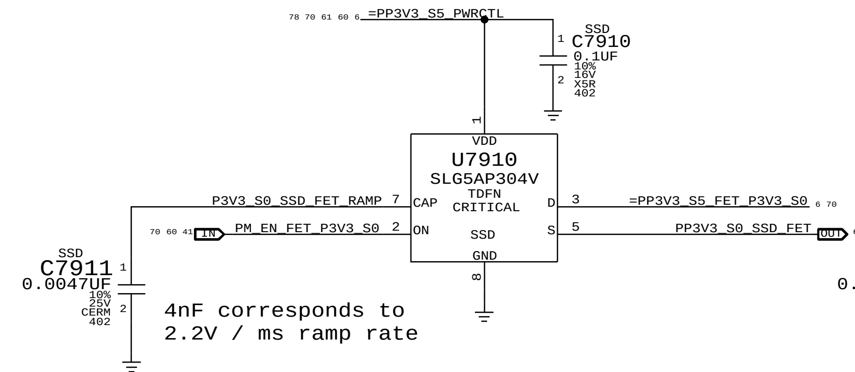
3.3V S0 FET

Max avg current: 1.7 A (budget)
Max peak current: 1.82 A (budget)



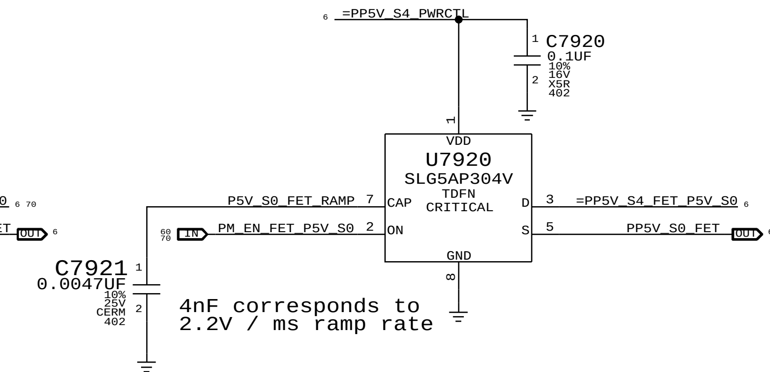
3V3 S0 SSD

Max avg current: 2.12 A (budget)
Max peak current: 3.03 A (budget)

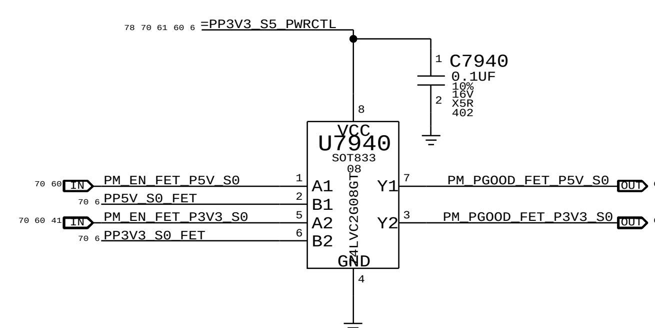


5V S0 FET

Max avg current: 1.26 A (budget)
Max peak current: 2.08 A (budget)

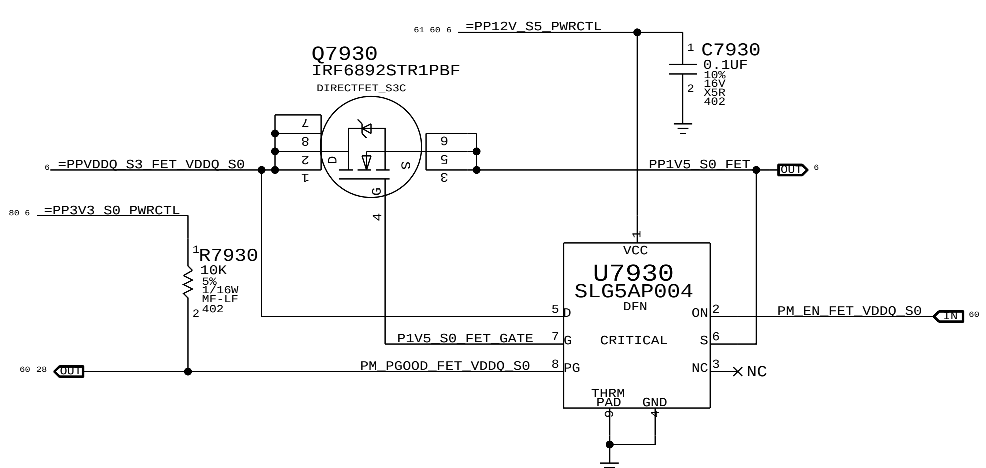


5V / 3V3 S0 PGOODs



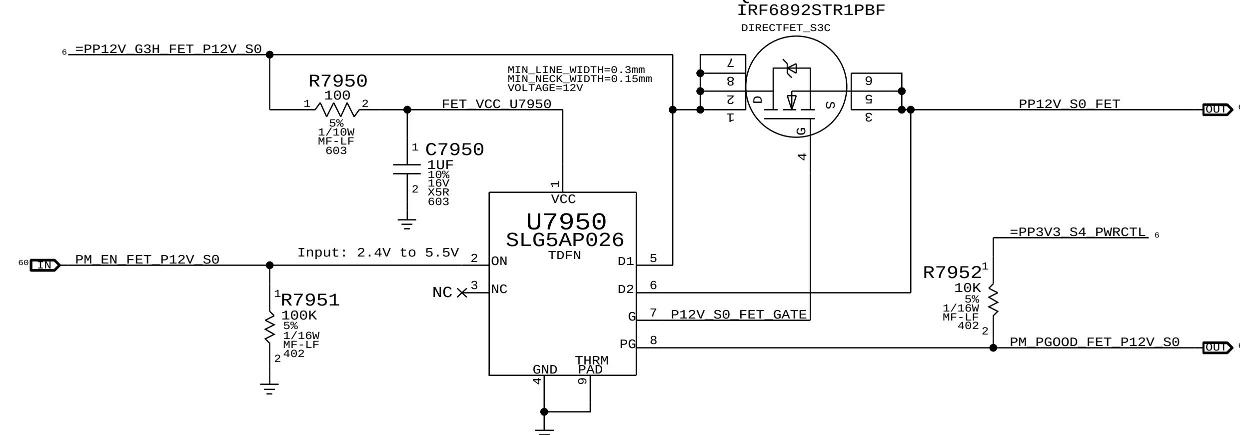
1.5V S0 FET

Max avg current: 1.27 A (budget)
Max peak current: 4.8 A (budget)



12V S0 FET

Max avg current: 14.3 A (budget)
Max peak current: 23 A (budget)



SYNC_MASTER=D7_NICK SYNC_DATE=01/04/2012

PAGE TITLE
FET-Controlled S0 and S4

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Page Notes

Power aliases required by this page:

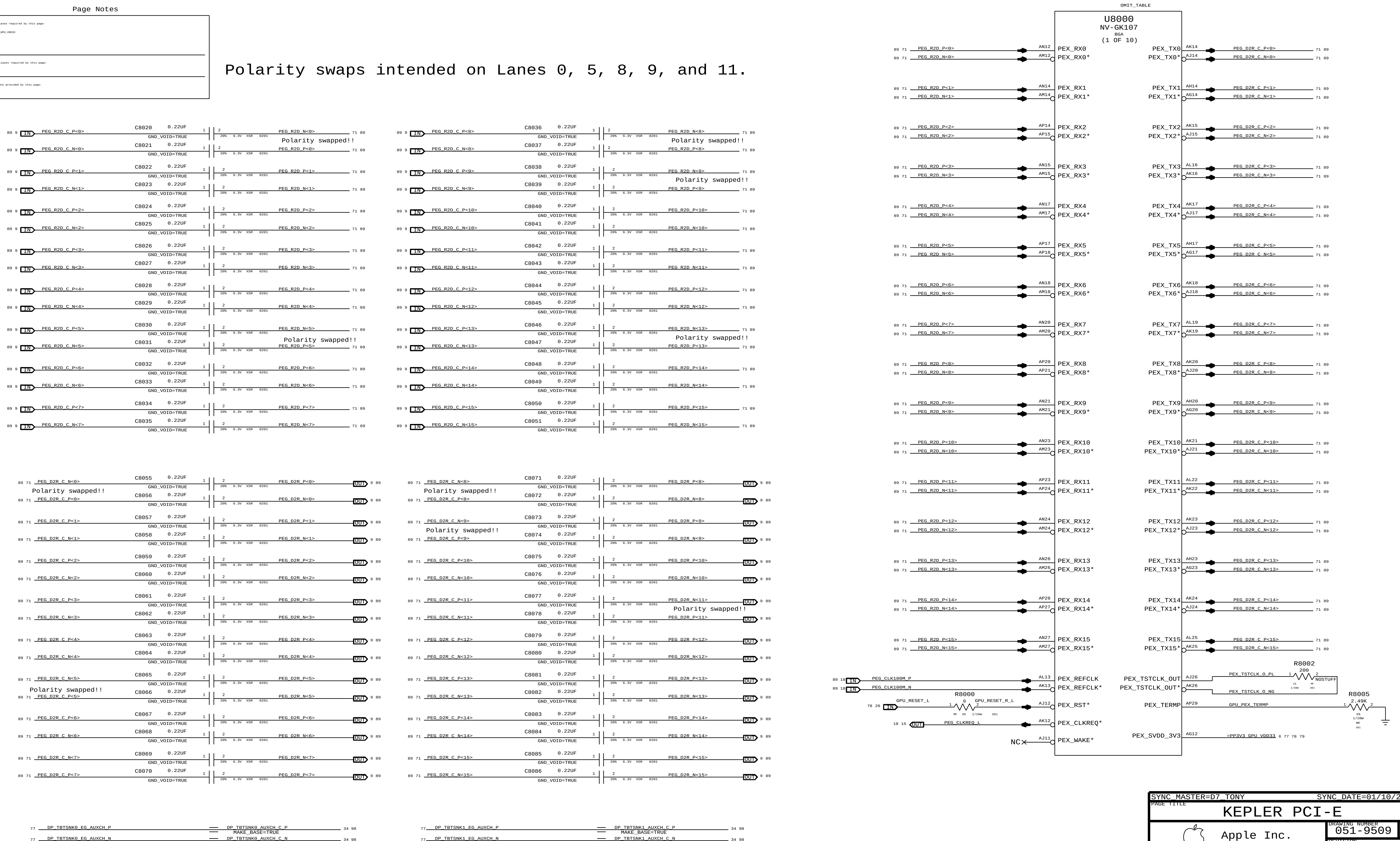
Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

Polarity swaps intended on Lanes 0, 5, 8, 9, and 11.



SYNC_MASTER=D7_TONY SYNC_DATE=01/10/2012

PAGE TITLE

KEPLER PCI-E

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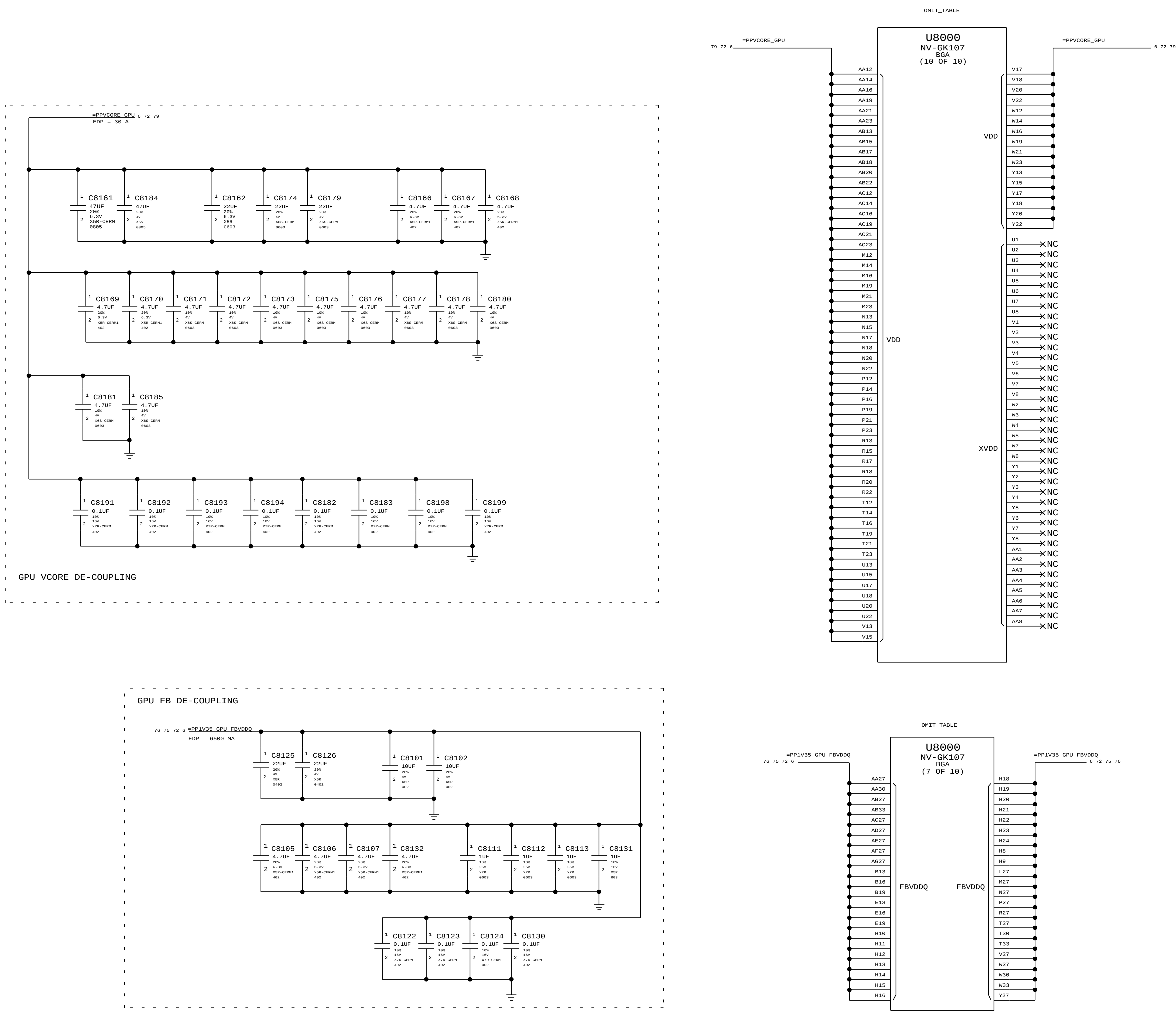

Page Notes

SYNC_MASTER=D7_TONY SYNC_DATE=01/10/2012

PAGE TITLE

KEPLER CORE/FB POWER

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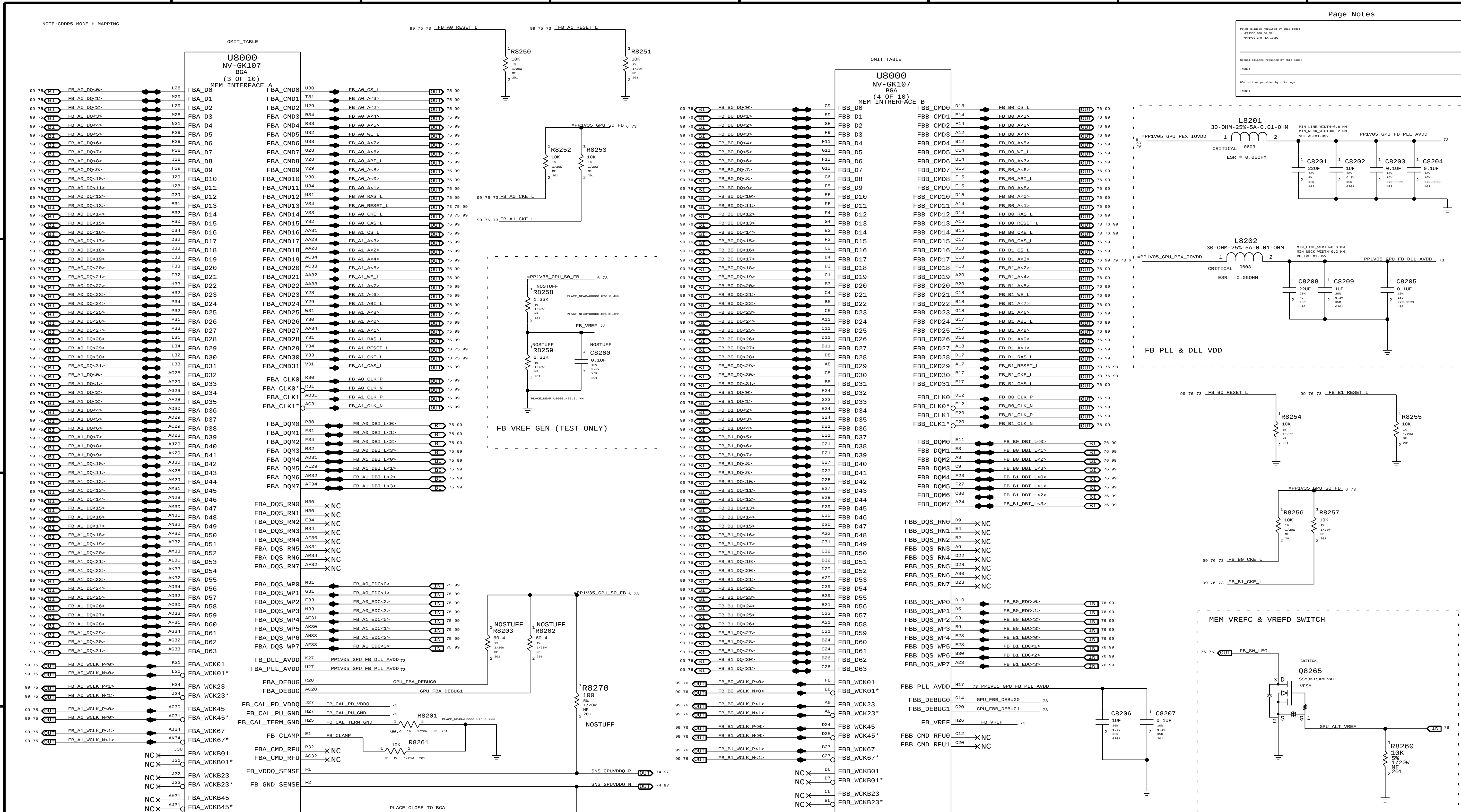

SYNC_MASTER=D7_TONY SYNC_DATE=01/10/2012

PAGE TITLE

KEPLER FRAME BUFFER I/F

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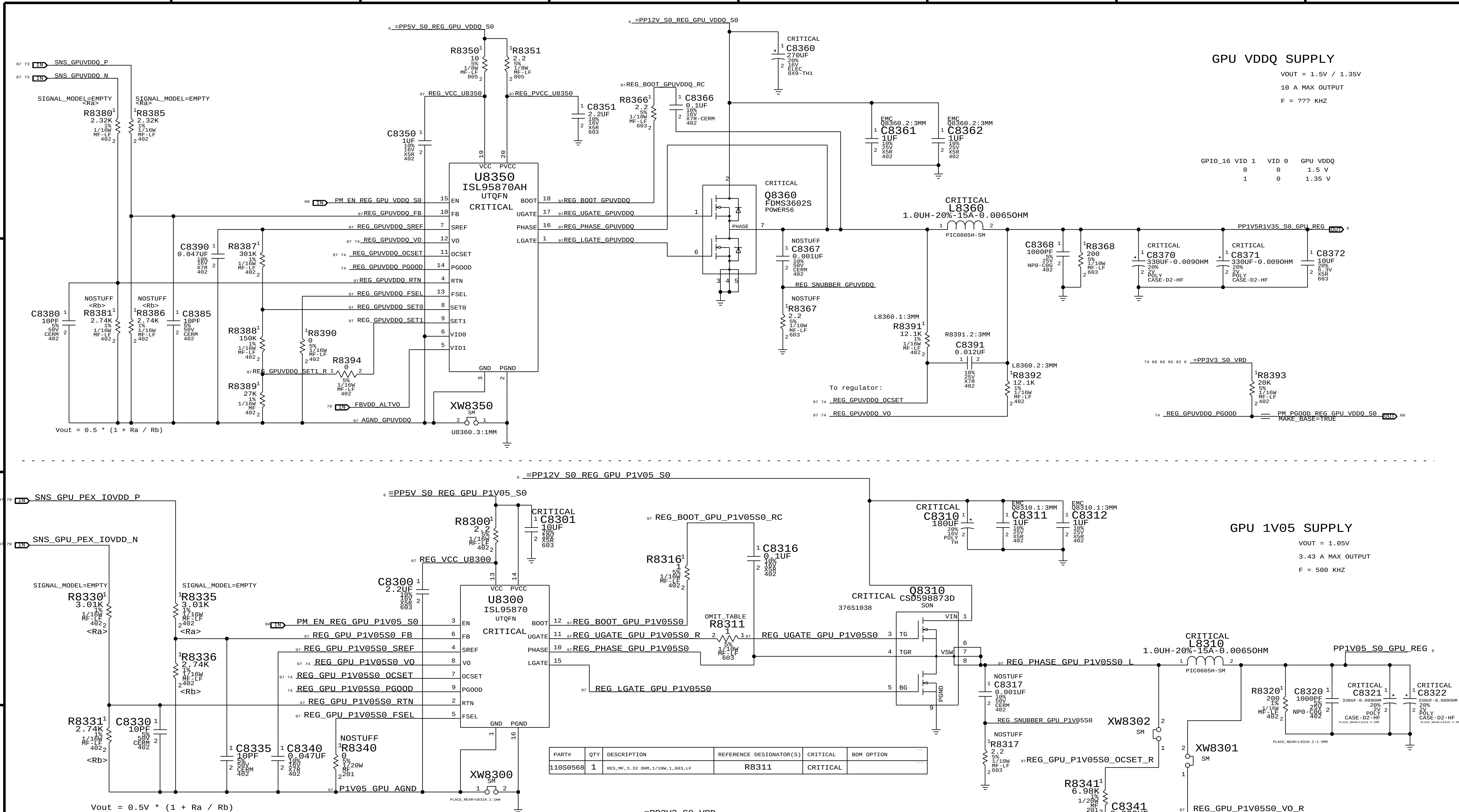

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
110S0568	1	RES,MF,3.32 OHM,1/10W,1,603,LF	R8311	CRITICAL	

SYNC_MASTER=D7_NICK SYNC_DATE=01/03/2012

PAGE TITLE

1V05 GPU POWER SUPPLY

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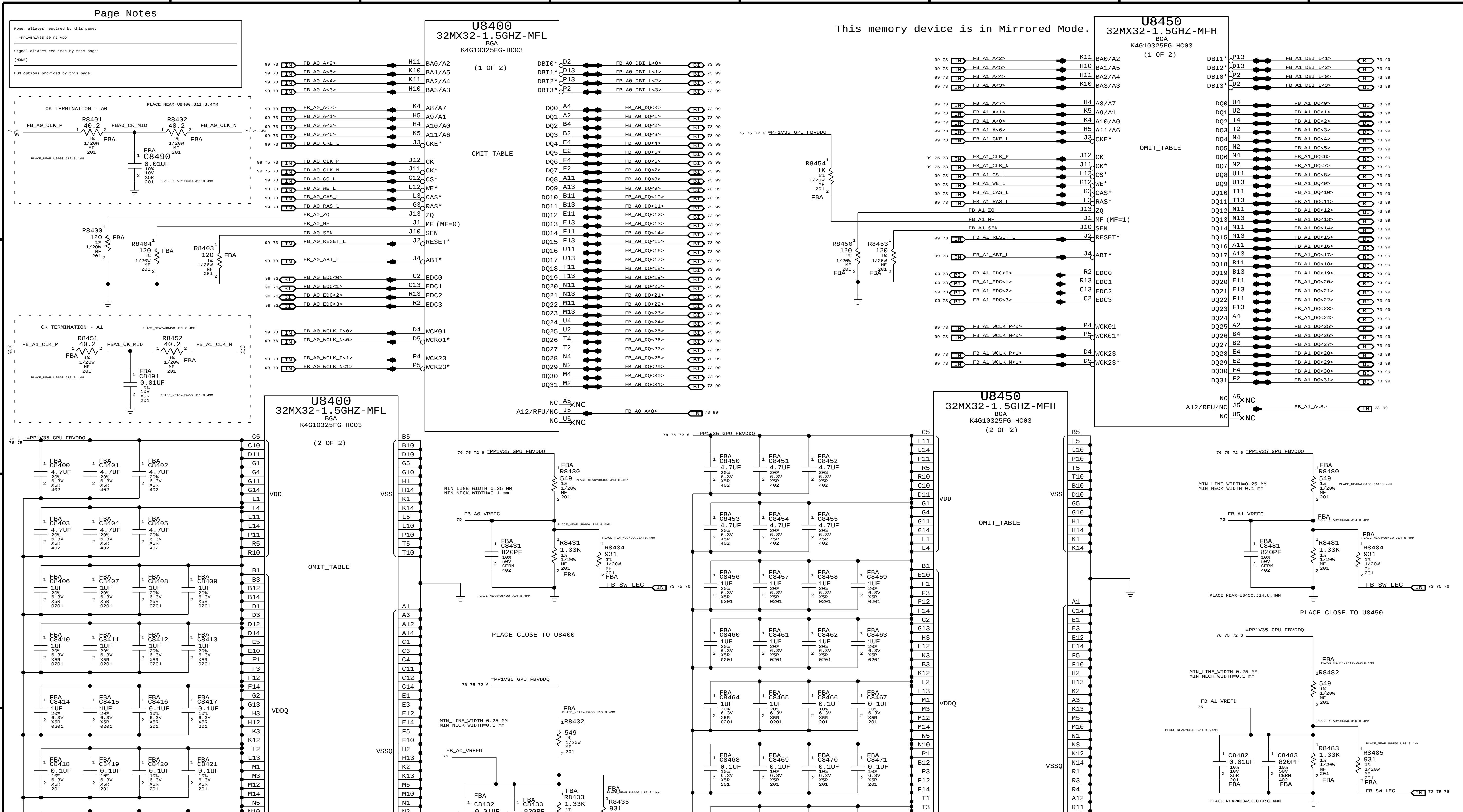

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PAGE TITLE

GDDR5 Frame Buffer A

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051-9509	D

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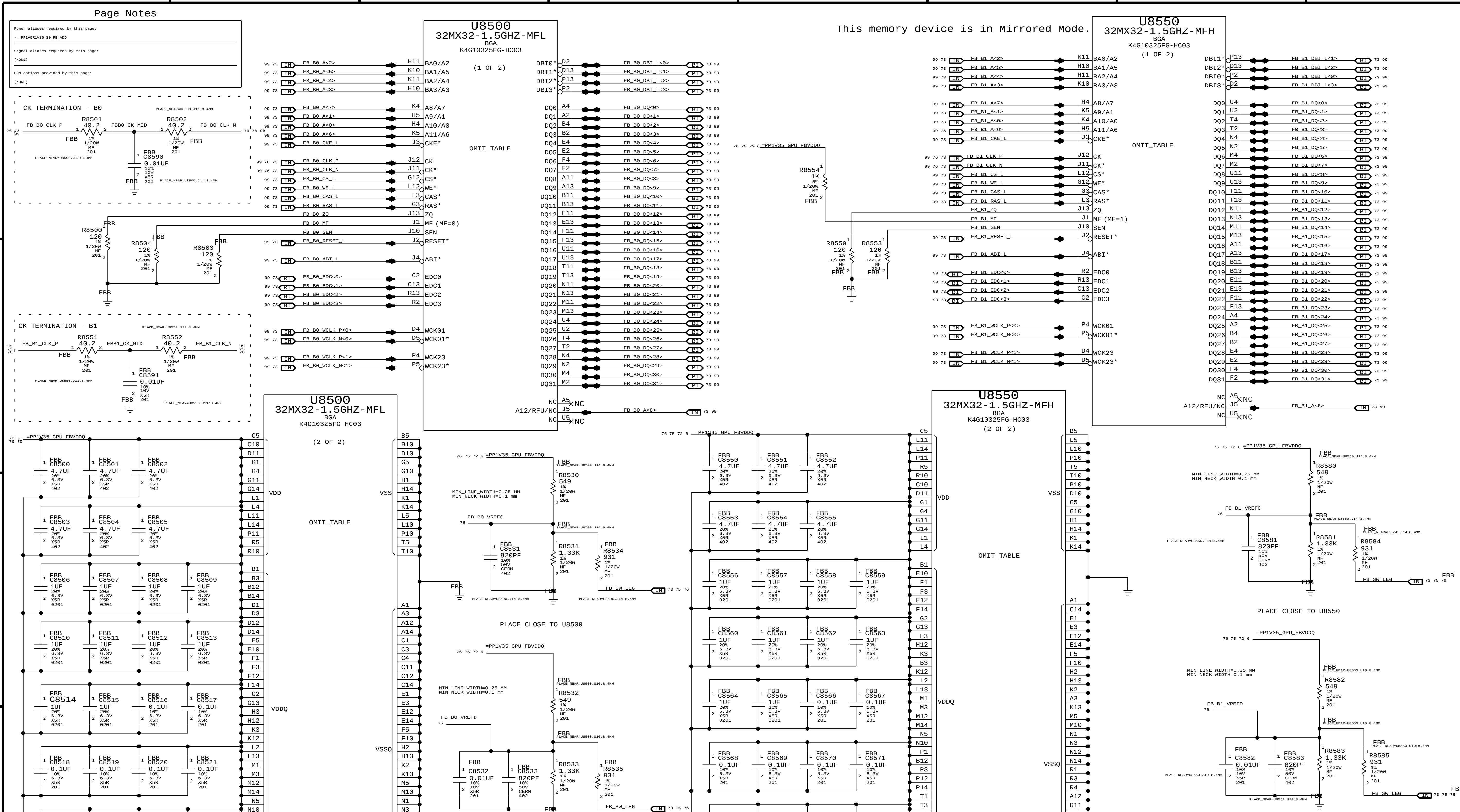

SYNC_MASTER=D7_TONY SYNC_DATE=12/13/2011

PAGE TITLE

GDDR5 Frame Buffer B

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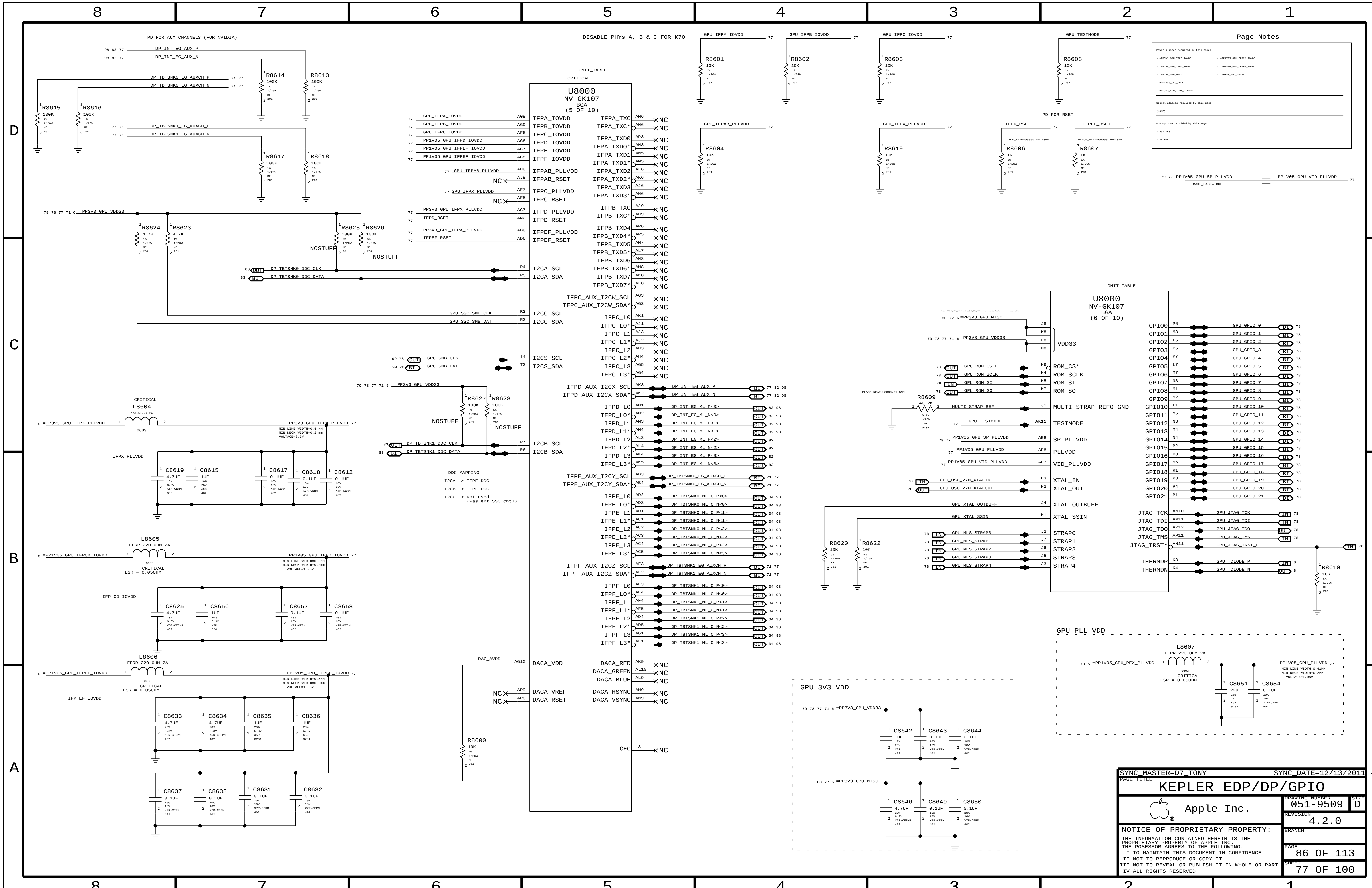

SYNC_MASTER=D7_TONY SYNC_DATE=12/13/2011

PAGE TITLE

KEPLER EDP/DP/GPIO

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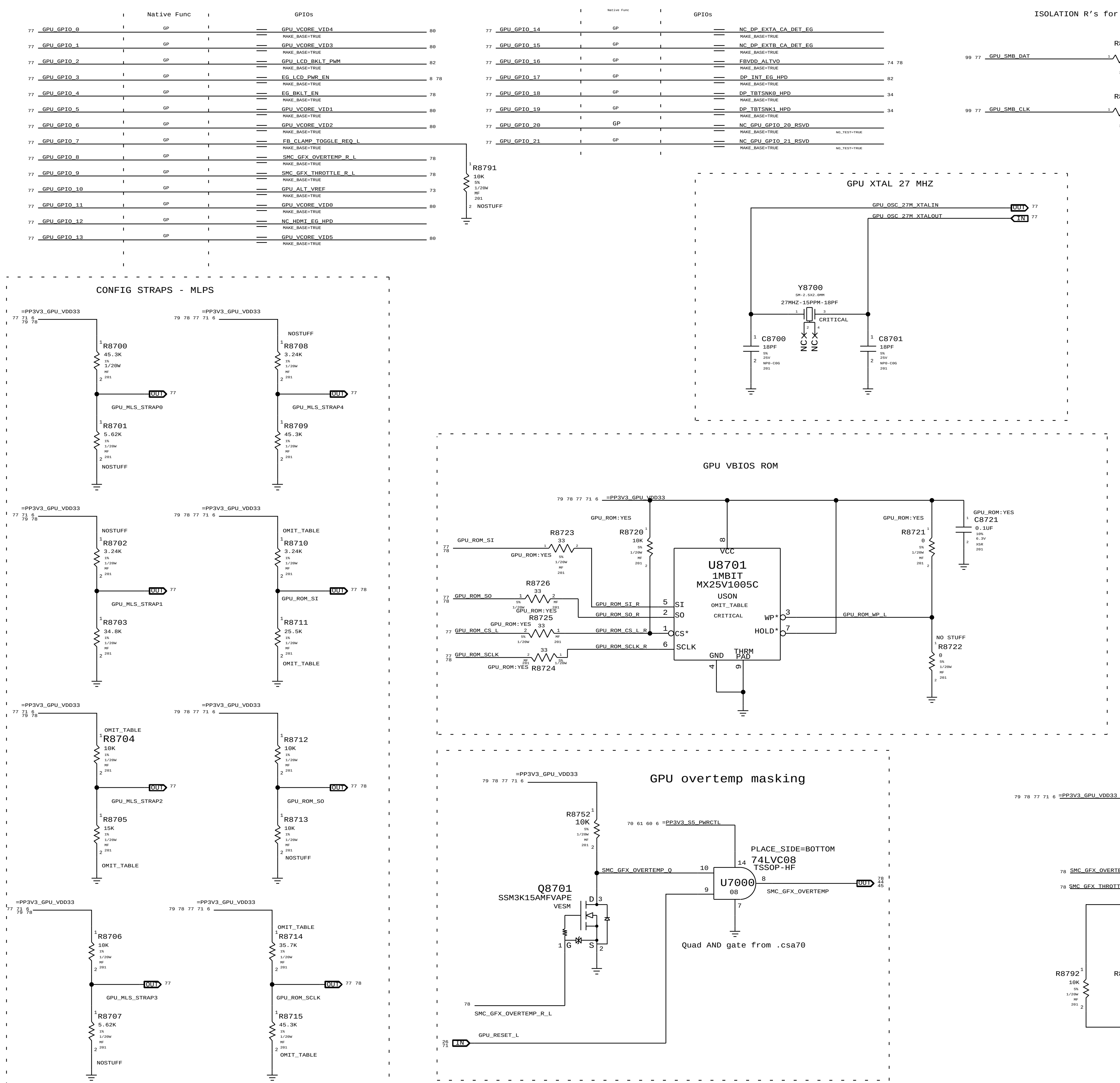
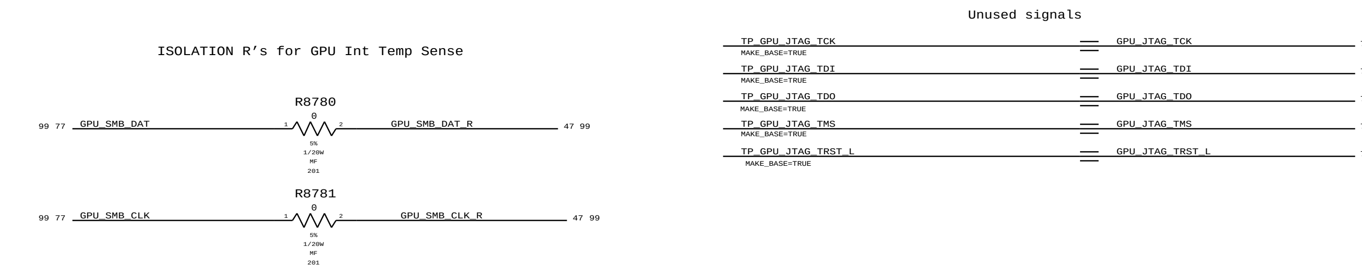
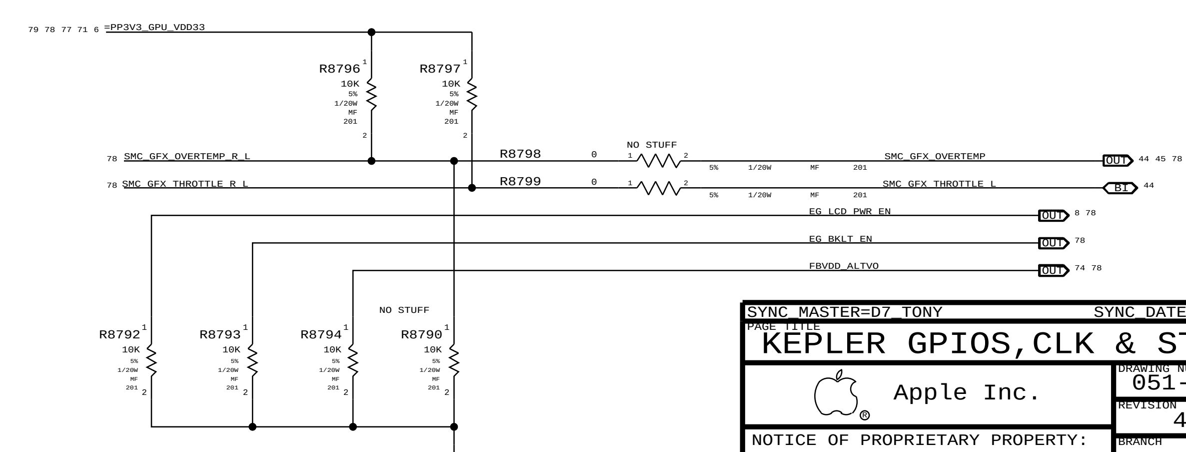




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8704	CRITICAL	GPU:107GTX
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R8704	CRITICAL	GS
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8704	CRITICAL	GPU:107EGE
118S0280	1	RES,MF,30.1k ,1,1/20W,0201	R8705	CRITICAL	GPU:107GTX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8705	CRITICAL	GS
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R8705	CRITICAL	GPU:107EGE
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:BOTH_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:BOTH_HYNIX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH1_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH2_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH1_HYNIX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH2_HYNIX
118S0175	1	RES,MF,20.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_HYNIX
118S0013	1	RES,MF,10.0k ,1,1/20W,0201	R8711	CRITICAL	FB:CH1_SAMSUNG
118S0013	1	RES,MF,10.0k ,1,1/20W,0201	R8711	CRITICAL	FB:CH2_SAMSUNG
118S0230	1	RES,MF,24.9k ,1,1/20W,0201	R8711	CRITICAL	FB:CH1_HYNIX
118S0230	1	RES,MF,24.9k ,1,1/20W,0201	R8711	CRITICAL	FB:CH2_HYNIX
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R8714	CRITICAL	GPU:107GTX
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R8714	CRITICAL	GS
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8714	CRITICAL	GPU:107EGE
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8715	CRITICAL	GPU:107GTX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8715	CRITICAL	GS
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8715	CRITICAL	GPU:107EGE



SYNC_MASTER=D7_TONY SYNC_DATE=01/10/2012

PAGE TITLE

KEPLER GPIOS,CLK & STRAPS

DRAWING NUMBER 051-9509 SIZE D

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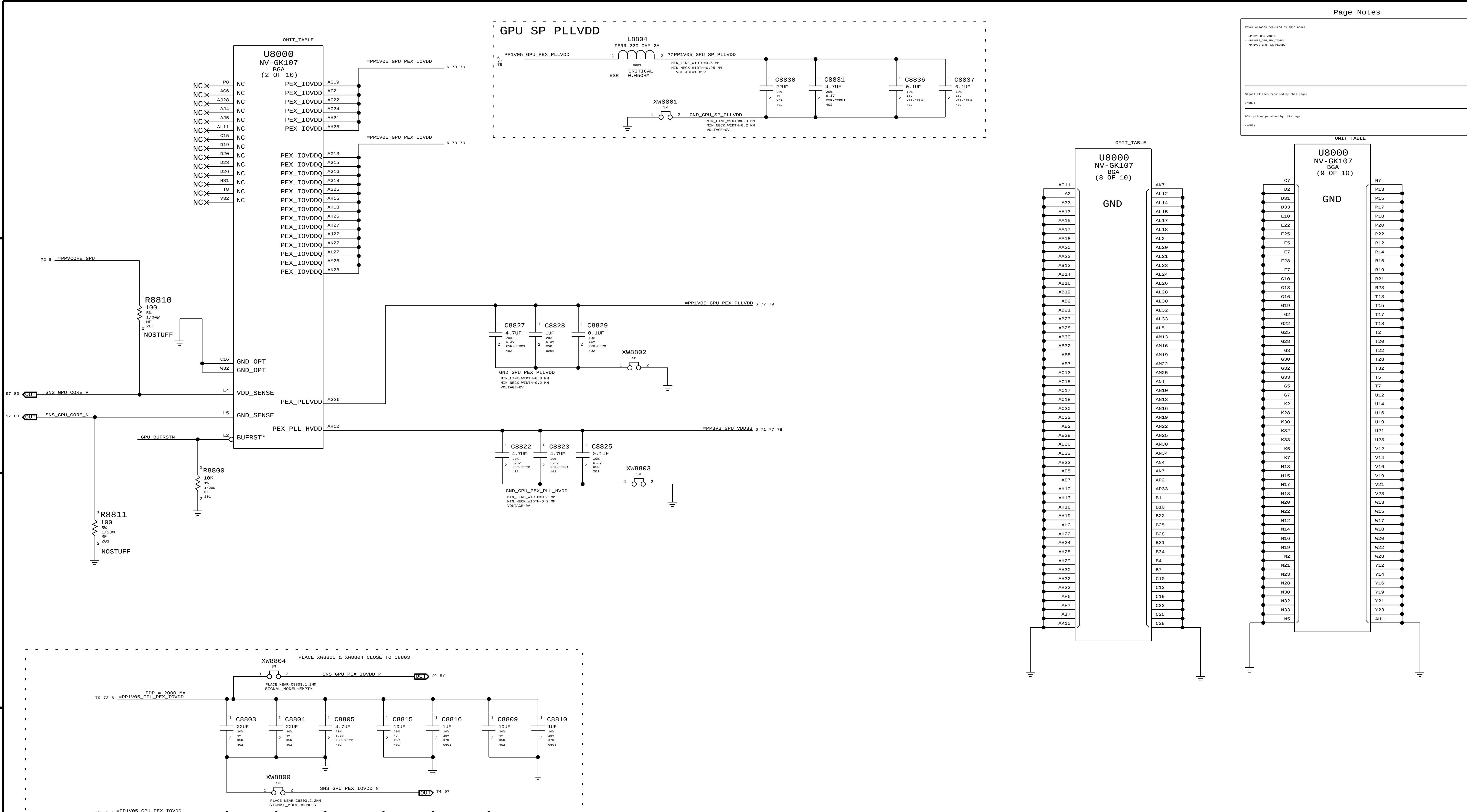

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PAGE TITLE

KEPLER PEX PWR/GNDS

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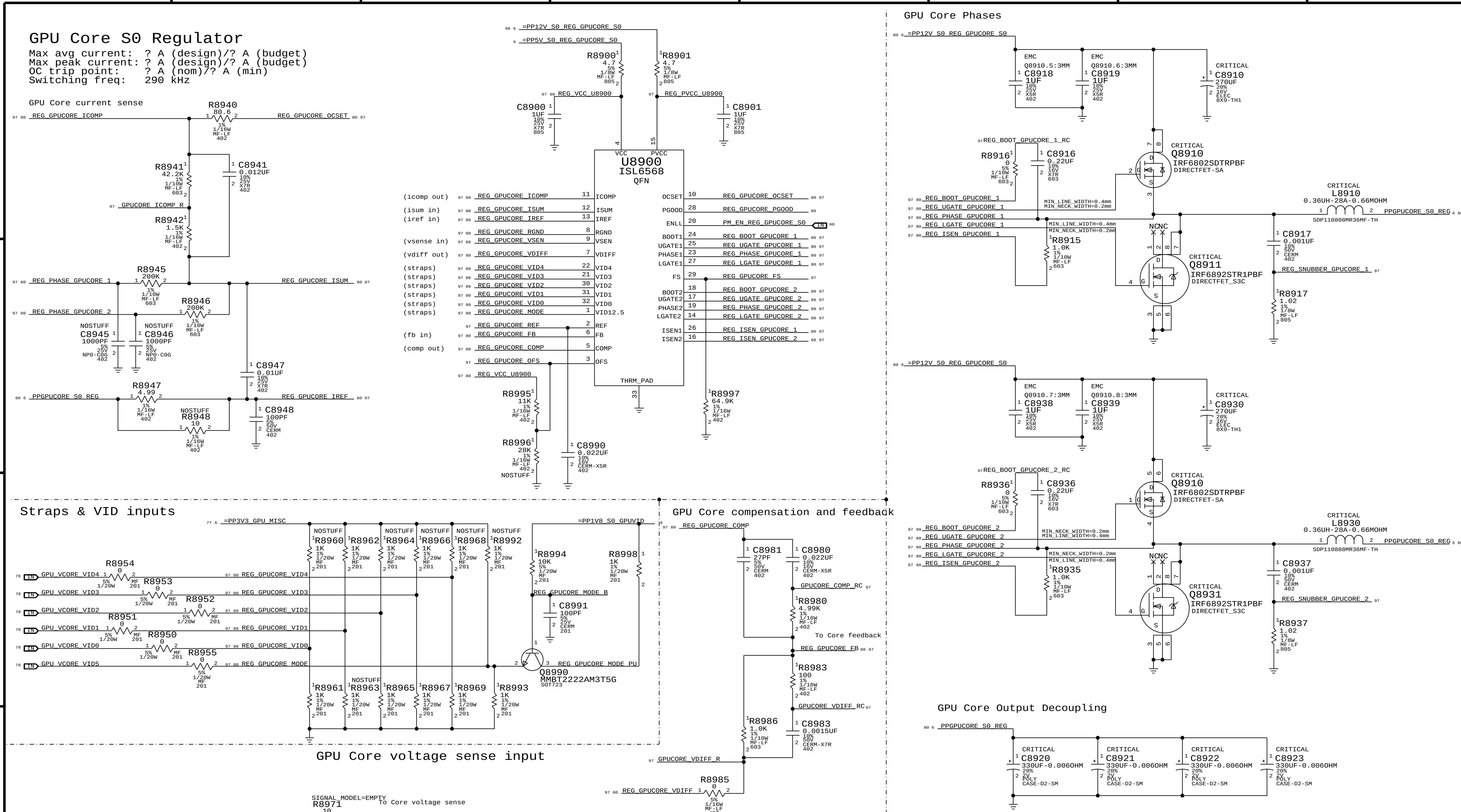

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VReg GPU Core

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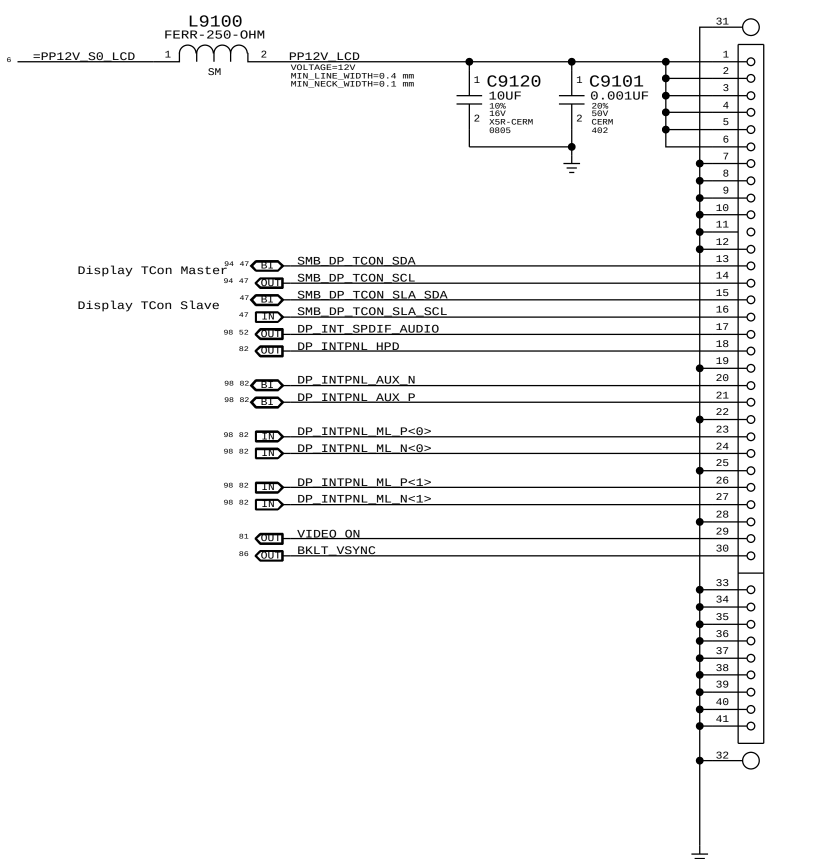
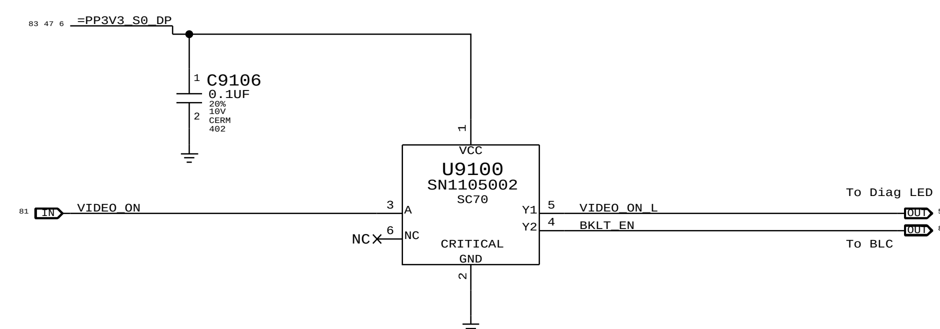
Internal DP Connector



Backlight Control

U9000 output Y2 is a non-inverted, delayed version of input A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay



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PAGE TITLE
Internal DP Support

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PAGE 91 OF 113
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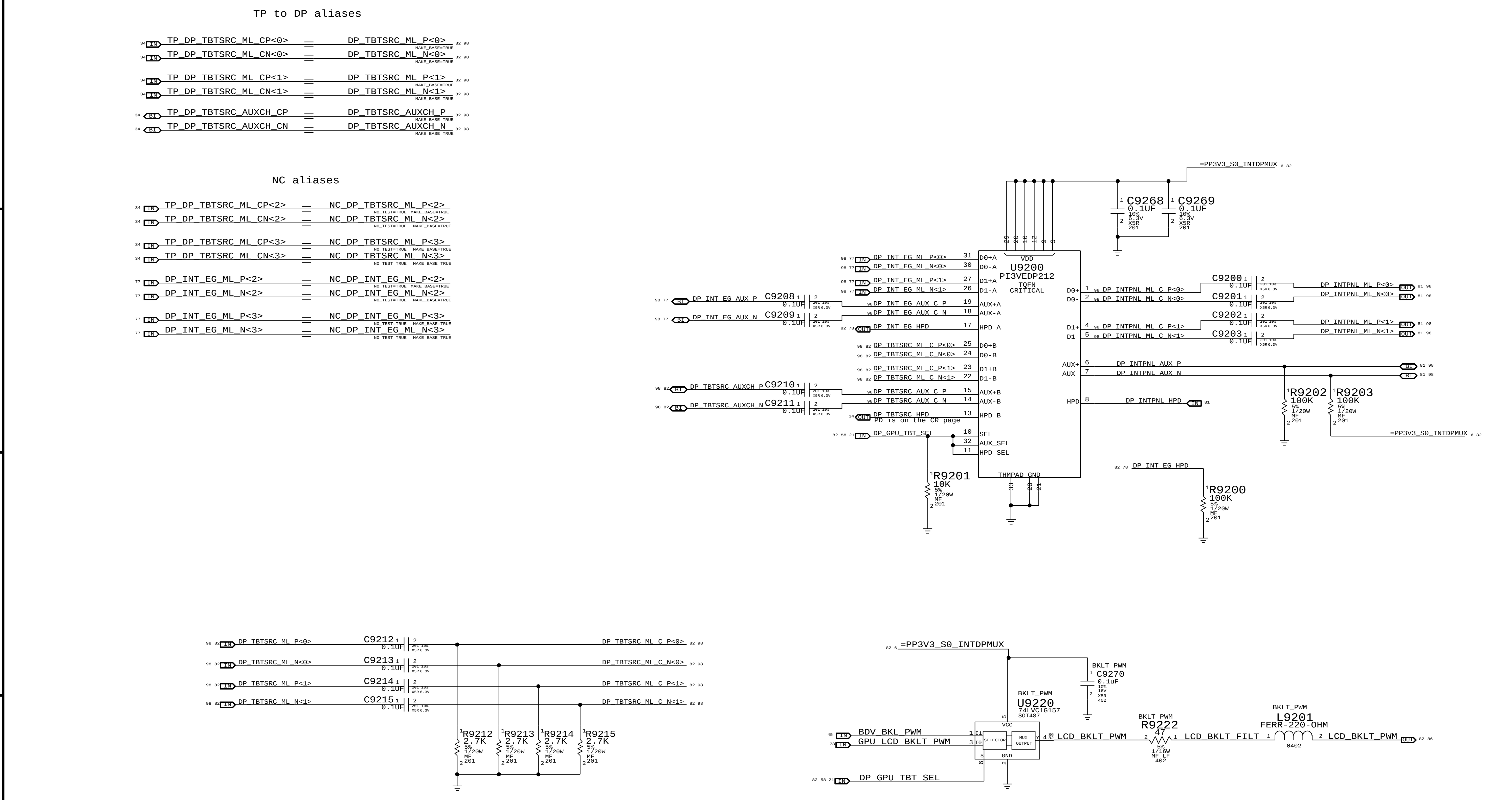

SYNC_MASTER=D7_NICK	SYNC_DATE=12/14/2011
PAGE TITLE: Internal DP MUXing	
Apple Inc.	DRAWING NUMBER: 051-9509 SIZE: D
	REVISION: 4.2.0
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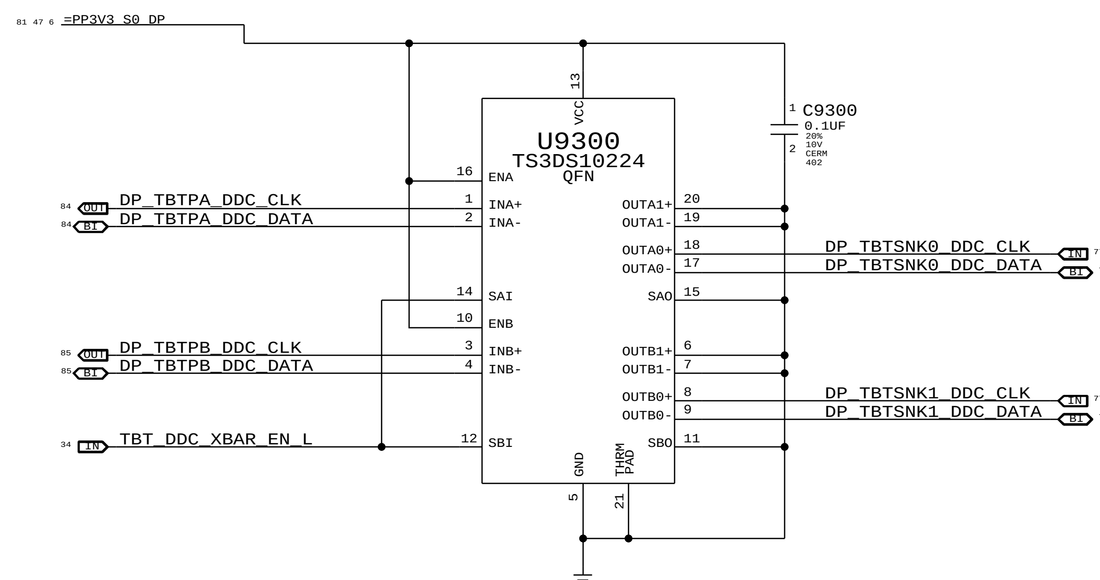
	SHEET: 82 OF 100

Dual-Port Host DDC Crossbar



SYNC_MASTER=D7_DOUG SYNC_DATE=12/15/2011

PAGE TITLE

TBT DDC Crossbar

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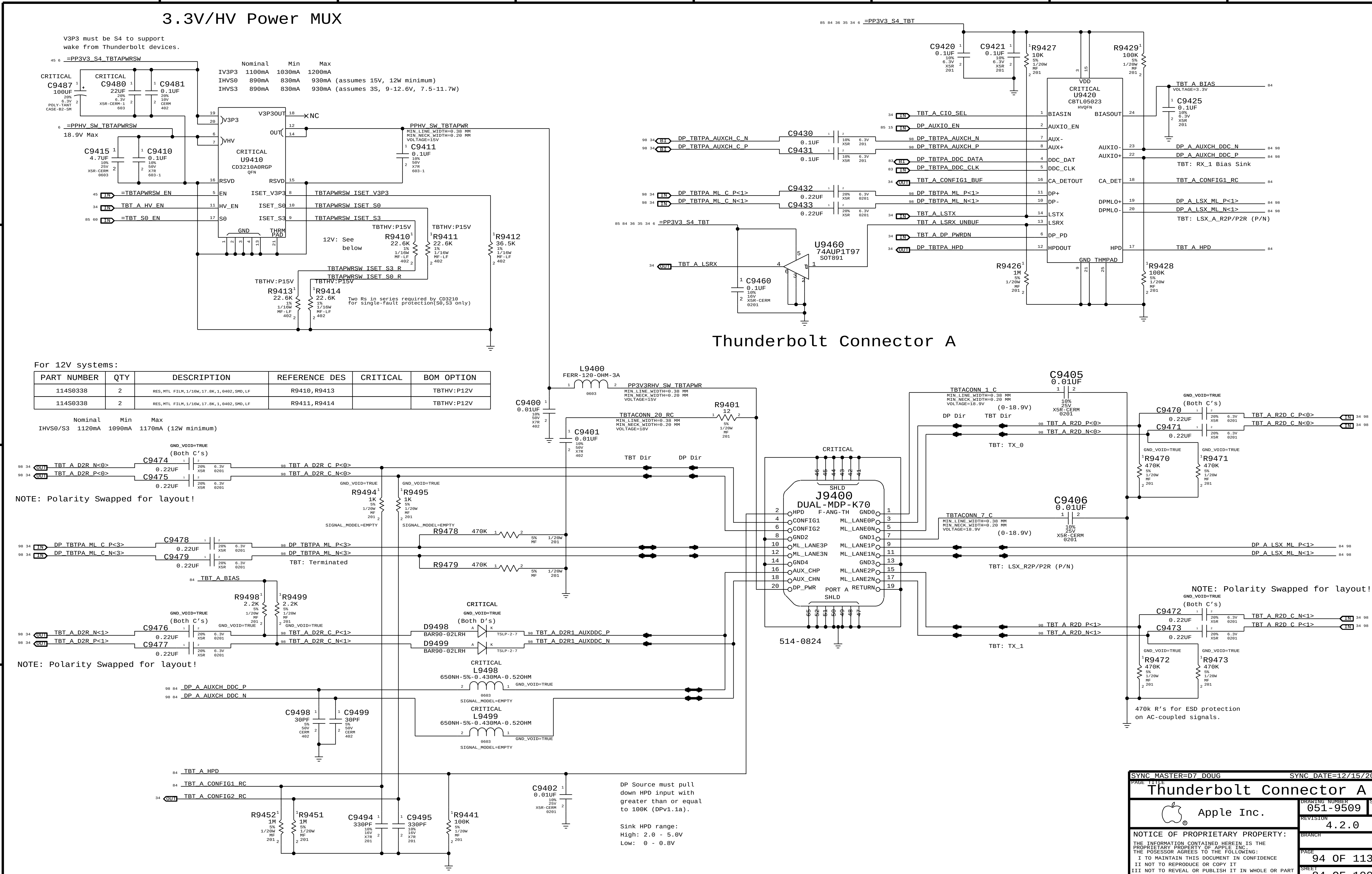
3.3V/HV Power MUX

Thunderbolt Connector A



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9410,R9413		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9411,R9414		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)

SYNC_MASTER=D7_DOUG SYNC_DATE=12/15/2011

PAGE TITLE: Thunderbolt Connector A

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3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)

12V: See below

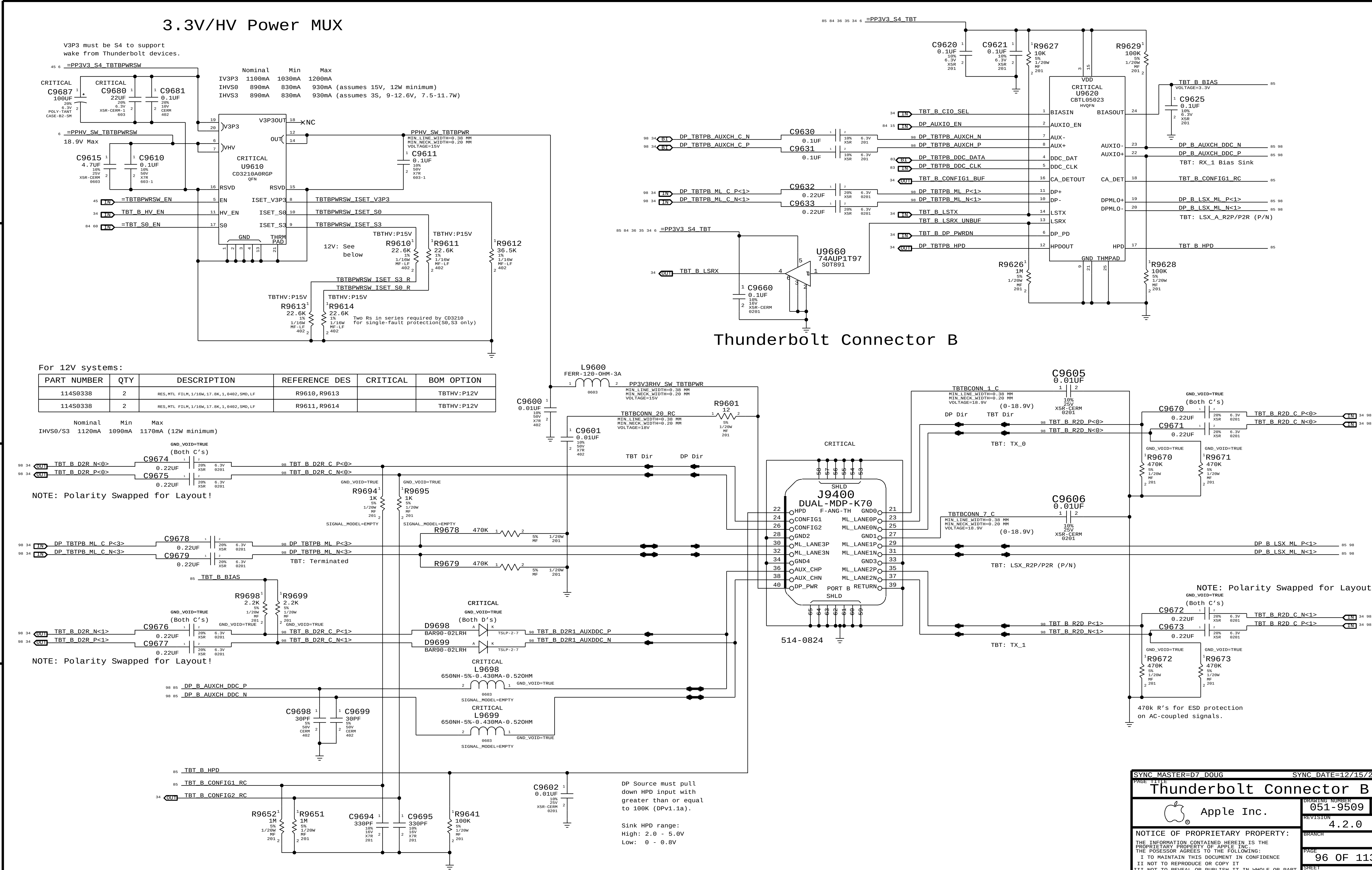
Two Rs in series required by CD3210 for single-fault protection(S0,S3 only)

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

Thunderbolt Connector B



SYNC_MASTER=D7_DOUG SYNC_DATE=12/15/2011

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

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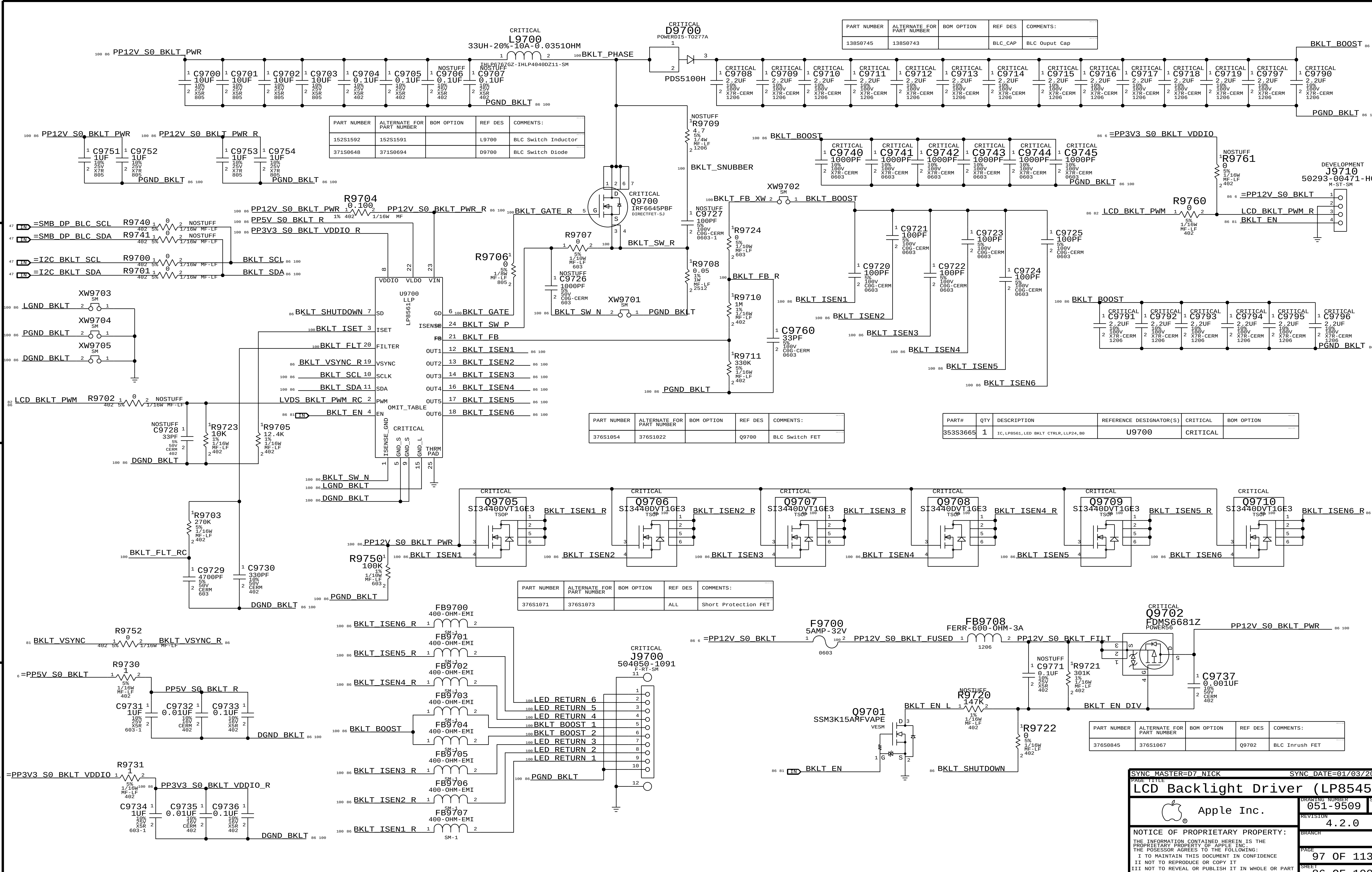

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0745	138S0743		BLC_CAP	BLC Ouput Cap

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S1592	152S1591		L9700	BLC Switch Inductor
371S0648	371S0694		D9700	BLC Switch Diode

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1054	376S1022		Q9700	BLC Switch FET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S3665	1	IC,LP8561,LED BKLT CTRLR,LLP24,B0	U9700	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1071	376S1073		ALL	Short Protection FET

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S0845	376S1067		Q9702	BLC Inrush FET

SYNC_MASTER=D7_NICK SYNC_DATE=01/03/2012

PAGE TITLE

LCD Backlight Driver (LP8545)

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K70 Board Specific Physical and Spacing Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	*	0.076 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

Board Stack-up

Finished board thickness: 1.58 mm

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 mm
2	Plane	1 oz
	Prepreg	0.076 mm
3	Signal	0.5 oz
	Prepreg	0.435 mm
4	Plane	1 oz
	Core	0.127 mm
5	Plane	1 oz
	Prepreg	0.435 mm
6	Signal	0.5 oz
	Prepreg	0.076 mm
2	Plane	1 oz
	Prepreg	0.071 mm
Btm	Signal	0.5 oz (Cu plated)

SYNC_MASTER=D7_DAVE SYNC_DATE=12/12/2011

PAGE TITLE
K70 Rule Definitions

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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

Minimum diff spacing is 4 mil
Table 3-5, Intel Doc# 473718

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?

Main Segment Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
3-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
3-3	8	9.84	12	13.78	
3-4	6	7.87	12	13.78	
3-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_A_DQS*	*	*	DDR_DATA_ISO	
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_B_DQS*	*	*	DDR_DATA_ISO	
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ	See Note (3)
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL	See Note (3)
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL	
DDR_A_*	DDR_B_*	*	DDR_CH2CH	See Note (2)

Note (1):
Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to contraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):
Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):
In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

	Electrical Contraint Set	Physical	Spacing		
	Channel A				
I178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	12 29
I179	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	12 29
I255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	8 12
I256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	8 12
I180	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	12 29
I181	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	12 29
I182	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	12 29
I247	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	8 12
I248	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	8 12
I249	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	8 12
I183	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	12 29
I184	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	12 29
I185	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	12 29
I186	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	12 29
I187	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	12 29
I188	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	12 31
I189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	12 31
I191	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	12 31
I190	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	12 31
I192	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	12 31
I193	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	12 31
I194	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	12 31
I195	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	12 31
I196	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	12 31
I197	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	12 31
I198	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	12 31
I199	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	12 31
I200	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	12 31
I201	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	12 31
I203	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	12 31
I202	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	12 31
I204	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	12 31
I205	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	12 31
I206	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	12 31
I207	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	12 31
I208	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	12 31
I209	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	12 31
I210	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	12 31
I211	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	12 31
	Channel B				
I212	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	12 30
I213	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	12 30
I254	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	8 12
I253	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	8 12
I214	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	12 30
I215	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	12 30
I216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	12 30
I250	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	8 12
I251	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	8 12
I252	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	8 12
I217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	12 30
I218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	12 30
I219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	12 30
I220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	12 30
I221	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	12 30
I222	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	12 31
I223	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	12 31
I225	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	12 31
I224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	12 31
I226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	12 31
I227	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	12 31
I228	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	12 31
I229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	12 31
I230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	12 31
I231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	12 31
I232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	12 31
I233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	12 31
I234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	12 31
I235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	12 31
I237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	12 31
I236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	12 31
I238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	12 31
I239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	12 31
I240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	12 31
I241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	12 31
I242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	12 31
I243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	12 31
I244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	12 31
I245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	12 31
	Reset				
I246		DDR_50S		MEM_RESET_L	28 29 30

SYNC_MASTER=D7_DAVE SYNC_DATE=12/12/2011

PAGE TITLE: DDR3 Constraints

Apple Inc.

DRAWING NUMBER: 051-9509 SIZE: D

REVISION: 4.2.0

BRANCH

PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing	Net	
I440	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<15>	71
I441	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<15>	71
I442		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<15>	9 71
I443		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<15>	9 71
I444	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<15>	9 71
I445	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<15>	9 71
I446		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<15>	71
I447		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<15>	71
I449	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<14>	71
I448	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<14>	71
I450		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<14>	9 71
I452		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<14>	9 71
I451	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<14>	9 71
I453	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<14>	9 71
I455		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<14>	71
I454		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<14>	71
I457	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<13>	71
I456	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<13>	71
I458		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<13>	9 71
I460		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<13>	9 71
I459	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<13>	9 71
I461	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<13>	9 71
I463		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<13>	71
I462		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<13>	71
I464	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<12>	71
I465	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<12>	71
I467		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<12>	9 71
I466		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<12>	9 71
I468	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<12>	9 71
I470	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<12>	9 71
I469		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<12>	71
I471		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<12>	71
I472	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<11>	71
I474	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<11>	71
I473		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<11>	9 71
I475		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<11>	9 71
I477	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<11>	9 71
I476	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<11>	9 71
I478		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<11>	71
I479		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<11>	71
I480	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<10>	71
I482	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<10>	71
I481		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<10>	9 71
I484		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<10>	9 71
I483	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<10>	9 71
I485	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<10>	9 71
I487		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<10>	71
I486		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<10>	71
I489	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<9>	71
I488	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<9>	71
I491		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<9>	9 71
I490		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<9>	9 71
I492	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<9>	9 71
I494	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<9>	9 71
I493		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<9>	71
I496		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<9>	71
I495	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<8>	71
I497	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<8>	71
I498		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<8>	9 71
I499		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<8>	9 71
I501	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<8>	9 71
I500	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<8>	9 71
I502		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<8>	71
I503		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<8>	71

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing	Net	
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<7>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<7>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<7>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<7>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<7>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<7>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<7>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<7>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<6>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<6>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<6>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<6>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<6>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<6>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<6>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<6>	71
[illegible]	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<5>	71
[illegible]	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<5>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<5>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<5>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<5>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<5>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<5>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<5>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<4>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<4>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<4>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<4>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<4>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<4>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<4>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<4>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<3>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<3>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<3>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<3>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<3>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<3>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<3>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<3>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<2>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<2>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<2>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<2>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<2>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<2>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<2>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<2>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<1>	71
[illegible]	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<1>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<1>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<1>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<1>	9 71
[illegible]	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<1>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<1>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<1>	71
[illegible]	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<0>	71
[illegible]	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<0>	71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<0>	9 71
[illegible]		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<0>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<0>	9 71
[illegible]	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<0>	9 71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<0>	71
[illegible]		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<0>	71

CPU PCIe Clocks

	Electrical Contraint Set	Physical	Spacing	Net	
[illegible]	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P	10 71
[illegible]	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N	10 71

CPU PCIe Compensation

	Electrical Contraint Set	Physical	Spacing	Net	
[illegible]		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_COMP	10

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CPU PCIe Constraints

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Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	50_OHM_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

SSD x2 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_SSD_R2D	PCIE_SSD_R2D	*	PCIE_SAME_DIR
PCIE_SSD_D2R	PCIE_SSD_D2R	*	PCIE_SAME_DIR
PCIE_SSD_D2R	PCIE_SSD_R2D	*	PCIE_ALT_DIR
PCIE_SSD_D2R	*	*	PCIE_ISO
PCIE_SSD_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PCIe (PCH)

	Electrical Contraint Set	Physical	Spacing		
	x4 Thunderbolt				
I132	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_P<3..0>	34
I133	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_N<3..0>	34
I134		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_P<3..0>	18 34
I135		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_N<3..0>	18 34
I136	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_P<3..0>	18 34
I137	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_N<3..0>	18 34
I138		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_P<3..0>	34
I139		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_N<3..0>	34
I140	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_P	18 34
I141	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_N	18 34
	x2 SSD				
I174	PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_P<1>	41
I175	PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_N<1>	41
I176		PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_C_P<1>	18 41
I177		PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_C_N<1>	18 41
I178	PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_P<1>	18 41
I179	PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_N<1>	18 41
I180		PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_C_P<1>	41
I181		PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_C_N<1>	41
I186	PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_P<0>	41
I187	PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_N<0>	41
I188		PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_C_P<0>	18 41
I189		PCIE_PHY	PCIE_SSD_R2D	PCIE_SSD_R2D_C_N<0>	18 41
I190	PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_P<0>	18 41
I191	PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_N<0>	18 41
I192		PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_C_P<0>	41
I193		PCIE_PHY	PCIE_SSD_D2R	PCIE_SSD_D2R_C_N<0>	41
I182	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_P	18 41
I183	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_N	18 41
	x1 AirPort				
I142	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_P	33
I143	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_N	33
I144		PCIE_PHY	PCIE	PCIE_AP_R2D_C_P	18 33
I146		PCIE_PHY	PCIE	PCIE_AP_R2D_C_N	18 33
I145	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_P	18 33
I147	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_N	18 33
I185		PCIE_PHY	PCIE	PCIE_AP_D2R_C_P	
I184		PCIE_PHY	PCIE	PCIE_AP_D2R_C_N	
I148	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_P	18 33
I149	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_N	18 33
	x1 Caesar IV				
I151	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_P	37
I150	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_N	37
I152		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_P	18 37
I154		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_N	18 37
I153	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_P	18 37
I155	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_N	18 37
I156		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_P	37
I157		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_N	37
I158	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_P	18 37
I159	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_N	18 37

DMI

	Electrical Contraint Set	Physical	Spacing		
	DMI				
I166	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_P<3..0>	10 19
I165	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_N<3..0>	10 19
I167	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_P<3..0>	10 19
I168	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_N<3..0>	10 19
I169	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_P	11 18
I170	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_N	11 18
	DMI Compensation				
I171		COMP_DMI_PHY	COMP_PCIE	PCH_DMI_COMP	19

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SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD
FDI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_DIFF_PHY	*	FDI_85D
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

FDI Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Imp	Design	Iso	Design	Comments
6-1/6-2	85	85	12	11.81	FDI main length

FDI Compensation Rules (mils)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	55	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

SATA

	Electrical Contraint Set	Physical	Spacing	Net	
	PCH SATA Port 0 (HDD)				
I90	SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_P	41
I92	SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_N	41
I88		SATA_PHY	SATA	SATA_HDD_R2D_C_P	18 41
I89		SATA_PHY	SATA	SATA_HDD_R2D_C_N	18 41
I94	SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_P	18 41
I95	SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_N	18 41
I91		SATA_PHY	SATA	SATA_HDD_D2R_C_P	41
I93		SATA_PHY	SATA	SATA_HDD_D2R_C_N	41
	PCH SATA Port 1 (SSD)				
I97	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_P	41
I99	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_N	41
I96		SATA_PHY	SATA	SATA_SSD_R2D_C_P	18 41
I98		SATA_PHY	SATA	SATA_SSD_R2D_C_N	18 41
I103	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_P	18 41
I102	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_N	18 41
I100		SATA_PHY	SATA	SATA_SSD_D2R_C_P	41
I101		SATA_PHY	SATA	SATA_SSD_D2R_C_N	41
	SSD PCIe/SATA Mux Output				
I115	PCIE_SATA_R2D_MUX_CONN	SATA_PHY	SATA	PCIE_SATA_SSD_R2D_P	41
I116	PCIE_SATA_R2D_MUX_CONN	SATA_PHY	SATA	PCIE_SATA_SSD_R2D_N	41
I117	PCIE_SATA_D2R_MUX_CONN	SATA_PHY	SATA	PCIE_SATA_SSD_D2R_P	41
I118	PCIE_SATA_D2R_MUX_CONN	SATA_PHY	SATA	PCIE_SATA_SSD_D2R_N	41
	PCH SATA Compensation				
I104		COMP_SATA_PHY	COMP_SATA	PCH_SATAICOMP	18
I106		COMP_SATA_PHY	COMP_SATA	PCH_SATA3COMP	18
I105		COMP_SATA_PHY	COMP_SATA	PCH_SATA3RBIAS	18

FDI

	Electrical Contraint Set	Physical	Spacing	Net	
	FDI				
I109	FDI_TX	FDI_DIFF_PHY	FDI	CPU_FDI_TX_P<7..0>	8 10
I110	FDI_TX	FDI_DIFF_PHY	FDI	CPU_FDI_TX_N<7..0>	8 10
I111		FDI_SE_PHY	FDI	CPU_FDI_FSYNC<1..0>	8 10
I112		FDI_SE_PHY	FDI	CPU_FDI_LSYNC<1..0>	8 10
I113		FDI_SE_PHY	FDI	CPU_FDI_INT	8 10
	FDI Compensation				
I114		COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO	10

XDP

	Electrical Contraint Set	Physical	Spacing	Net	
	CPU XDP				
I37		XDP_PHY	XDP	XDP_BPM_L<7..0>	11 25
I38		XDP_PHY	XDP	CPU_CFG<17..0>	8 10 15 25
I39	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P	11 15
I40	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N	11 15
I41		CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_P	15 18 25
I43		CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_N	15 18 25
I42		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P	25
I44		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N	25
I45		XDP_PHY	CLK_JTAG	XDP_CPU_TCK	11 25
I46		XDP_PHY	XDP	XDP_CPU_TMS	11 25
I47		XDP_PHY	XDP	XDP_CPU_TDI	11 25
I48		XDP_PHY	XDP	XDP_CPU_TDO	11 25
	PCH XDP				
I49		XDP_PHY	CLK_JTAG	XDP_PCH_TCK	18 25
I50		XDP_PHY	XDP	XDP_PCH_TMS	18 25
I51		XDP_PHY	XDP	XDP_PCH_TDI	18 25
I52		XDP_PHY	XDP	XDP_PCH_TDO	18 25

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PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCI Clock				
I333	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	18 26
I334	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	20 26

LPC

Electrical Contraint Set	Physical	Spacing	Net	Pages
LPC				
I336	LPC_55S	LPC	LPC_AD<3..0>	18 44 46
I335	LPC_55S	LPC	LPC_R_AD<3..0>	18
I338	LPC_55S	LPC	LPC_FRAME_L	18 44 46
I337	LPC_55S	LPC	LFRAME_L	18
LPC Clocks				
I339	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	26 46
I340	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 26
I341	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	26 44
I342	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	20 26

PCH Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCH Reference Clock				
I385	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB	26
I406	CLK_PCH_55S	CLK_PCH	PCH_CLK25M_XTALIN	18 26
PCH Ref Clock Comp				
I346	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP	18
PCH RTC 32K				
I347	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 26
I348	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 26
I352	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	26
SMC 32K				
I349	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	19 45
I350	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	44 45

25 MHz Reference Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
25M Reference Crystal				
I379	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	26
I380	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	26
I381	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	26
25M Reference Clocks				
I384	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET	26 37
I386	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R	26
I388	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT	26 34
I387	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R	34

HDA

Electrical Contraint Set	Physical	Spacing	Net	Pages
HDA				
I353	HDA_55S	HDA	HDA_BIT_CLK	18 52
I361	HDA_55S	HDA	HDA_BIT_CLK_R	18
I363	HDA_55S	HDA	HDA_RST_L	18 52
I362	HDA_55S	HDA	HDA_RST_R_L	18
I364	HDA_55S	HDA	HDA_SDOUT	18 52
I367	HDA_55S	HDA	HDA_SDOUT_R	15 18
I366	HDA_55S	HDA	HDA_SYNC	15 18 52
I368	HDA_55S	HDA	HDA_SYNC_R	18
I365	HDA_55S	HDA	HDA_SDIN0	18 52
I369	HDA_55S	HDA	AUD_SDI_R	52
SPDIF				
I370		HDA	AUD_SPDIF_CHIP	52
I371		HDA	AUD_SPDIF_OUT	52 56

SPI Bootrom

Electrical Contraint Set	Physical	Spacing	Net	Pages
SPI ROM				
I391	SPI_50S	SPI	SPI_CLK_R	18 46
I392	SPI_50S	SPI	SPI_CLK	46
I393	SPI_50S	SPI	SPI_ALT_CLK	46
I408	SPI_50S	SPI	SPI_SMC_CLK	44 45
I407	SPI_50S	SPI	SPI_MLB_CLK	45 46
I394	SPI_50S	SPI	SPI_CS0_R_L	18 46
I395	SPI_50S	SPI	SPI_CS0_L	46
I396	SPI_50S	SPI	SPI_ALT_CS_L	46
I409	SPI_50S	SPI	SPI_SMC_CS_L	44 45
I397	SPI_50S	SPI	SPI_MLB_CS_L	45 46
I399	SPI_50S	SPI	SPI_MOSI_R	18 46
I400	SPI_50S	SPI	SPI_MOSI	46
I401	SPI_50S	SPI	SPI_ALT_MOSI	46
I410	SPI_50S	SPI	SPI_SMC_MOSI	44 45
I411	SPI_50S	SPI	SPI_MLB_MOSI	45 46
I404	SPI_50S	SPI	SPI_MISO	18 46
I403	SPI_50S	SPI	SPI_ALT_MISO	46
I412	SPI_50S	SPI	SPI_SMC_MISO	44 45
I413	SPI_50S	SPI	SPI_MLB_MISO	45 46
I405	SPI_50S	SPI	SPIROM_USE_MLB	21 46

SYNC_MASTER=D7_DAVE SYNC_DATE=12/12/2011

PAGE TITLE

PCH and BR Constraints

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

8 7 6 5 4 3 2 1

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D

USB-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=3:1_SPACING	?
USB2_ISO	TOP,BOTTOM	=3:1_SPACING	?
USB3_ISO	*	=5.5:1_SPACING	?
USB3_ISO	TOP,BOTTOM	=5.5:1_SPACING	?

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
12.2.1	90	90	12	11.81	USB 2.0
13.3.1	85	85	20	21.65	USB 3.0

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB3	*	*	USB3_ISO

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S
CIV_SPI	*	SPI_55S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
ENET_DIFF	*	*	ENET_DIFF_ISO	
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF	
ENET_TRANS	*	*	ENET_TRANS_ISO	2 kV isolation
COMP_ENET	*	*	COMP_ENET_ISO	
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF	

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

Camera Processor-to-Camera Sensor I/F (SMIA/MIPI)

Camera Processor's SMIA Interface Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D

Camera Processor's SMIA Interface Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF

USB 3.0 and USB 2.0 Trixies Muxing

	Electrical Contraint Set	Physical	Spacing	Net	Pages
External Port A (J4600)					
I320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_P	42
I321	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_N	42
I391		USB3_PHY	USB3	USB3_EXTA_RX_F_P	20 42
I392		USB3_PHY	USB3	USB3_EXTA_RX_F_N	20 42
I413	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_P	20 42
I414	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_N	20 42
I415		USB3_PHY	USB3	USB3_EXTA_TX_F_P	42
I416		USB3_PHY	USB3	USB3_EXTA_TX_F_N	42
I493		USB3_PHY	USB3	USB3_EXTA_TX_C_P	42
I494		USB3_PHY	USB3	USB3_EXTA_TX_C_N	42
I417	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_P	20 42
I418	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_N	20 42
I419		USB2_PHY	USB2	USB2_EXTA_MUXED_P	42
I420		USB2_PHY	USB2	USB2_EXTA_MUXED_N	42
I393		USB2_PHY	USB2	USB2_EXTA_P	42
I394		USB2_PHY	USB2	USB2_EXTA_N	42
External Port B (J4610)					
I421	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P	42
I422	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N	42
I423		USB3_PHY	USB3	USB3_EXTB_RX_F_P	20 42
I424		USB3_PHY	USB3	USB3_EXTB_RX_F_N	20 42
I425	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P	20 42
I427	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N	20 42
I426		USB3_PHY	USB3	USB3_EXTB_TX_F_P	42
I428		USB3_PHY	USB3	USB3_EXTB_TX_F_N	42
I495		USB3_PHY	USB3	USB3_EXTB_TX_C_P	42
I496		USB3_PHY	USB3	USB3_EXTB_TX_C_N	42
I429	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P	20 42
I430	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N	20 42
I436		USB2_PHY	USB2	USB_PCH_9_P	20 42
I435		USB2_PHY	USB2	USB_PCH_9_N	20 42
I432		USB2_PHY	USB2	USB2_EXTB_MUXED_P	42
I431		USB2_PHY	USB2	USB2_EXTB_MUXED_N	42
I434		USB2_PHY	USB2	USB2_EXTB_P	42
I433		USB2_PHY	USB2	USB2_EXTB_N	42
External Port C (J4700)					
I437	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_P	43
I439	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_N	43
I438		USB3_PHY	USB3	USB3_EXTC_RX_F_P	20 43
I440		USB3_PHY	USB3	USB3_EXTC_RX_F_N	20 43
I441	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_P	20 43
I442	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_N	20 43
I444		USB3_PHY	USB3	USB3_EXTC_TX_F_P	43
I443		USB3_PHY	USB3	USB3_EXTC_TX_F_N	43
I498		USB3_PHY	USB3	USB3_EXTC_TX_C_P	43
I497		USB3_PHY	USB3	USB3_EXTC_TX_C_N	43
I446	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P	20 43
I445	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N	20 43
I450		USB2_PHY	USB2	USB2_EXTC_P	43
I452		USB2_PHY	USB2	USB2_EXTC_N	43
External Port D (J4710)					
I453	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P	43
I455	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N	43
I454		USB3_PHY	USB3	USB3_EXTD_RX_F_P	20 43
I456		USB3_PHY	USB3	USB3_EXTD_RX_F_N	20 43
I457	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P	20 43
I458	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N	20 43
I460		USB3_PHY	USB3	USB3_EXTD_TX_F_P	43
I459		USB3_PHY	USB3	USB3_EXTD_TX_F_N	43
I499		USB3_PHY	USB3	USB3_EXTD_TX_C_P	43
I500		USB3_PHY	USB3	USB3_EXTD_TX_C_N	43
I462	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P	20 43
I461	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N	20 43
I463		USB2_PHY	USB2	USB_PCH_10_P	20 43
I464		USB2_PHY	USB2	USB_PCH_10_N	20 43
I465		USB2_PHY	USB2	USB2_EXTD_MUXED_P	43
I467		USB2_PHY	USB2	USB2_EXTD_MUXED_N	43
I466		USB2_PHY	USB2	USB2_EXTD_P	43
I468		USB2_PHY	USB2	USB2_EXTD_N	43
Camera (J3510)					
I324	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P	40
I326	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N	40
PCH USB Compensation					
I384		PCH_55S	COMP_PCH	PCH_USB_RBIAS	20

RMH Love

	Electrical Contraint Set	Physical	Spacing	Net	Pages
USB 2.0 Hub					
I401	USB2_HUB	USB2_PHY	USB2	USB_PCH_7_P	20 27
I402	USB2_HUB	USB2_PHY	USB2	USB_PCH_7_N	20 27
I409	USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_P	27 33
I410	USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_N	27 33
I505	USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_MUX_P	33
I506	USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_MUX_N	33
I376	USB2_HUB_RES	USB2_PHY	USB2	USB_SMC_P	27 44
I378	USB2_HUB_RES	USB2_PHY	USB2	USB_SMC_N	27 44
I469		USB2_PHY	USB2	USB_HUB_2_P	27
I470		USB2_PHY	USB2	USB_HUB_2_N	27
USB 2.0 Hub Compensation					
I471		PCH_55S	COMP_PCH	USB_HUB_RBIAS	27
USB 2.0 Hub Crystal					
I472		CLK_XTAL	XTAL	USB_HUB_XTAL1	27
I473		CLK_XTAL	XTAL	USB_HUB_XTAL2	27

Et tu Brute?

	Electrical Contraint Set	Physical	Spacing	Net	Pages
Ethernet					
I474	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>	37 38
I475	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>	37 38
I476		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>	38
I477		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>	38
I519			ENET_TRANS	ENETCONN_MCT0	38
I520			ENET_TRANS	ENETCONN_MCT1	38
I521			ENET_TRANS	ENETCONN_MCT2	38
I522			ENET_TRANS	ENETCONN_MCT3	38
I523			ENET_TRANS	ENETCONN_MCT_BS	38
I478		ENET_COMP_PHY	COMP_ENET	ENET_RDAC	37
SD					
I479	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>	37
I480		SD_PHY	SD	SDCONN_DATA<7..0>	37 39
I515		SD_PHY	SD	SDCONN_DATA_R_<7..0>	39
I481	SD_CMD	SD_PHY	SD	ENET_SD_CMD	37
I482		SD_PHY	SD	SDCONN_CMD	37 39
I516		SD_PHY	SD	SDCONN_CMD_R	39
I483	SD_CLK	SD_PHY	SD	ENET_SD_CLK	37
I484		SD_PHY	SD	SDCONN_CLK	37 39
I486		SD_PHY	SD	SDCONN_CLK_R	39
I485		SD_PHY	SD	ENET_MEDIA_SENSE	15 18 37
I488		SD_PHY	SD	ENET_SD_DETECT_L	37 39
I487		SD_PHY	SD	SDCONN_WP	37 39
CIV SPI					
I490		CIV_SPI	SPI	ENET_SCLK	37
I489		CIV_SPI	SPI	ENET_MISO	37
I492		CIV_SPI	SPI	ENET_MOSI	37
I491		CIV_SPI	SPI	ENET_CS_L	37

Camera Processor-Camera Sensor I/F

	Electrical Contraint Set	Physical	Spacing	Net	Pages
I501	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P	40
I502	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N	40
I526	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_F_P	40
I527	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_F_N	40
I504	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P	40
I503	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N	40
I525	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_F_P	40
I524	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_F_N	40
I507		SPI_50S	SPI	CAM_SF_CLK	40
I508		SPI_50S	SPI	CAM_SF_CLK_R	40
I509		SPI_50S	SPI	CAM_SF_DIN	40
I510		SPI_50S	SPI	CAM_SF_DIN_R	40
I511		SPI_50S	SPI	CAM_SF_CS_L	40
I513		SPI_50S	SPI	CAM_SF_WP_L	40
I512		SPI_50S	SPI	CAM_SF_DOUT	40
I514		SPI_50S	SPI	CAM_SF_DOUT_R	40
I517		SMB_PHY	SMB	I2C_CAMSENSOR_SDA	40
I518		SMB_PHY	SMB	I2C_CAMSENSOR_SCL	40

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMBus

	Electrical Contraint Set	Physical	Spacing	Net	
	SMC				
I10		SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	44 47
I9		SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	44 47
I11		SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	44 47
I12		SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	44 47
I13		SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	44 47
I14		SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	44 47
I15		SMB_PHY	SMB	SMBUS_SMC_3_SCL	44 47
I16		SMB_PHY	SMB	SMBUS_SMC_3_SDA	44 47
I19		SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	44 45
I20		SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	44 45
	PCH				
I21		TBT_I2C_55S	TBT_I2C	SMBUS_PCH_CLK	18 47
I52		TBT_I2C_55S	TBT_I2C	SMBUS_PCH_DATA	18 47
I53		SMB_PHY	SMB	SML_PCH_0_CLK	18 47
I54		SMB_PHY	SMB	SML_PCH_0_DATA	18 47
	Display TCon				
I55		SMB_PHY	SMB	SMB_DP_TCON_SCL	47 81
I56		SMB_PHY	SMB	SMB_DP_TCON_SDA	47 81

Temperature Sense

	Electrical Contraint Set	Physical	Spacing	Net	
	EMC1414-1 (Production)				
I57	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_P	50
I58	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_N	50
I59	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_2_P	50
I60	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_2_N	50
I65	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_P	6 50
I66	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_N	6 50
	TMP423 (Development)				
I67	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_P	50
I68	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_N	50
I69	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_P	50
I70	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_N	50
I61	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_SKIN_P	50
I62	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_SKIN_N	50
I71	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_P	50
I72	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_N	50
	HDD Out-of-Band				
I73			SENSE	SMC_HDD_OOB_TEMP	
I74			SENSE	HDD_OOB_TEMP_CONN	
I75			SENSE	HDD_OOB_TEMP_FILT	
I76			SENSE	HDD_OOB_TEMP_R	
	SSD Out-of-Band				
I77			SENSE	SMC_SSD_OOB_TEMP	
I78			SENSE	SMC_SSD_TEMP_CTL	
I79			SENSE	SSD_OOB_TEMP	

SMC

	Electrical Contraint Set	Physical	Spacing	Net	
	SMC				
I1		CLK_XTAL	XTAL	SMC_XTAL	44 45
I2		CLK_XTAL	XTAL	SMC_EXTAL	44 45

Current/Voltage Sense

	Electrical Contraint Set	Physical	Spacing	Net	
	Common				
I3			SENSE	GND_SMC_AVSS	44 45 48 49
	12V S5 (System Total)				
I4	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_P	48
I5	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_N	48
I6			SENSE	ISNS_P12VG3H_R	48
I7			SENSE	ISNS_P12VG3H	45 48
I8			SENSE	VSNS_P12VG3H	45 48
	12V S0 (GPU Core)				
I22	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_P	48
I23	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_N	48
I24			SENSE	ISNS_P12VS0_GPUCORE_R	48
I25			SENSE	ISNS_P12VS0_GPUCORE	45 48
I26			SENSE	VSNS_P12VS0_GPUCORE	45 48
	HDD				
I80	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_P	48
I82	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_N	48
I81			SENSE	ISNS_HDDS0_R	48
I83			SENSE	ISNS_HDDS0	45 48
I84			SENSE	VSNS_HDDS0	45 48
	SSD				
I85	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_P	49
I87	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_N	49
I86			SENSE	ISNS_SSDS0_R	49
I88			SENSE	ISNS_SSDS0	45 49
I89			SENSE	VSNS_SSDS0	45 49
	VDDQ S3 (DDR)				
I33	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_P	49
I32	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_N	49
I35			SENSE	ISNS_VDDQS3_DDR_R	49
I34			SENSE	ISNS_VDDQS3_DDR	45 49
I36			SENSE	VSNS_VDDQS3_DDR	45 49
	VDDQ S0 (GPU Uncore)				
I37	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUUC_P	48
I38	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUUC_N	48
I39			SENSE	ISNS_P12VS0_GPUUC_R	48
I40			SENSE	ISNS_P12VS0_GPUUNCORE	45 48
I41			SENSE	VSNS_P12VS0_GPUUNCORE	45 48
	CPU Core				
I42	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_CPUCORE_P	48
I43	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_CPUCORE_N	48
I44			SENSE	ISNS_CPUCORE_FB	48
I45			SENSE	ISNS_CPUCORE	45 48
I46			SENSE	VSNS_CPUCORE	45 48
	CPU AXG				
I47	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_CPUAXG_P	48
I48	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_CPUAXG_N	48
I49			SENSE	ISNS_CPUAXG_FB	48
I50			SENSE	ISNS_CPUAXG	45 48
I51			SENSE	VSNS_CPUAXG	45 48

DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

VDDQ S3 (1.5V)/VTT S0

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I43	POWER	POWER	5V			REG_V5IN_U7700	68
Local Ground							
I44	GND	GND	0V			AGND_VDDQS3	68
VDDQ S3							
I45	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3	68
I47	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L	68
I46	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3	68
I48	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC	68
I49	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3	68
I50	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R	68
I52	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3	68
I51	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3	68
I54	POWER	POWER	1.5V			PPVDDQ_S3_SENSE	
I53	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS	68
I55	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF	68
I56	VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN	68
I57	VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE	68
I58	VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP	68
I59	VR_CTL_PHY	VR_CTL				LDO_DDRVTTS0_SNS	68
Output Bus							
I60	POWER	POWER	1.5V			PPVDDQ_S3	6
I61	POWER_DDR	POWER_DDR	0.75V			PPDDRVTT_S3	6
I62	POWER_DDR	POWER_DDR	0.75V			PPDDRVTT_S0	6
FET Switched							
I63	POWER	POWER	1.5V			PP1V5_S0	6
Sensed							
I64	POWER	POWER	1.5V			PPVDDQ_S3_DDR	6

CPU VccIO/ PCH 1.05V S0

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I1	POWER	POWER	5V			REG_VCC_U7400	65
I2	POWER	POWER	5V			REG_PVCC_U7400	65
Local Ground							
I3	GND	GND	0V			AGND_P1V05S0	65
1.05V S0							
I5	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0	65
I4	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L	65
I7	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0	65
I6	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC	65
I8	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0	65
I9	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R	65
I10	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0	65
I12	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V05S0	65
I11	VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET	65
I14	VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO	65
I13 VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_P	13 65
I15 VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_N	13 65
I16	SNS_DIFF_PHY	SENSE				SNS_P1V05S0_XW_P	65
I17	SNS_DIFF_PHY	SENSE				SNS_P1V05S0_XW_N	65
I19		SENSE				REG_P1V05S0_FB	65
I18		SENSE				REG_P1V05S0_RTN	65
I21	VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF	65
I20	VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL	65
Output Bus							
I22	POWER	POWER	1.05V			PP1V05_S0	6
FET Switched							
I23	POWER	POWER	1.05V			PP1V05_TBTLC	6
I66	POWER	POWER	1.05V			PP1V05_TBTCIO	6

CPU VccSA

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I24	POWER	POWER	5V			REG_VCC_U7500	66
I25	POWER	POWER	5V			REG_PVCC_U7500	66
Local Ground							
I26	GND	GND	0V			AGND_VCCSAS0	66
VCCIO							
I27	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0	66
I28	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VCCSAS0	66
I29	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VCCSAS0_RC	66
I31	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0	66
I30	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VCCSAS0	66
I33	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VCCSAS0	66
I32	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSET	66
I35	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO	66
I34		SENSE				SNS_CPU_VCCSA	13 66
I37 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_P	66
I36 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_N	66
I38		SENSE				REG_VCCSAS0_FB	66
I39		SENSE				REG_VCCSAS0_RTN	66
I40	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF	66
I41	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL	66
Output Bus							
I42	POWER	POWER	0.925V			PPVCCSA_S0	6

SYNC_MASTER=D7_DAVE SYNC_DATE=12/12/2011

PAGE TITLE
VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

CPU Core Phases

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	12V			PP12V_S0_CPUCORE_FLT	62 63 64
	POWER	POWER	5V			REG_VCC_U7100	62
Local Ground							
	GND	GND	0V			AGND_CPU	62 63 64
Phase 1							
	POWER	POWER	12V			REG_LVCC_U7210	63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	62 63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	62
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUCORE_1	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUCORE_1	63
	POWER	POWER	1.1V			PPCPUCORE_S0_SENSE_1	63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_P	62 63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_N	63
		SENSE				REG_ISENCORE_1_NR	62 63
Phase 2							
	POWER	POWER	12V			REG_LVCC_U7230	63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	62 63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	62
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUCORE_2	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUCORE_2	63
	POWER	POWER	1.1V			PPCPUCORE_S0_SENSE_2	63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_P	62 63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_N	63
		SENSE				REG_ISENCORE_2_NR	62 63
Phase 3							
	POWER	POWER	12V			REG_LVCC_U7250	63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	62 63
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	62
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	63
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUCORE_3	63
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUCORE_3	63
	POWER	POWER	1.1V			PPCPUCORE_S0_SENSE_3	63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_P	62 63
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_N	63
		SENSE				REG_ISENCORE_3_NR	62 63

CPU AXG Phase and Core Controller

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
AXG							
	POWER	POWER	12V			REG_LVCC_U7330	64
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	62 64
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	62
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	64
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	64
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	64
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	64
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUAXG	64
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUAXG	64
	POWER	POWER	1.1V			PPCPUAXG_S0_SENSE	64
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_P	64
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_N	64
		SENSE				REG_ISENAXG_PR	62 64
		SENSE				REG_ISENAXG_NR	62 64
ISL6364							
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	62
	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	62
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	62
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	62
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	62
	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	62
VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_P	13 62
VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_N	13 62
	SNS_DIFF_PHY	SENSE				SNS_VCORE_R_P	62
	SNS_DIFF_PHY	SENSE				SNS_VCORE_R_N	62
	SNS_DIFF_PHY	SENSE	1.1V			SNS_VCORE_XW_P	62
	SNS_DIFF_PHY	SENSE	0V			SNS_VCORE_XW_N	62
		SENSE				REG_CPUCORE_VSEN	62
		SENSE				REG_CPUCORE_RGND	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	48 62
	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FDVID	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	62
	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	62
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	62
	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	62
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	62
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	62
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	62
VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_P	13 62
VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_N	13 62
	SNS_DIFF_PHY	SENSE				SNS_VAXG_R_P	62
	SNS_DIFF_PHY	SENSE				SNS_VAXG_R_N	62
	SNS_DIFF_PHY	SENSE	1.1V			SNS_VAXG_XW_P	62
	SNS_DIFF_PHY	SENSE	0V			SNS_VAXG_XW_N	62
		SENSE				REG_CPUAXG_VSEN	62
		SENSE				REG_CPUAXG_RGND	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	48 62
	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	62
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	62
	VR_VID_PHY	VR_VID				CPU_VIDSCLK	13 62
	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	13
	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	13 62
	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	13
	VR_VID_PHY	VR_VID				CPU_VIDSOUT	13 62
	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	13
Output Bus							
	POWER	POWER	1.1V			PPVCORE_S0_CPU	6
	POWER	POWER	1.1V			PPVAXG_S0	6

SYNC_MASTER=D7_DAVE SYNC_DATE=12/12/2011

PAGE TITLE
CPU VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

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GPU Core Phases and Controller

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	12V			REG_PVCC_U8900	80
	POWER	POWER	5V			REG_VCC_U8900	80
Phase 1							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_1	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_ISEN_GPUCORE_1	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_GPUCORE_1	80
Phase 2							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_2	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_2	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_ISEN_GPUCORE_2	80
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_GPUCORE_2	80
ISL6568							
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ICOMP	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_OCSET	80
	VR_CTL_PHY	VR_CTL				GPUCORE_ICOMP_R	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_IREF	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ISUM	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_COMP	80
	VR_CTL_PHY	VR_CTL				GPUCORE_COMP_RC	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_VDIFF	80
	VR_CTL_PHY	VR_CTL				GPUCORE_VDIFF_R	80
	VR_CTL_PHY	VR_CTL				GPUCORE_VDIFF_RC	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_FB	80
VSNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_GPU_CORE_P	79 80
VSNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_GPU_CORE_N	79 80
		SENSE				REG_GPUCORE_VSEN	80
		SENSE				REG_GPUCORE_RGND	80
	VR_VID_PHY	VR_VID				REG_GPUCORE_VID4	80
	VR_VID_PHY	VR_VID				REG_GPUCORE_VID3	80
	VR_VID_PHY	VR_VID				REG_GPUCORE_VID2	80
	VR_VID_PHY	VR_VID				REG_GPUCORE_VID1	80
	VR_VID_PHY	VR_VID				REG_GPUCORE_VID0	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_MODE	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_REF	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_OFS	80
	VR_CTL_PHY	VR_CTL				REG_GPUCORE_FS	80
Output Bus							
	POWER	POWER	1.0V			PPVCORE_S0_GPU	6

GPU 1.05V S0

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	5V			REG_VCC_U8300	74
Local Ground							
	GND	GND	0V			P1V05_GPU_AGND	74
1.05V S0							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPU_P1V05S0	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPU_P1V05S0_L	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPU_P1V05S0	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPU_P1V05S0_RC	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPU_P1V05S0	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPU_P1V05S0_R	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPU_P1V05S0	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_OCSET_R	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_VO_R	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_OCSET	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_VO	74
VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_P	74 79
VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_N	74 79
		SENSE				REG_GPU_P1V05S0_FB	74
		SENSE				REG_GPU_P1V05S0_RTN	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_SREF	74
	VR_CTL_PHY	VR_CTL				REG_GPU_P1V05S0_FSEL	74
Output Bus							
	POWER	POWER	1.05V			PP1V05_S0_GPU	6

GPU VDDQ

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	5V			REG_VCC_U8350	74
	POWER	POWER	5V			REG_PVCC_U8350	74
Local Ground							
	GND	GND	0V			AGND_GPUVDDQ	74
GPU VDDQ							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUVDDQ	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUVDDQ	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUVDDQ_RC	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUVDDQ	74
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUVDDQ	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_OCSET	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_VO	74
VSNS_GPU_VDDQ	SNS_DIFF_PHY	SENSE				SNS_GPUVDDQ_P	73 74
VSNS_GPU_VDDQ	SNS_DIFF_PHY	SENSE				SNS_GPUVDDQ_N	73 74
		SENSE				REG_GPUVDDQ_FB	74
		SENSE				REG_GPUVDDQ_RTN	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SREF	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_FSEL	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET0	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET1	74
	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET1_R	74
Output Bus							
	POWER	POWER	1.5V			PPVDDQ_S0_GPU	6

3.3V S5/5V S4

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	12V			REG_VIN_U7600	67
	POWER	POWER	5V			REG_VCC1_U7600	67
	POWER	POWER	5V			REG_VCC2_U7600	67
3.3V S5							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P3V3S5	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P3V3S5_RC	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P3V3S5	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R	67
	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB	67
5V S3							
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P5VS4	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P5VS4_RC	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P5VS4	67
	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R	67
	VR_CTL_PHY	VR_CTL				REG_P5VS4_FB	67
Output Bus							
	POWER	POWER	5V			PP5V_S5	6
	POWER	POWER	5V			PP5V_S4	6
	POWER	POWER	3.3V			PP3V3_S5	6
FET Switched							
	POWER	POWER	5V			PP5V_S0	6
	POWER	POWER	3.3V			PP3V3_S4	6
	POWER	POWER	3.3V			PP3V3_S0	6
	POWER	POWER	3.3V			PP3V3_S0_SSD	6
	POWER	POWER	3.3V			PP3V3_ENET	6
	POWER	POWER	3.3V			PP3V3_TBTLC	6
Sensed							
	POWER	POWER	3.3V			PPSSD_S0	6

DDR3 Vref

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Memory Vref							
	POWER	POWER	3.3V			PP3V3_S4_VREFMRGN_DAC	32
	POWER	POWER	3.3V			PP3V3_S4_VREFMRGN_CTRL	32
	POWER_DDR	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_A_S3	6
	POWER_DDR	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_B_S3	6
	POWER_DDR	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_A_S3	6
	POWER_DDR	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_B_S3	6
	POWER_DDR	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_A	11 32
	POWER_DDR	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_B	11 32
	POWER_DDR	POWER_DDR	0.75V			CPU_DDR_VREF	11

3.42V G3H

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
3.42V G3H							
	POWER	VR_SWITCH	12V			P3V42G3H_BOOST	69
	POWER	VR_SWITCH	12V			P3V42G3H_SW	69
	VR_CTL_PHY	VR_CTL				P3V42G3H_FB	69
	VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L	69
Output Bus							
	POWER	POWER	3.425V			PP3V42_G3H	6

1.8V S0

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
1.8V S0							
	POWER	VR_SWITCH	5V			REG_PHASE_P1V8S0	68
	VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB	68
	VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNCH	68
Output Bus							
	POWER	POWER	1.8V			PP1V8_S0	6

HDD S0

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
HDD S0							
	POWER	POWER	5V			PPHDD_S0	6

12V

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER	POWER	12V			PP12V_ACDC	6
FET Switched							
	POWER	POWER	12V			PP12V_S5	6
	POWER	POWER	12V			PP12V_S0	6
Sensed							
	POWER	POWER	12V			PP12V_G3H	6
	POWER	POWER	12V			PP12V_S0_GPUCORE	6
	POWER	POWER	12V			PP12V_S0_GPUUNCORE	6

Ground/Common

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net
Common						
	GND	GND	0V			GND

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP,BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.
Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

	Electrical Contraint Set	Physical	Spacing	Net	
I511		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	34 77
I513		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	34 77
I512	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	34
I514	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	34
I516		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	34 71
I515		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	34 71
I517	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	34
I518	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	34
I519		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	34 77
I521		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	34 77
I520	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	34
I523	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	34
I522		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	34 71
I524		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	34 71
I526	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	34
I525	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	34

	Electrical Contraint Set	Physical	Spacing	Net	
I603	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_P<3..0>	82
I604	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_N<3..0>	82
I686	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	82
I687	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	82
I529	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_P	82
I530	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_N	82
I601		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_P	82
I602		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_N	82

*

	Electrical Contraint Set	Physical	Spacing	Net	
I531		TBT_I2C_55S	TBT_I2C	=I2C_TBTRTR_SCL	34 47
I532		TBT_I2C_55S	TBT_I2C	=I2C_TBTRTR_SDA	34 47
I533	TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK	34
I534	TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI	34
I538	TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO	34
I535	TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L	34

*: Only used on hosts supporting T29 video-in

DisplayPort

Graphics Source

	Electrical Contraint Set	Physical	Spacing	Net	
I642	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<1..0>	77 82
I643	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<1..0>	77 82
I644	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	77 82
I645	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	77 82
I646		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_P	82
I647		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_N	82

Internal Panel

	Electrical Contraint Set	Physical	Spacing	Net	
I605		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_P<3..0>	82
I606		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_N<3..0>	82
I607	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	81 82
I608	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	81 82
I609	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P	81 82
I610	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N	81 82

Internal DP SPDIF

	Electrical Contraint Set	Physical	Spacing	Net	
I611			HDA	DP_INT_SPDIF_AUDIO	52 81

TBT/DP Net Properties

Port A

	Electrical Contraint Set	Physical	Spacing	Net	
I541	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1..0>	34 84
I542	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1..0>	34 84
I544		TBTDP_90D	TBTDP	TBT_A_R2D_P<1..0>	84
I543		TBTDP_90D	TBTDP	TBT_A_R2D_N<1..0>	84
I595	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	34 84
I596	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	34 84
I680	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	34 84
I681	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	34 84
I545		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	84
I547		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	84
I678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	84
I679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	84
I546	DP_A_LSX	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	84
I548	DP_A_LSX	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	84
I549		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1..0>	84
I550		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1..0>	84
I564	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	34 84
I563	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	34 84
I675	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	34 84
I674	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	34 84
I599	TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P	34 84
I600	TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N	34 84
I565		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P	84
I566		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N	84
I648	DP_A_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_P	84
I649	DP_A_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_N	84
I650		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P	84
I651		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N	84

Port B

	Electrical Contraint Set	Physical	Spacing	Net	
I652	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1..0>	34 85
I653	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1..0>	34 85
I655		TBTDP_90D	TBTDP	TBT_B_R2D_P<1..0>	85
I654		TBTDP_90D	TBTDP	TBT_B_R2D_N<1..0>	85
I657	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	34 85
I658	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	34 85
I685	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	34 85
I684	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	34 85
I683		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	85
I682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	85
I656		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	85
I660		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	85
I659	DP_B_LSX	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	85
I662	DP_B_LSX	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	85
I661		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1..0>	85
I663		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1..0>	85
I665	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	34 85
I664	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	34 85
I676	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	34 85
I677	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	34 85
I666	TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P	34 85
I667	TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N	34 85
I669		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P	85
I670		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N	85
I668	DP_B_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_P	85
I672	DP_B_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_N	85
I671		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P	85
I673		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N	85

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PAGE TITLE
TBT/DP Constraints

Apple Inc.

DRAWING NUMBER 051-9509 SIZE D

REVISION 4.2.0

BRANCH

D C B A

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

	Trace-to-Trace				Isolation				
Table	Micro	Design	Strip	Design	Micro	Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=5x_DIELECTRIC	?
GDDR_ISO	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=5x_DIELECTRIC	?
GDDR_CLK2CLK	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=7x_DIELECTRIC	?
GDDR_EDC_ISO	TOP,BOTTOM	=7x_DIELECTRIC	?
GDDR_EDC2EDC	*	=7x_DIELECTRIC	?
GDDR_EDC2EDC	TOP,BOTTOM	=7x_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=5x_DIELECTRIC	?
GDDR_WCK2WCK	TOP,BOTTOM	=5x_DIELECTRIC	?

Constraints (x in {A, B}, y in {0, 1})

Memory Address: MAxy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

Data: DQxy[31:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DQ	*	*	GDDR_ISO
GDDR_*_*_DQ	=SAME	*	GDDR_DQ2DQ

Error Detection: EDCxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_EDC	*	*	GDDR_EDC_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_EDC2EDC

Data Dynamic Bus Inversion: DDBIxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DBI	*	*	GDDR_ISO
GDDR_*_*_DBI	=SAME	*	GDDR_DBI2DBI

Forwarded Clock: WCKxy[1:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_WCK	*	*	GDDR_ISO
GDDR_*_*_WCK	=SAME	*	GDDR_WCK2WCK

GDDR5 Frame Buffer A

	Electrical Contraint Set	Physical	Spacing		
	Memory Address				
I642	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<8..0>	73 75
I643	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<8..0>	73 75
	Address Dynamic Bus Inv				
I644	GDDR_A0_ADBI	GDDR_ADBI_PHY	GDDR_A_0_ADBI	FB_A0_ABI_L	73 75
I645	GDDR_A1_ADBI	GDDR_ADBI_PHY	GDDR_A_1_ADBI	FB_A1_ABI_L	73 75
	Control				
I647	GDDR_A0_CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CKE_L	73 75
I648	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CS_L	73 75
I649	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_WE_L	73 75
I650	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CAS_L	73 75
I651	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_RAS_L	73 75
I652	GDDR_A1_CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CKE_L	73 75
I654	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CS_L	73 75
I653	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_WE_L	73 75
I655	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CAS_L	73 75
I656	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_RAS_L	73 75
	Clock				
I657	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_P	73 75
I658	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_N	73 75
I660	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_P	73 75
I659	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_N	73 75
	Data				
I661	GDDR_A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<7..0>	73 75
I663	GDDR_A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<15..8>	73 75
I662	GDDR_A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<23..16>	73 75
I664	GDDR_A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<31..24>	73 75
I666	GDDR_A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<7..0>	73 75
I665	GDDR_A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<15..8>	73 75
I667	GDDR_A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<23..16>	73 75
I668	GDDR_A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<31..24>	73 75
	Error Detection				
I670	GDDR_A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<0>	73 75
I669	GDDR_A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<1>	73 75
I671	GDDR_A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<2>	73 75
I672	GDDR_A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<3>	73 75
I673	GDDR_A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<0>	73 75
I675	GDDR_A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<1>	73 75
I674	GDDR_A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<2>	73 75
I676	GDDR_A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<3>	73 75
	Data Dynamic Bus Inv				
I677	GDDR_A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<0>	73 75
I678	GDDR_A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<1>	73 75
I680	GDDR_A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<2>	73 75
I679	GDDR_A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<3>	73 75
I682	GDDR_A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<0>	73 75
I681	GDDR_A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<1>	73 75
I683	GDDR_A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<2>	73 75
I684	GDDR_A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<3>	73 75
	Forwarded Clock				
I685	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<0>	73 75
I687	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<0>	73 75
I686	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<1>	73 75
I688	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<1>	73 75
I689	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<0>	73 75
I690	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<0>	73 75
I692	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<1>	73 75
I691	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<1>	73 75

GDDR5 Frame Buffer B

	Electrical Contraint Set	Physical	Spacing		
	Memory Address				
I695	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	73 76
I694	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	73 76
	Address Dynamic Bus Inv				
I696	GDDR_B0_ADBI	GDDR_ADBI_PHY	GDDR_B_0_ADBI	FB_B0_ABI_L	73 76
I697	GDDR_B1_ADBI	GDDR_ADBI_PHY	GDDR_B_1_ADBI	FB_B1_ABI_L	73 76
	Control				
I698	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	73 76
I701	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	73 76
I700	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	73 76
I702	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	73 76
I703	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	73 76
I704	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	73 76
I705	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	73 76
I706	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	73 76
I708	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	73 76
I707	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	73 76
	Clock				
I693	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	73 76
I710	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	73 76
I709	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	73 76
I711	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	73 76
	Data				
I713	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	73 76
I712	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	73 76
I714	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	73 76
I715	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	73 76
I716	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	73 76
I718	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	73 76
I717	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	73 76
I719	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	73 76
	Error Detection				
I720	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	73 76
I721	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	73 76
I723	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	73 76
I722	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	73 76
I725	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	73 76
I724	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	73 76
I726	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	73 76
I727	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	73 76
	Data Dynamic Bus Inv				
I729	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	73 76
I728	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	73 76
I730	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	73 76
I731	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	73 76
I732	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	73 76
I733	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	73 76
I734	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	73 76
I735	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	73 76
	Forwarded Clock				
I736	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	73 76
I737	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	73 76
I739	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	73 76
I738	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	73 76
I740	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	73 76
I741	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	73 76
I743	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	73 76
I742	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	73 76

GPU Misc.

	Electrical Contraint Set	Physical	Spacing		
	SMB				
I564		SMB_PHY	SMB	GPU_SMB_CLK	77 78
I563		SMB_PHY	SMB	GPU_SMB_DAT	77 78
I752		SMB_PHY	SMB	GPU_SMB_CLK_R	47 78
I754		SMB_PHY	SMB	GPU_SMB_DAT_R	47 78

Frame Buffer Reset

	Electrical Contraint Set	Physical	Spacing		
	Reset				
I744	GDDR_A0_RESET	GDDR_50S	GDDR_CTRL	FB_A0_RESET_L	73 75
I745	GDDR_A1_RESET	GDDR_50S	GDDR_CTRL	FB_A1_RESET_L	73 75
I746	GDDR_B0_RESET	GDDR_50S	GDDR_CTRL	FB_B0_RESET_L	73 76
I750	GDDR_B1_RESET	GDDR_50S	GDDR_CTRL	FB_B1_RESET_L	73 76

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GDDR5/GPU Constraints

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Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions
BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	0.45mm	1000

Constraints
BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_CTL	*	BLC_CTL_ISO
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Is it chel'oh or sel'oh?

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus							
I749	POWER	POWER	12V			PP12V_S0_BKLT_FUSED	86
I750	POWER	POWER	12V			PP12V_S0_BKLT_FILT	86
I772	POWER	POWER	12V			PP12V_S0_BKLT_PWR	86
I794	POWER	POWER	12V			PP12V_S0_BKLT_PWR_R	86
I795	POWER	POWER	5V			PP5V_S0_BKLT_R	86
I796	POWER	POWER	3.3V			PP3V3_S0_BKLT_VDDIO_R	86
Local Ground							
I751	BLC_CTL_PHY	BLC_PHASE	0V			PGND_BKLT	86
I799	BLC_CTL_PHY	BLC_PHASE	0V			DGND_BKLT	86
I800	BLC_CTL_PHY	BLC_PHASE	0V			LGND_BKLT	86
Backlight							
I752	POWER_BLC	BLC_PHASE	80V	TRUE		BKLT_PHASE	86
I753	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_GATE	86
I754	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_GATE_R	86
I755	BLC_CTL_PHY	BLC_PHASE	80V	TRUE	TRUE	BKLT_SNUBBER	86
I757	BLC_CTL_PHY	BLC_PHASE	12V	TRUE		BKLT_SW_R	86
I759	BLC_CTL_PHY	BLC_CTL				BKLT_ISET	86
I763	BLC_CTL_PHY	BLC_CTL				BKLT_FLT	86
I765	BLC_CTL_PHY	BLC_CTL				BKLT_FLT_RC	86
I797	SNS_DIFF_PHY	SENSE				BKLT_SW_P	86
I798	SNS_DIFF_PHY	SENSE				BKLT_SW_N	86
I762		SENSE				BKLT_FB	86
I801		BLC_HV	67V			BKLT_FB_XW	86
I764		BLC_HV	67V			BKLT_FB_R	86
I776	POWER_BLC_RET	BLC_CTL				BKLT_ISEN1	86
I775	POWER_BLC_RET	BLC_CTL				BKLT_ISEN2	86
I778	POWER_BLC_RET	BLC_CTL				BKLT_ISEN3	86
I777	POWER_BLC_RET	BLC_CTL				BKLT_ISEN4	86
I780	POWER_BLC_RET	BLC_CTL				BKLT_ISEN5	86
I779	POWER_BLC_RET	BLC_CTL				BKLT_ISEN6	86
I789	POWER_BLC_RET	BLC_HV				BKLT_ISEN1_R	86
I788	POWER_BLC_RET	BLC_HV				BKLT_ISEN2_R	86
I790	POWER_BLC_RET	BLC_HV				BKLT_ISEN3_R	86
I791	POWER_BLC_RET	BLC_HV				BKLT_ISEN4_R	86
I792	POWER_BLC_RET	BLC_HV				BKLT_ISEN5_R	86
I793	POWER_BLC_RET	BLC_HV				BKLT_ISEN6_R	86
I783	POWER_BLC_RET	BLC_HV				LED_RETURN_1	86
I782	POWER_BLC_RET	BLC_HV				LED_RETURN_2	86
I784	POWER_BLC_RET	BLC_HV				LED_RETURN_3	86
I785	POWER_BLC_RET	BLC_HV				LED_RETURN_4	86
I786	POWER_BLC_RET	BLC_HV				LED_RETURN_5	86
I787	POWER_BLC_RET	BLC_HV				LED_RETURN_6	86
Output Bus							
I770	POWER_BLC	BLC_HV	67V			BKLT_BOOST	86
I773	POWER_BLC	BLC_HV	67V			BKLT_BOOST_1	86
I774	POWER_BLC	BLC_HV	67V			BKLT_BOOST_2	86

Cello Miscellaneous

Electrical Contraint Set	Physical	Spacing	Net	
SPI				
I564	SMB_PHY	SMB	BKLT_SCL	86
I563	SMB_PHY	SMB	BKLT_SDA	86

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BLC Constraints

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